

BANARAS HINDU UNIVERSITY

B.Voc. Semester I Examination, 2022-23

Paper: BVCA-101 — Introduction To Computers

Time: Three Hours

Maximum Marks: 70

Instructions: Attempt all questions. Questions of Section A, B and C carry 15, 5 and 1 marks each respectively.

Question 1. Discuss the generations of computers based on their characteristics and processing speed.

OR

Based on block diagram of computer discuss its different components.

Question 2. What is Compiler ? Describe how compiler differs from interpreter ?

OR

Define operating system and give a detailed account on different types of operating system.

SECTION — B |)

Question 3. What is DOS ? Describe the internal and external commands of DOS.

Question 4. Provide the differences between the procedural and object oriented programming.
|

Question 5. Write short notes of the following :

(a) Overloading

(i) Polymorphism

Question 6. Define mail merge in MS-Office.

Question 7. What are the advantages and disadvantages of WAN ?

Question 8. What are the differences between primary | and secondary memory ?

Question 9. Explain the different types of 4 common network topologies.

Question 10. What is formatting in Excel ? Explain the formatting features available in Excel for the numeric data.

SECTION — C

Note: Answer in briefest possible way. |

Question 11. Who is the father of Computers ?

(a) James Gosling

- (b) Charles Babbage
- (c) Dennis Ritchie
- (d) Bjarne Stroustrup

Question 12. Which of the following is the correct abbreviation of COMPUTER ?

- (a) Commonly Occupied Machines Used in Technical and Educational Research
- (b) Commonly Operated Machines Used in Technical and Environmental Research
- (c) Commonly Oriented Machines Used in Technical and Educational Research
- (d) Commonly Operated Machines Used in | Technical and Educational Research

Question 13. What is the full form of CPU ?

- (a) Computer Processing Unit
- (b) Computer Principle Unit
- (c) Central Processing Unit
- (d) Control Processing Unit

Question 14. Which of the following computer language is written in binary codes only ?

- (a) Pascal
- (b) machine language
- (c) C
- (d) C#

Question 15. Which of the following is the smallest unit of data in a computer ?

- (a) Bit
- (b) KB
- (c) Nibble
- (d) Byte

Question 16. Which of the following service allows a user to log in to another computer somewhere on the Internet ?

- (a) e-mail
- (b) UseNet
- (c) Telnet
- (d) FIP

Question 17. The software designed to perform a specific task:

- (a) Synchronous Software
- (b) Package Software
- (c) Application Software
- (d) System Software

Question 18. What do you call a specific instruction designed to do a task ?

- (a) Command
- (b) Process
- (c) Task
- (d) Instruction

Question 19. Public domain software is usually :

- (a) System supported
- (b) Source supported
- (c) Community supported
- (d) Programmer supported

Question 20. OSS stands for :

- (a) Open System Service
- (b) Open Source Software
- (c) Open System Software
- (d) Open Synchronized Software

BANARAS HINDU UNIVERSITY

B.Voc. Semester I Examination, 2022-23

Paper: GEN-102 — Management Concepts Strategic Management

Time: Three hours

Maximum Marks: 70

*Instructions: Attempt all questions. Symbols have their usual meanings.***SECTION — A**

Long Answer type questions: To be answered in about 750 words each: 15x2=30 Marks

Question 1. What are the various steps that a manager is required to take while making a decision? Also enumerate the different kinds of decision that managers need to take.

OR

Question 2. What do you mean by delegation of authority? What are the various steps involved in - delegation of authority? Is it essential to delegate authority in any organization?

Question 3. Environmental-Scanning is necessary for strategy formulation. In the light of the above statement state the relevance of environmental scanning. Also write a brief note about any four techniques of environmental scanning.

OR

Question 4. Explain the meaning of strategic choice and strategic implementation. What are the steps involved in strategy implementation? How is strategic implementation relevant in the present corporate world?

SECTION — B

Short Answer type questions: To be answered in about 550 words each. Answer any Six: 5x6=30 Marks

Question 1. 'Planning and Controlling are closely related to each other.' Do you agree with the statement? Discuss.

Question 2. Explain the process of 'Management by Objectives (MBO)'. How does it differ from traditional management?

Question 3. What are the various techniques of direction? Which technique do you think is the best one?

Question 4. What are the different types of incentives that a manager can make use of to motivate his subordinates?

Question 5. What do you understand by 'Strategy' in business? Explain any two types of strategy.

Question 6. 'Vision' provides a roadmap to company's future. How is it different from the 'Mission' of the company?

Question 7. What are the different elements about a company's situation to be considered for performing 'Company's situation/position analysis'?

Question 8. What do you mean by 'Strategic choice'? What are the various issues in strategic implementation?

SECTION — C

Answer the following questions in a word: 1x10=10 Marks

Question 1. is an important technique of controlling

Question 2. Means One subordinate should be commanded by one and only one superior at a time.

Question 3. is the 'Father of Scientific Management.'

Question 4. Organization is a mixture of Project organization and Functional organization.

Question 5. is the number of people a manager can effectively manage?

Question 6. 'BCG matrix' is a model for analysis.

Question 7. The managerial exercise of putting a strategy into action is called

Question 8. Political environment is a component of Environment.

Question 9. is an expression of company's vision?

Question 10. The decision to select among the grand strategies identified is called Strategic

BANARAS HINDU UNIVERSITY

B.Voc. Semester I Examination, 2022-23

Paper: BVCA-102 — Fundamentals of Artificial Intelligence

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer All Sections. Candidates are required to***SECTION — A**

Note: Long Answer type questions to be answered in about 750 words each. (Any two) 15x 2=30

Question 1. (a) What is Artificial Intelligence ? Discuss five applications of Artificial Intelligence.

(a) What are the challenges faced by Artificial Intelligence ?

Question 2. (a) What are agents and environment ? What is an Ideal Rational Agent ?

(a) Discuss different types of agents in Artificial Intelligence. P. T.O.

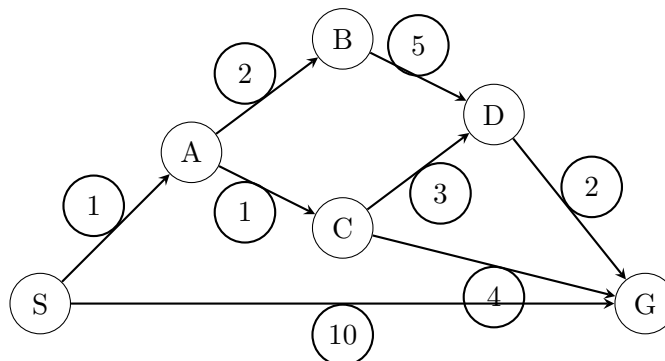
Question 3. (a) What are Heuristic Search techniques ? Discuss Hill Climbing Algorithm. |

(a) How Steepest Ascent Hill Climbing algorithm differs from Hill Climbing algorithm ? |

Question 4.

(a) Explain AO* algorithm with example. [7]

(b) Discuss briefly A* algorithm. Find the optimal path and cost from Start node *S* to Goal node *G* for the given graph using A* algorithm. (The path cost is given on the edges of the graph and the table shows the values of heuristics cost). [8]



State	$h(n)$
S	5
A	3
B	4
C	2
D	6
G	0

SECTION — B

Note: Short Answer type questions to be answered in about 550 words each. (Any six)
($5 \times 6 = 30$)

Question 5. What are the steps for building an Expert System ?

Question 6. Discuss any five features of Environment.

Question 7. What do you understand by PEAS ? Design PEAS information for Self-driving cars and Medical Diagnostic System.

Question 8. What is Uninformed Search technique ? Discuss about Depth First Search algorithm.

Question 9. Define Facts, Rules and Queries in terms of Prolog Programming. Write a Prolog program to find maximum of two numbers.

Question 10. What is Means End Analysis ? Explain with example.

Question 11. Differentiate between Forward Chaining and Backward Chaining in the context of Expert Systems.

Question 12. What are Lists in prolog ? Define any three list operations. (3) P. T.O.

SECTION — C

Question 13. Select the correct option:

($1 \times 10 = 10$)

(i) What is state space?

(a) The whole problem

(b) Your Definition to a problem

(c) Problem you design

(d) Representing your problem with variables and parameters

(ii) What is the space complexity of Depth-first search?

(a) $O(b)$

(b) $O(b^l)$

(c) $O(m)$

- (d) $O(bm)$
- (iii) When is breadth-first search optimal?
 - (a) When there is less number of nodes
 - (b) When all step costs are equal
 - (c) When all step costs are unequal
 - (d) None of the mentioned
- (iv) The father of Artificial Intelligence is:
 - (a) Alan Turing
 - (b) John McCarthy
 - (c) Charles Babbage
 - (d) None of the above
- (v) A technique that was developed to determine whether a machine could or could not demonstrate artificial intelligence is known as:
 - (a) Boolean Algebra
 - (b) Turing Test
 - (c) Logarithm
 - (d) Algorithm
- (vi) The component of an Expert system is:
 - (a) Knowledge Base
 - (b) Inference Engine
 - (c) User Interface
 - (d) All of the above
- (vii) An AI agent perceives and acts upon the environment using:
 - (a) Sensors
 - (b) Perceiver
 - (c) Actuators
 - (d) Both (a) and (c)
- (viii) The main function of a problem-solving agent is to:
 - (a) Solve the given problem and reach the goal
 - (b) Find out which sequence of actions will get it to the goal state

(c) Both (a) & (b)

(d) None of the above

(ix) Which of the following is not a variable in Prolog?

(a) Paul

(b) a_123

(c) _123

(d) Hello

(x) What would the output of the following Prolog query be?

`?- f(a, b) = f(Y, X)`

(a) `X=b, Y=a.`

(b) This expression will not compile

(c) `X=a, Y=b.`

(d) `X=X, Y=Y.`

BANARAS HINDU UNIVERSITY

B.Voc. Semester I Examination, 2022-23

Paper: BVCA-103 — Applied Mathematics

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer all Sections. Candidates are required to***SECTION — A: Long Answer Questions**

[15 × 2 = 30]

*Instructions: Long answer type questions each have to be solved stepwise.***Question 1.** Prove that:

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

OR**Question 2.** Apply Gauss elimination method to solve the equations:

$$\begin{aligned} x + 4y - z &= -5 \\ x + y - 6z &= -12 \\ 3x - y - z &= 4 \end{aligned}$$

Question 3. Find the eigenvalues and verify the Cayley-Hamilton theorem for the given matrix:

$$A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$$

OR**Question 4.** Define with examples:

(a) Set types:

- (i) Singleton Set
- (ii) Finite Set
- (iii) Infinite Set
- (iv) Null Set
- (v) Subset

(b) If $n(A) = 50$, $n(B) = 60$, and $n(C) = 25$, then find $n(A \cup B \cup C)$ if A, B, C are disjoint sets.

SECTION — B

Note: Attempt any six out of eight. Short answer type questions to be solved in required steps: $(5 \times 6 = 30)$

Question 1. Calculate the number of subsets for the given set $A = \{0, 1, 2\}$ and write all the possible subsets of the given set.

Question 2. Find the probability of getting Head and Tail both on one toss of a single coin.

Question 3. Write the characteristic equation of the given matrix:

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$$

Question 4. Write the statement of Cayley-Hamilton theorem and verify this theorem for an Identity matrix of order 3×3 .

Question 5. Define Derivation Tree.

Question 6. Apply Gauss-Seidel iteration method to solve the equations:

$$20x + y - 2z = 17$$

$$3x + 20y - z = -18$$

$$2x - 3y + 20z = 25$$

Perform this operation up to two iterations.

Question 7. Define set of set and write the power set of the given set $A = \{1, 2, 3\}$.

Question 8. Define the sample space and write the sample space of the outcome of a throw of a dice, and find the probability that we get 6 in a single throw of dice.

SECTION — C

Note: Attempt all questions. Very short answer type questions or one-word answer type questions: $(10 \times 1 = 10)$

1. Can we calculate the determinant value of a row matrix?
2. Verify if the Identity matrix of order 2×2 is a non-singular matrix.
3. Why is the system of Homogeneous equations called consistent?
4. Can we refer to a null set as an empty set?
5. Calculate the value of the given determinant:

$$A = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \\ 1 & 1 & 2 \end{vmatrix}$$

6. Write what type of set is $A = \{\}$.
7. Write the definition of 'Mutually Exclusive Event'.
8. Calculate the number of terms in the expansion of a 3×3 determinant.
9. What is an Upper Triangular Matrix?
10. Define Function.

BANARAS HINDU UNIVERSITY

B.Voc. Semester I Examination, 2022-23

Paper: BVCA-103b — Applied Mathematics

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer all Sections. Candidates are required to***SECTION — A: Long Answer Questions**

[15 × 2 = 30]

*Instructions: Long answer type questions each have to be solved stepwise.***Question 1.** Prove that:

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

OR**Question 2.** Apply Gauss elimination method to solve the equations:

$$x + 4y - z = -5$$

$$x + y - 6z = -12$$

$$3x - y - z = 4$$

Question 3. Find the eigenvalues and verify the Cayley-Hamilton theorem for the given matrix:

$$A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$$

OR**Question 4.** Define with examples:

(a) Set types:

(i) Singleton Set

(ii) Finite Set

(iii) Infinite Set

(iv) Null Set

(v) Subset

(b) If $n(A) = 50$, $n(B) = 60$, and $n(C) = 25$, then find $n(A \cup B \cup C)$ if A, B, C are disjoint sets.

SECTION — B

Note: Attempt any six out of eight. Short answer type questions to be solved in required steps: $(5 \times 6 = 30)$

Question 1. Calculate the number of subsets for the given set $A = \{0, 1, 2\}$ and write all the possible subsets of the given set.

Question 2. Find the probability of getting Head and Tail both on one toss of a single coin.

Question 3. Write the characteristic equation of the given matrix:

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$$

Question 4. Write the statement of Cayley-Hamilton theorem and verify this theorem for an Identity matrix of order 3×3 .

Question 5. Define Derivation Tree.

Question 6. Apply Gauss-Seidel iteration method to solve the equations:

$$20x + y - 2z = 17$$

$$3x + 20y - z = -18$$

$$2x - 3y + 20z = 25$$

Perform this operation up to two iterations.

Question 7. Define set of set and write the power set of the given set $A = \{1, 2, 3\}$.

Question 8. Define the sample space and write the sample space of the outcome of a throw of a dice, and find the probability that we get 6 in a single throw of dice.

SECTION — C

Note: Attempt all questions. Very short answer type questions or one-word answer type questions: $(10 \times 1 = 10)$

1. Can we calculate the determinant value of a row matrix?
2. Verify if the Identity matrix of order 2×2 is a non-singular matrix.
3. Why is the system of Homogeneous equations called consistent?
4. Can we refer to a null set as an empty set?
5. Calculate the value of the given determinant:

$$A = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \\ 1 & 1 & 2 \end{vmatrix}$$

6. Write what type of set is $A = \{\}$.
7. Write the definition of 'Mutually Exclusive Event'.
8. Calculate the number of terms in the expansion of a 3×3 determinant.
9. What is an Upper Triangular Matrix?
10. Define Function.

BANARAS HINDU UNIVERSITY

B.Voc. Semester I Examination, 2022-23

Paper: BVCA-104 — Problem Solving C

Time: Three Hours

Maximum Marks: 70

Instructions: Answer all Sections. Candidates are required to

Paper : BVCA-104 (Skill Training in R) eee EE Mar FIO RR — Note: Answer all Sections. Candidates are required to write their answers in their own words as far as practicable.

SECTION — A

| Note: Long Answer type questions to be answered in | about 750 words each : 15x 2=30

Question 1. List out the different control structures in R Programming with general syntaxes and brief explanation. Write R code to the function by using if-else command. | foyy)=x+y if x<y =(1-x-y) if x>y 0 if x=y P.T.00. **OR**

Question 2. Define nested if-else and nested loop statements with general syntaxes and explanation. Write R codes for finding a maximum number from the given three numbers. a

Question 3. | Write a brief note on R. Write R codes for computing matrix inverse for 3 x 3 matrix. ! **OR**

Question 4. Illustrate arithmetic, logical and relational operators | with an example for each. Write R programme for the | factorial function using a for-loop.

SECTION — B

Note: Short Answer type questions to be answered in about 550 words each (any six): . 5x6=30 !

Question 1. Write general syntax of R functions that are used to | create bar and pie charts. Show the percentage of | students registered in Data Science, R Programming | and Machine learning subjects using these charts. | The percentages are 30%, 40% and 30% in data i science, R programming and Machine learning, respectively.

Question 2. Giving a suitable example for each, explain the uses of the following operators "%x%", "%*%", "%%", "%/%" and "%in%".

Question 3. Using function() command, write R codes to enter the following mathematical expressions in R:

(a) $y = 10^x + \sqrt{x} \log_{10}(x)$

(b) $y = e^{x/2} - \sin(x) \tan(|x|^4)$

Question 4. State important features of R.

Question 5. Differentiate between vector, list, Matrix and Data frame with general syntaxes and examples. .

Question 6. Write all R codes to perform correlation and regression analysis for the following data set.

Height (inches)	60	62	65	68	70	72
Weight (lbs)	110	120	140	160	170	190

Question 7. Write R syntaxes to apply t-test (Single sample and Two sample) and Chi-square test. For illustration, you may take data of your choice.

Question 8. Explain the use of next break and stop functions. Give an example for each.

SECTION — C

Note: Answer the following questions in a word : 10x1=10

Question 1. How do you enter integer, complex, character and numeric data in R?

Question 2. Write R code to enter the following data in R using seq() and rep(); 1.0, 1.0, 1.0, 1.0, 1.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0. [3]

Question 3. Write general syntax of the import/export of CSV file.

Question 4. Write a code to compute sum, mean and product of a given vector of elements.

Question 5. Write R code to add a legend in R plot.

Question 6. How do you set the margin of the plot in R ?

Question 7. Risan implementation of Language.

Question 8. Write R code to extract a column of a data frame.

Question 9. Write down the output of the codes : N=1: 15; (which(N>=6 & N<10)).

Question 10. Write down the output of the codes : x = letters[1:10]; x[seq(2, 10, by=2)]

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) I Semester Examination, 2016-17

Subject: Computer Science

**Paper: CS-101 (Problem Solving through C Programming) /
BCS-101 (Introduction to Computer Programming Through C)**

Time: Three hours

Maximum Marks: 70

Instructions: Answer any five questions, including Question No. 1, which is compulsory. All undefined symbols have their usual meanings.

Question 1. Attempt all questions:

[2 × 9 = 18]

- (a) Define Algorithm. What are its characteristics?
- (b) What is the difference between '=' operator and '==' operator?
- (c) When is the register storage class most useful? Give a suitable example.
- (d) What is the role of a Preprocessor in 'C' programming?
- (e) Why is the '&' operator not used with array name in a `scanf()` statement?
- (f) What do you mean by '→' operator in C? When is it used?
- (g) What will be the output of the following code?

```
void main()
{
    int a=3, b=4;
    b %= 3+4;
    a *= a+5;
    printf("%d_%d", b, a);
}
```

- (h) Consider the following code segment and write its final output:

```
void main()
{
    int x=3, z;
    z = x / ++x;
    printf("x=%d_z=%d", x, z);
}
```

- (i) What is the output of the following code?

```
void main()
{
    int a=30, b=0, c;
    c = (a != 10) && (b = 0);
    printf("c=_%d", c);
}
```

Question 2.

- (a) What is the difference between top-down approach and bottom-up approach of programming? [2]
- (b) Draw a flowchart to find the roots of a quadratic equation $ax^2 + bx + c = 0$. [4]
- (c) List the rules for naming a variable in C. [2]
- (d) Explain the difference between nested if-else clause and else if clause used in C language. [5]

Question 3.

- (a) Define function. What do you mean by function prototype? Is it always compulsory to write function prototype statement while using a function in a 'C' program? Justify your answer with one example. [5]
- (b) Describe the two ways of passing arguments to function? When do you prefer to use each of them? [4]
- (c) Write a program using function to find m^n where $m, n \geq 0$. [4]

Question 4.

- (a) State the difference between automatic and static storage class variables by using appropriate example. [4]
- (b) Write your own function for the following operations on string: [6]
 - (i) Comparison of two strings
 - (ii) Copy one string into another string
- (c) What is the use of enumerated data type in 'C' programming? [3]

Question 5.

- (a) How are 2D array elements stored in computer memory in 'C' programming? [2]
- (b) Write a 'C' program to find out transpose of a given matrix. [4]
- (c) What is meant by the following terms? Give examples of use of each of them. [7]
 - (i) Array of pointers
 - (ii) Pointer to Array

Question 6.

- (a) Explain the difference between a 'Structure' and a 'Union' with suitable examples. [4]
- (b) How do you pass parameters to main() function in a 'C' program? Explain with a suitable example. [4]

- (c) Can a structure be used within another structure? Give appropriate example to support your answer. [5]

Question 7.

- (a) What is the need of dynamic memory allocation? Explain the role of `realloc()` function with its proper syntax. [5]
- (b) Define Macro. In what situation is it better to use Macro instead of using Function? [3]
- (c) What is conditional compilation? How does it help a programmer? Explain with an example. [5]

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester I Examination, 2023-24

Paper: CS-101 — Problem Solving through C Programming

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions including Question 1 which is compulsory. Write your Roll No. at the top immediately on receipt of this question paper.*

Question 1. Attempt any *nine*.

[2×9 = 18]

- (a) Differentiate between `float` and `double` data types in C.
- (b) What is the difference between the `=` operator and the `==` operator in C?
- (c) What is the role of the `break` keyword in C programming?
- (d) What is the difference between a string and an array of characters?
- (e) What is the use of the `#ifndef` preprocessor directive in C programming?
- (f) What is the difference between `scanf()` and `gets()` functions?
- (g) Write the syntax and usage of the `->` operator in C.
- (h) Define the concept of *pointer to pointer* with a suitable example.
- (i) What will be the output of the following code? (Explain with reason.)

```
void main() {  
    int x = 10, y = 5, p, q;  
    p = x - 19;  
    q = (p++) && (y = 3);  
    printf("p=%d y=%d", p, q);  
}
```

- (j) What will be the output of the following code? (Explain with reason.)

```
void main() {  
    int x = 3, y = 2, z;  
    z = x++ + ++y;  
    printf("x=%d y=%d z=%d", x, y, z);  
}
```

Question 2.

- (a) What is an *algorithm*? Draw the flowchart to print the real roots of the quadratic equation $ax^2 + bx + c = 0$ using the Sridharacharya formula. [4]
- (b) Explain the difference between a `while` loop and a `do...while` loop with a suitable example. [5]

- (c) Explain the `switch...case` statement with a suitable example. [4]

Question 3.

- (a) Discuss the *logical operators* in C with a suitable example. [4]
- (b) Differentiate between *actual* and *formal* arguments when using a function. Explain *call by reference* with a suitable example. [5]
- (c) Define *recursion*. Write a program to find the factorial of a given number using recursion. [4]

Question 4.

- (a) State the difference between `register` and `static` storage class variables with appropriate examples. [4]
- (b) Write your own functions in C for the following string operations: [6]
- (i) Copy one string into another.
 - (ii) Reverse a given string.
- (c) What are *command line arguments*? Why are they used? [3]

Question 5.

- (a) How are elements of a 2D array stored in computer memory? Give one example. [2]
- (b) Differentiate between the concepts of *pointer to an array* and *array of pointers* with suitable examples. [6]
- (c) How can a pointer be used to access the elements of a one-dimensional array? Explain with an example. [5]

Question 6.

- (a) Explain the different possible *modes* for opening a file in C. Write a program to copy the contents of one file into another. [5]
- (b) What do you mean by *dynamic memory allocation* in C? Explain the role of the `realloc()` function with a suitable example. [4]
- (c) Define *structure* with a suitable example. How is memory allocation done for structure variables? Give one example. [4]

Question 7.

- (a) What is the role of the *Preprocessor* in C programming? Differentiate between the two forms of include directives: `#include<mylib.h>` and `#include"mylib.h"`. [4]
- (b) Explain the concept of *nested structure* with a suitable example. [5]

- (c) Define *union* with a suitable example. Give one real-life example where a union can be used. [4]

BANARAS HINDU UNIVERSITY

B.Voc. Semester I Examination, 2022-23

Paper: GEN-101 — Communication Skills

Time: Three hours

Maximum Marks: 70

Instructions: Attempt all questions. Questions of Section A, B and C Carry 15, 5 and 1 marks each

COMPUTER APPLICATION GEN - 101 : Communication Skills Note: Attempt all questions. Questions of Section A, B and C Carry 15, 5 and 1 marks each respectively. ,

SECTION — A

Note: Answer approximately in 550-750 words.

Question 1. Define Communication. What are the reasons behind miscommunication? Suggest some ways to overcome miscommunication.

OR

Define Business Correspondence. What are the stages in writing a business letter? What are the essential qualities of a good business letter? 2, Define Listening. What is the importance of listening skills in a business organization? How are | those skills useful in dealing with customer complaint?

OR

Explain SWOT Analysis. Briefly discuss Advantages and Limitations of SWOT analysis.

SECTION — B

Note: Answer approximately in 350-450 words. Attempt any Six from the following.

Question 3. What are various bases of classification of business report?

Question 4. Describe in details various types of conferences.

Question 5. Write a note on Telephoning.

Question 6. How should an effective interviewer behave during interview? What pitfall effective interviewers avoid? Explain the types of patterned interview.

Question 7. Discuss briefly, in proper sequence, the steps followed in planning an effective communication. |

Question 8. Discuss the importance of Non-Verbal Communication.

Question 9. Discuss five elements (factors) of the process of communication.

Question 10. Why business communication is called, “Life blood” of an organization. P.T.O.
(2) |

SECTION — C

Note: Answer in briefest possible way.

Question 1. refers to the special language of a trade. |

- (a) Jargon
- (b) Expression. ‘
- (c) Colloquialism.
- (d) Suggestion. |

Question 2. Audio-Visual communication is most suitable for mass and mass

- (a) Publicity& education.
- (b) Policies& political.
- (c) Save& store. |
- (d) Education & political.

Question 3. Interpretation of data is followed by ——— . See eee

- (a) Recommendation.
- (b) Suggestion.
- (c) Conclusions. |
- (d) Complaint.

Question 4. Gray colour refers :

- (a) Confidence & wisdom. |
- (b) Cheerfulness & vigour.
- (c) Life & coolness. |
- (d) Danger.

Question 5. Communication in an organization should ideally flow

- (a) From top to bottom.
- (b) From bottom to top. |
- (c) Both ways.
- (d) Horizontally. |

Question 6. Statutory Report is :

- (a) An informal report.

- (b) Not having legal sanction.
- (c) Always a statistical report.
- (d) Formal report.

Question 7. Is the main part of the report.

- (a) Conclusion.
- (b) Recommendations.
- (c) Description.
- (d) References.

Question 8. The main objective of communication is:

- (a) Information and persuasion.
- (b) Skill and personality development.
- (c) Control and management.
- (d) Need.

Question 9. An effort to influence the attitude and feelings of others is

- (a) Persuasion. oO
- (b) Suggestion.
- (c) Advice. : :
- (d) Appreciation.

Question 10. The communication cycle, the process of re translation of signals into ideas is called:

- (a) Encoding.
- (b) Decoding.
- (c) Response.
- (d) Feedback.

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14

Paper: BCS-201 — Digital Logic Circuits

Time: Three hours

Maximum Marks: 70

*Instructions: Answer any **five** questions, including Question 1, which is compulsory.***Question 1.**

- (a) Perform the following conversions. Show the detailed calculation being done for conversions: [3×2 = 6]
- (i) $(673.23)_{10} = (\quad)_2$
- (ii) $(1101101.111)_2 = (\quad)_{10}$
- (iii) $(2AC5.D)_{16} = (\quad)_8$
- (b) Represent the decimal number 3620 in BCD and excess-3 code. [2]
- (c) What is race around condition? [2]
- (d) List the names of any four digital logic families. [2]
- (e) What is the advantage of PLA over ROM? [2]

Question 2.

- (a) Perform the following binary operations: [3]
- (i) 111×1101
- (ii) $11011001 \div 1100$
- (b) Subtract 3 from -12 using 2's complement method. [2]
- (c) Perform the addition of the following numbers in the given base without converting to decimal: [4]
- (i) $(1230)_4 + (23)_4$
- (ii) $(296)_{12} + (57)_{12}$
- (d) What do you understand by floating point representation? Represent $(4.21875)_{10}$ in binary floating point notation. [5]

Question 3.

- (a) Simplify the following Boolean expressions to a minimum number of literals by using Boolean algebra theorems: [4]

(i) $xyz - x'y - xyz'$

(ii) $BC + AC' + AB + BCD$

- (b) What is the difference between canonical form and standard form? Which form is preferable when implementing a Boolean function with gates? Which form is obtained when reading a function from a truth table? [5]

- (c) Express the following function in a sum of minterms and product of maxterms:

$$F(w, x, y, z) = y'z + wxy' + wxz' + w'x'z$$

[5]

Question 4.

- (a) What are the differences between combinational circuit and sequential circuit? Explain with their block diagrams. List the names of any two combinational and any two sequential circuits. [4]

- (b) Implement the following function with NAND gates (with no more than six gates, each having three inputs). Assume that both the normal and complement inputs are available.

$$F = BD + BCD + AB'C'D' + A'B'CD'$$

[5]

- (c) Simplify the Boolean function F using the don't care condition d , in its minimum sum of product form:

$$F(w, x, y, z) = w'x'y'z' + w'x'yz' + w'xy'z + w'xyz'$$

$$d(w, x, y, z) = w'x'y'z + yz' + wyz$$

[5]

Question 5.

- (a) Design a combinational circuit that accepts a three bit number and generates an output binary number equal to the square of the input number. [5]

- (b) What is a decoder? Design a full adder circuit with a decoder and two OR gates. [4]

- (c) What is meant by a priority encoder? How does a priority encoder differ from an ordinary encoder? Explain with suitable example. [5]

Question 6.

- (a) Define Multiplexer. Implement an 8x1 multiplexer using two 4x1 multiplexers and a 2x1 multiplexer. [5]

- (b) Implement a 3-bit binary to gray code converter using PLA as follows:

- (i) Make a truth table. [2]

- (ii) Simplify using K-map. [2]

(iii) Design circuit using PLA; and [3]

(iv) Specify the size of PLA. [2]

Question 7.

(a) Implement JK flip flop using D-flip flop. [4]

(b) Define irregular counter. Design a counter with the following binary sequence: 0, 1, 3, 7, 6, 4 and repeat. Use T flip-flops. [5]

(c) Define register. By using registers, design a sequential circuit whose state table is listed as follows: [5]

Present State		Input	Next State		Output
A_1	A_2	x	A_1	A_2	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	0	0	1
1	0	0	1	0	0
1	0	1	0	1	0
1	1	0	1	1	0
1	1	1	0	0	1

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14

(Ancillary Subject)

Paper: BCS-203 — Fundamentals of Computers

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions including Question 1, which is compulsory.***Question 1. (Compulsory)**

- (a) Why do we use secondary memory? [3.5]
- (b) What are the two main components of CPU in a computer system? List the main functions of each component. [3.5]
- (c) Why are peripheral devices necessary for a computer system? List any five peripheral devices. [3.5]
- (d) Mention the different types of software used in computer systems. Explain each with an example. [3.5]

Question 2.

- (a) Draw the block diagram of basic organization of a computer system. Explain each component with an example. [7]
- (b) What is primary storage device? Explain RAM and ROM. [5]
- (c) How many number of bytes are in a Kilobyte, Megabyte, Gigabyte and Terabyte? [2]

Question 3.

- (a) Write short notes on the Generations of computers. [6]
- (b) Convert the following numbers according to the specified bases: [8]
 - (i) $(11010011)_2 = (?)_{16}$
 - (ii) $(111011.10)_2 = (?)_{10}$
 - (iii) $(A6B4)_{16} = (?)_8$
 - (iv) $(123)_8 = (?)_{10}$

Question 4.

- (a) Explain the applications of computer in various fields. [6]
- (b) What is an operating system? What are the main functions of an operating system? [8]

Question 5.

- (a) What are the major components of database management systems? Describe each of those components. [8]
- (b) What are input devices? Explain any three input devices. [6]

Question 6.

- (a) Answer the following multiple choice questions: [2×4 = 8]
- (i) Which access method is used for magnetic tape?
(i) Direct (ii) Random (iii) Sequential (iv) None of the above
- (ii) The language that the computer can understand and execute is called
(i) Machine Language (ii) Application software (iii) System program (iv) None of the above
- (iii) Which of the following is used as a primary storage device?
(i) Magnetic tape (ii) PROM (iii) Floppy disk (iv) None of the above
- (iv) Operating system is
(i) A collection of hardware components (ii) A collection of software routines
(iii) A collection of Input-Output device (iv) None of the above
- (b) Write the differences between serial access memory and random access memory. List any two secondary storage devices with their characteristics. [6]

Question 7. Write short notes on any **two** of the following:

[7×2 = 14]

- (a) Instruction Cycle
- (b) Cache Memory
- (c) BCD code

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Examination, 2016–17

Paper: BCS-301 — Numerical Computing

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Use of calculator is allowed. All undefined symbols have their usual meanings. Each question carries **14 marks**.*

Question 1. (Compulsory — attempt all parts)

[2 × 7 = 14]

- (a) Explain different types of errors with an example.
- (b) If the first significant figure of a number is k and the number is correct to n significant figures, prove that the relative error is less than $\frac{1}{2k \times 10^{n-1}}$.
- (c) Prove that the Newton–Raphson process has second-order (quadratic) convergence.
- (d) Given that the equation $x^3 - 7 = 69$ has a root between 5 and 8, apply the method of Regula-Falsi (False Position) to determine it.
- (e) Define Finite Differences. Explain Forward, Backward, and Central differences with examples.
- (f) Evaluate $\Delta \tan^{-1} x$.
- (g) State the conditions under which the Gauss–Seidel iteration method converges.

Question 2.

[6 + 8 = 14]

- (a) Find a real root of the equation $x^3 - 2x - 5 = 0$ using the Bisection Method, correct to three decimal places.
- (b) From the following table, estimate the number of students who obtained marks between 40 and 45 using Newton’s Forward Interpolation Formula:

Marks	30–40	40–50	50–60	60–70	70–80
No. of Students	31	42	51	35	31

Question 3.

[6 + 8 = 14]

- (a) Apply the Gauss Elimination method to solve the following system of equations, correct to three decimal places:

$$x + y + z = 9, \quad 2x - 3y + 4z = 13, \quad 3x + 4y + 5z = 40$$

- (b) Apply the Gauss–Jordan method to solve the following system, correct to three decimal places:

$$x + 4y - z = -5, \quad x + y - 6z = -12, \quad 3x - y - z = 4$$

Question 4. The table gives the distance y (in nautical miles) of the visible horizon for given heights x (in feet) above the earth's surface: [6 + 8 = 14]

x (ft)	100	150	200	250	300	350	400
y (mi.)	10.63	13.03	15.04	16.81	18.42	19.90	21.27

Find the values of y when:

- (a) $x = 218$ ft
- (b) $x = 410$ ft

Question 5. Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ using the following rules and compare with the actual value: [3 + 4 + 4 + 3 = 14]

- (a) Trapezoidal Rule
- (b) Simpson's $\frac{1}{3}$ Rule
- (c) Simpson's $\frac{3}{8}$ Rule
- (d) Weddle's Rule

Question 6. Using the Runge-Kutta method of fourth order, solve $\frac{dy}{dx} = x + y$ with $y(0) = 1$ at $x = 0.2$ and $x = 0.4$. [14]

Question 7. [6 + 8 = 14]

- (a) Apply Milne's method to find a solution of the differential equation $y' = x - y^2$ in the range $0 \leq x \leq 1$ for the boundary condition $y = 0$ at $x = 0$.
- (b) Given $\frac{dx}{dy} = x^2(1 + y)$ and $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$, $y(1.3) = 1.979$, evaluate $y(1.4)$ by the Adams-Bashforth method.

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2021-22

Paper: BVCA-201 — Introduction To Databases

Time: Three Hours

Maximum Marks: 70

Instructions: Answer the following (each question carry

REFNO 052213 — Paper : BVCA-201 | Introduction to Databases _ Part - A | (Short Questions)

Question 3. Note : Answer the following (each question carry | 2 marks) : $10 \times 2 = 20$: 1. Define the following terms : . | (a) Data model | b Database schema . | (c) Client/server architecture) . (d Owner entity type | (ce) Weak entity type | (f Entity REFNO 052213 (gz) Attribute Value " . (h) Relation state

(a) Degree of a relation

(b) Normalization PART — B Note: All compulsory : Either or Type Questions. $5 \times 4 = 20$

Question 2. What is the difference between logical data independence and physical data independence ? Which one is harder to achieve ? Why ?
Discuss the different data types of SQL.

OR

Question 3. Discuss the entity integrity and referential integrity constraints. Why is each considered important ? . Discuss Functional dependency with suitable example.

Question 4. What is a relationship type ? Explain the : differences among a relationship instance, a relationship type, and a relationship set. REFNO 052213

What is a participation role ? When is it necessary to use role names in the description of relationship types ? : .

Question 5. What is meant by a recursive relationship type ? Give some examples of recursive relationship types.

OR

Define SQL ? Write the Syntax of create table and Insert table in the data ? .

Question 6. Why are duplicate tuples not allowed in a relation ? | OR | What is the difference between a key and a | super key ? : Part - C i Note: Open choice - 3 out of 4 questions. | ' $3 \times 10 = 30$

Question 7. Describe the three-schema architecture. Why . do we need mappings between scheme a levels ? How do different schema definition languages support this architecture ? Fi . 3 PLEO: REFNO 052213 & | . '

Question 8. Discuss the different type of user-friendly ' interfaces and types of users who typically use each.

Question 9. Discuss the various reasons that lead to the occurrence of NULL values in relations. |

Question 10. Discuss the role of a high-level data model, in the database design process. ' KKK , 4 " 300

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2022-23

Paper: BVCA-201 — Introduction To Databases

Time: Three Hours

Maximum Marks: 70

SECTION — A: Long Answer Questions

[12 × 2 = 24]

Instructions: Answer any two questions. Each question carries 12 marks.

Question 1. Draw an E-R Diagram for a hospital management system in detail and explain it.

Question 2. What is an E-R Diagram? Explain its components.

Question 3. What is normalization? Explain it with different normal forms.

Question 4. What is the difference between a File System and a DBMS? Explain any six applications of Database Management Systems.

SECTION — B: Short Answer Questions

[6 × 6 = 36]

Instructions: Answer any six questions. Each question carries 6 marks.

Question 5. What is a Database and a DBMS? Explain with a suitable example.

Question 6. What is Data Abstraction in DBMS and what are its three levels?

Question 7. What are the different types of SQL commands?

Question 8. What is the difference between TRUNCATE, DELETE, and DROP statements?

Question 9. What are UNION, MINUS, and INTERSECT commands in DBMS?

Question 10. What is a Trigger in DBMS?

Question 11. What are the advantages of a DBMS?

Question 12. What is the difference between a primary key and a foreign key?

Question 13. What is a data model? List different types of data models and explain.

SECTION — C: Multiple Choice Questions

[1 × 10 = 10]

Instructions: Answer all questions. Each question carries 1 mark.

Question 14. Answer the following multiple choice questions:

- (a) What do you mean by a one-to-many relationship?
- (i) One class may have many teachers.
 - (ii) One teacher can have many classes.
 - (iii) Many classes may have many teachers.
 - (iv) Many teachers may have many classes.
- (b) A Database Management System is a type of _____ software.
- (i) System software.
 - (ii) Application software.
 - (iii) General software.
 - (iv) Both (a) and (c).
- (c) Which of the following refers to the level of data abstraction that describes exactly how the data is actually stored?
- (i) Conceptual Level.
 - (ii) Physical Level.
 - (iii) File Level.
 - (iv) Logical Level.
- (d) Which one of the following is a type of Data Manipulation Command?
- (i) Create.
 - (ii) Alter.
 - (iii) Delete.
 - (iv) All of the above.
- (e) The term “Data independence” refers to:
- (i) Data is defined separately and not included in the programs.
 - (ii) Programs are not dependent on the logical attributes of the data.
 - (iii) Programs are not dependent on the physical attributes of the data.
 - (iv) Both (b) and (c).
- (f) What is the relational calculus?
- (i) It is a kind of procedural language.
 - (ii) It is a non-procedural language.
 - (iii) It is a high-level language.
 - (iv) It is a Data Definition language.

- (g) Which one of the following refers to the total view of the database content?
- (i) Conceptual view.
 - (ii) Physical view.
 - (iii) Internal view.
 - (iv) External view.
- (h) The Database Management Query language is generally designed to:
- (i) Support end-users who use English-like commands.
 - (ii) Specifying the structure of the database.
 - (iii) Support in the development of complex application software.
 - (iv) All of the above.
- (i) Which one of the following is commonly used to define the overall design of the database?
- (i) Application program.
 - (ii) Data definition language.
 - (iii) Schema.
 - (iv) Source code.
- (j) Which of the following keys is generally used to represent the relationships between the tables?
- (i) Primary key.
 - (ii) Foreign key.
 - (iii) Secondary key.
 - (iv) None of the above.

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2021-22

Paper: BVCA-202 — Problem Solving AI

Time: Three Hours

Maximum Marks: 70

Instructions: Answer the following (each question carry

Problem Solving using Artificial Intelligence

PART — ANote : Answer the following (each question carry 2 marks) : $10 \times 2 = 20$ **Question 1.** i) Briefly explain Turing Test.

- (a) Discuss Chinese Room Argument.
- (b) Describe PEAS model on which an AI agent " works upon. | REFNO 052214 |
- (c) Transform the following English sentences into logical notation : : (i) Not all integers are odd.
- (d) Some numbers are not real. ‘
- (e) Write the following rules of inference related to language of propositions : i (i) Modus Ponens |
- (f) Hypothetical Syllogism : (vi) Which is not a logic programming: | (i) Prolog . : (ii) ROOP —_ OO (iii) Janus 7 | (iv) C | (vii) Which of the following is not a heuristic search :
- (a) Best First Search |
 - (i) Hill Climbing
 - (ii) Depth First Search
 - (iii) Which is not a proposition : _ 8
- (b) $X=3$. ° , ii) Moon is made up of cheese. .
 - (i) $X > 6$
 - (ii) $3+3=6$. | REFNO 052214 y (ix) Which is false, where the underlying universe of discourse is set of integers :
- (c) $\forall x[x < x+1]$. | : : (ii) $\exists x[x < x+1]$ - 8 (iii) $\exists x[x = x+1]$
 - (i) $\exists x[x=3]$
 - (ii) Which of the following technique is not structured knowledge representation technique :

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2022-23

Paper: BVCA-202 — Problem Solving AI

Time: Three Hours

Maximum Marks: 70

*Instructions: Section-A: Attempt any 2 out of 4 questions and**| . Paper : BVCA-202 Problem Solving using Artificial Intelligence | Note: Section-A: Attempt any 2 out of 4 questions and each question is of 12 marks.***SECTION — B: Attempt any 6 out of 9 questions and**

each question is of 6 marks.

SECTION — C: All questions are compulsory and*| are of 1 mark each. | Section -A | Note: Long - Answer questions to be Answered in 500 words each. 12x2=24 a 7***Question 1.** Explain different types of Agents defined based on their degree of perceived intelligence and capability.**Question 2.** Explain how resolution principle is used in Prolog language by an example. :**Question 3.** Write Best First Search algorithm. What is . ' difference between Greedy versus A* Best first, search Algorithm.**Question 4.** What do you understand by structure knowledge representation techniques ? Explain any one of these techniques.**SECTION — B**

Note: Attempt any six (Answer in about 200 words each. 6x6=36

Question 1. Give example of any search algorithm.**Question 2.** Explain Hill climbing algorithm with their types... | . 2 | REFNO 065181**Question 3.** If horses fly or cows eat grass, then the | mosquito is the national bird. If the mosquito is the national bird, then peanut butter tastes | good on hot dogs. But peanut tastes terrible | on hot dogs. Therefore, cows don't eat grass. Prove or disprove it using resolution principle. . | 4. Prove or disprove the validity of the following | argument with respect to given statement: i) If today is Tuesday then have test in CS or Economics.

(a) If my economics professor is sick then I . : will not have test in economics.

(b) Today is Tuesday and my economic , professor is sick. Conclusion : Therefore, I have a test in CS. :

Question 5. Consider the following graph. The numbers written on edges represent the distance between the nodes. The number written on node represents the heuristic value. Find the most cost — effective path to reach from start state A to final state - J using A* Algorithm.

Question 6. Write Pseudo code for Min - max algorithm! and explain its working by an example. Write advantages and disadvantages of logical representation technique of knowledge representation.

Question 8. Write a Prolog program to create the following knowledge base and return the answers for the following queries: Knowledge Base: Mary is female. Mary is sister of John. John is parent of Joseph. If x is female and x is sister of z and z is parent of y then x is aunt of y. Queries: Is Mary aunt of Joseph?

Question 9. Discuss the concept of rationality in AI agents. How does it relate to decision making?

SECTION — C :

Note: Attempt all the questions in a sentence each: 10x1=10.

Question 1. Explain Artificial Intelligence.

Question 2. What is Turing test? 5 P.T.O,

Question 3. What is meant by AI agent?

Question 4. Transform the following English sentences into logical notation: Not all integers are positive.

Question 5. Transform the following English sentences into logical notation: Some number are not even.

Question 6. Which is not a logic programming?

(a) Prolog ii ROOP iii Janus

(i) C

Question 7. Which of the following is not heuristic search? (i) Best First Search

(a) Hill Climbing (iii)/Depth First Search

Question 8. Which is not a proposition?

(a) $X=3$. (ii) Moon is made up of cheese. : (iii) $X > 6$

(i) $3 + 3=6$

Question 9. Which is false, where the underlying universe of discourse is set of integers?

(a) $\forall x[x < x+1]$

(i) $\exists x[x < x-4]$

(ii) $Sx[x = x+1]$ | (iv) $S!x[x = 3]$

Question 10. Which of the following technique is not structured knowledge representation technique.

(a) Semantic Nets ‘ ii) Production Rules - (iii) Scripts iv) Frames 7 200

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2021-22

Paper: BVCA-203 — Technical Documentation Communication

Time: Three Hours

Maximum Marks: 70

Instructions: Attempt all questions: 10x2=20

REFNO 052215 | Paper ; BVCA-203 | Technical Documentation and Communication

PART — A _____

Question 1. Define Technical Writing. ' 2. Define Websites.

Question 3. Paragraph Unity are 6°

Question 4. Linking Pronouns are

Question 5. Steps involved in Theses are

Question 6. Define Verb. REFNO 052215 OO, Dangling Modifiers. | . 8. Define Ornate Language. |

Question 9. Difference between Hyphens and Dashes.

Question 10. Full form of MLA style.

PART — B || _____

Note* Answer any four questions of the following . in 200 words. 5*x4=20

Question 1. Explain American Psychological Association ' (APA) Author - Date style. | 2. Explain Audience type and level of expertise.

Question 3. Explain Task- Oriente Documents.

Question 4. Describe Adjective and Conjunction with proper Examples.

Question 5. Describe Writers Resources. :

Question 2. | , REFNO,052215 ' | Part -C

Question 1. Describe all the steps involved in planning and producing documents. | 2. Explain the elements of Technical Documents. | 3. Describe Graphs and Figures used in technical : writing. | ° 4. Describe Words and Punctuation in technical | | ' | 3 . 300

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BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2021-22

Paper: BVCA-204 — Tech Entrepreneurship IPR

Time: Three Hours

Maximum Marks: 70

Instructions: Attempt all the following questions :

__ Paper: BVCA-204 oo Technical Entrepreneurship and IPR (Write your Roil No. at the top immediately on the . Part - A - "& 10x2=20

Question 1. (a) What is business environment scanning ?

(a) Who is a social entrepreneur ?

(b) What is the term of copyright in Indian system?

(c) Define Intellectual Property. __ __ (ec) What is meant by exclusive license ?

(d) What protects the intellectual property created by artists ? 3

(e) What do you mean by infringement of Intellectual property right ? ‘ ,

(f) Define entrepreneurial journey.

(g) SWOT analysis is a tool for self analysis. — Write true or false with reason. | (ij) What is geographical indication . | - intellectual property ? :

PART — B —

Note: Attempt any of the four following questions. , Each question carries equal marks. . 4x5=20.

Question 2. Differentiate between entrepreneur and Intrapreneur. ° OR ’ Differentiate between various types of . entrepreneurs. .**Question 3.** Explain intellectual property ‘search | techniques. | 2
What do you mean by business model ? Explain . ’ its types.**Question 4.** What are the exit strategies available to an entrepreneur ? .- ‘ : . OR Elucidate the process of business model review.**Question 5.** Explain the different types of trademarks with | examples. Differentiate between Trademark and design. oR : Define Industrial Design and discuss the need . to protect it as an intellectual property. .**Question 6.** How is licensing different from assignment ? | OR __Define Intellectual property right. How is its important ? 3 PTO, . Part - C Note: Long answer type questions. Attempt only three questions. Each question carries equal marks. ; 3*10=30

Question 7. What are the various functions and roles of an entrepreneur ? Explain. 8 What do you mean by Entrepreneurial creativity ? Explain its importance and types. 9, What are the various laws available for enforcement of intellectual property right in . India ?

Question 10. What is a licensing agreement ? Explain the advantage and disadvantages for licensor and licensee, with example. eK . 4 ; 300

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2022–23

Paper: BVCA-204 — Technology Entrepreneurship & IPR

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer all Sections. Each section carries marks as indicated. Section A —***Long Answer Questions***[15 × 2 = 30] Answer any**two questions. Each question carries 15 marks. Question 1. What is Entrepreneurship?**Explain the characteristics of a successful entrepreneur. Discuss the role of technology in modern entrepreneurship. Question 2. What is Intellectual Property Rights (IPR)? Explain**the different types of IPR (Patents, Copyrights, Trademarks, Trade Secrets) with suitable examples. Question 3. What is a Business Plan? Explain the key components of a business**plan. Why is a business plan important for a tech startup? Question 4. Explain the**concept of Technology Transfer. What are the various mechanisms of technology transfer? Discuss its role in economic development. Section B — Short Answer Questions**[5 × 6 = 30] Answer any six questions. Each question carries 5 marks. Question 5. What**is a Patent? What are the criteria for patentability? Question 6. Distinguish between a**copyright and a trademark. Question 7. What is SWOT analysis? Explain with an example.**Question 8. What is venture capital? How does it differ from angel investment? Question**9. Explain the term “Start-up India” initiative and its objectives. Question 10. What**is product lifecycle? Explain its stages briefly. Question 11. What is e-Commerce? List**any four advantages of e-Commerce. Question 12. Explain the term “Digital India” and**its significance for technology entrepreneurship. Section C — Very Short Answer***Questions***[1 × 10 = 10]**Answer all questions. Each carries 1 mark. Question 13. Define the term “Innovation”.**Question 14. What does IPR stand for? Question 15. Name any one international body**governing patents. Question 16. What is a “Sole Proprietorship”? Question 17. Define**“Bootstrapping” in entrepreneurship. Question 18. What does MSME stand for? Question**19. Name the government scheme for startup funding in India. Question 20. What is a**“Non-Disclosure Agreement (NDA)”? Question 21. Define “Incubator” in the context of**startups. Question 22. What is the full form of WIPO?*

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BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2021-22

Paper: BVCA-205 — Skill Training Python

Time: Three Hours

Maximum Marks: 70

Instructions: Attempt all questions. Symbols have their usual meanings.

Paper : BVCA-205 : Skill Training in Python

PART — A

: Multiple Choice Type Questions and Short | Questions) 10x2=20

Question 1. Which of the following is a Python tuple ?

(a) (1, 2, 3

(b) 4

1, 2, 3

(c) (2, 2, 3)

Question 2. Which one of the following is not a keyword in Python language ?

(a) Pass ,

(b) eval :

(c) essert

(d) nonlocal |

Question 3. In object oriented programming, what is the : relation of an object to a class ?
|

(a) class is an instance of an object |

(b) object is a child of a class

(c) object is a method of a class -__ |

(d) object is an instance of a class |)

Question 4. What will be the output of the following python : code snippet ? print (type (5/2)) print (type (5//2)) a

(a) float and int

(b) float and float .

- (c) int and int
- (d) int and float 2 S. What will be the output of the following python code snippet ? ' "
print (2**4+(5+5)** (1+1))
- (e) Error
- (f) 28
- (g) 118
- (h) 116

Question 6. What will be the output of the following Python code ? . i=1 | while True: f to
if i% 3.==0: break print (i) it=1 °

- (a) 123
- (b) Error ° * (c 12
- (c) None of the mentioned s,

Question 7. Which one of the following has the same | precedence level ? |

- (a) Addition and Subtraction — ' (b) Multiplication, Division and Addition
- (b) Multiplication, Division, Addition and . Subtraction
- (c) Addition and Multiplication |

Question 8. What will be the datatype of the test in the . below code snippet ? ° test = 10
print (type (test)) | test = "testbook" print (type(test))

- (a) str and int
- (b) int and int SO :
- (c) str and str
- (d) int and str

Question 9. What is the meaning »> in Python Language ? é

- (a) Interpreter is ready to take instruction |
- (b) Compiler is ready to take instruction
- (c) Three right shift |
- (d) Three left shit 4

Question 10. What is output of print (math.pow (3,2)) ?

- (a) 9.0
- (b) None

(c) 9

(d) None of the mentioned oo | Part - B (Internal Choice - Either or Type Questions) 5x4=20

Question 11. What are the common built - in data types in , Python ? : . OR . How to read and write into python files ? | Explain with the program.

Question 12. Explain the concept of overriding the methods . with the program. : 7 OR Write a python program to create a lambda function that returns a square value of a given program. .

Question 13. Write a python program to create a static method that calculates the square root value of a given number. : a OR | Explain values and items method used in dictionary with example.

Question 14. What is the difference between Python Arrays, - lists and tuple with some proper example ?

OR

Write a python program to define student class. | and create an object to it. Also, display the ! students details.

Question 15. What are the key features of Python over the | . other programming language ? OR — | Write a function that accept two values and | find their sum. | : * 6 , 7 Part - C | (Open choice - 3 out of 4 Questions) 3x 10=30

Question 16. List the operators that python supports. Explain the relational and logical operators along with their precedence while evaluating an expression.

Question 17. Write short notes on :

(a) Debugging

(b) Exception handling .,

Question 18. Write a python program to retrieve the elements from 2D' array and display them using two different loop. .

Question 19. Discuss the technical strengths of python. What is the difference between list and tuples in Python ? kK v 300 4 | F 4 ' | ' t ' ' x | ° 4 |

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BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2022-23

Paper: BVCA-205 — Skill Training Python

Time: Three hours

Maximum Marks: 70

Instructions: Attempt all questions. Symbols have their usual meanings.

x o BVCA - 205 : Skill Training in Python , : Answer All Sections.” , Candidates are required to write their answers in their own words as far as practicable. | Section-A Long Answer type questions to be answered in about 500 words each: 12x2=24 Marks Attempt any two: ,

Question 1. What is a ‘programming language? Explain the features of Python programming language. , :

Question 2. What do you understand by iteration? What are various loop structures supported by Python programming language. ‘

Question 3. What is searching? Explain the Binary search algorithm with a suitable program. — 3 4. What are operators? Explain all the operators supported by Python programming language. — . , . : , , Section- B . __ Attempt any six (Answer in about 200 words each): “6x65 36 Marks

Question 1. Write a python program to find the sum and average of N natural numbers 3 ee ee “2. Write a python program to print Fibonacci series upto N. :

Question 3. Write a python program to find factorial of a given number using recursion.

Question 4. write a python program to check whether a given number is prime or not. , 5, Write a python program to create a list with different types of elements and access : elements of the list using loops. . : . 6. Write a python program to find the largest element in the array. . . 7. Write a python program to implement Selection sort. . ‘ 8. Write a python program to perform various operation on strings such as copy, concatenate and compare. 7 | . 9. Write a python program to perform matrix addition. . . . ‘ : Section- C Attempt all the questions: 10 x 1=10 Marks | . 1. Which of the following is the correct extension of the Python file? . a) .python . so

(a) pl . . d).p . a , - . @) a

Question 2. What will be the value of the following Python expression? . 4+3%5

(a) 7 : , . b2 : . 4 :

(b) 2 . a 3 Which keyword is used for function in Python language? : . ‘

(c) Function , . ; a |

(d) def . : ae : : (Fun .

(e) Define . . : . . 4. What will be the output of the following Python function? . len("hello",2, 4, 6)) . , a) Pfor : . . mo, . ‘ b6 ; . . :

(f) 4 d)3 ; co , . : 5. How is a code block indicated in Python? ~ : . - a. Brackets.

(g) indentation. .

(h) Key. . : __ o. None of the above. . .

Question 6. What will be the output of the following code snippet? ot __ ee es ~ = — example = ["Sunday", "Monday", "Tuesday", "Wednesday"; de! example/2] : . . print(example) , _ a. ['Sunday', 'Monday', 'Tuesday', 'Wednesday'] — : __ b. ['Monday', 'Tuesday', 'Wednesday' c. ['Sunday', 'Monday', 'Wednesday'] , an . .

uesday', 'Wednesday' , . ' . . : __ 7, What will be the output of the following Python program? . : ' ' i=0 . while i< 5: : : . print(i) ' : , i+=1 if i==3: . . break . . . ' else: : print(0) . . a) error . . b) 120 7 . c) 012 .

(a) none of the mentioned

Question 8. What will be the result of the following expression in Python "2 ** 345 ** 2"? .

(a) 65536 , | . c. 169 : | d. None of these , :

Question 9. Which of the following statements are used in Exception Handling in Python? - , t

(a) try .

(b) except . . :

(c) finally . .

(d) All of the above , "10. Which of the following is a Python tuple? .

(e) , 2,3]

(f) (1, 2, 3) . .

(g) 2, 2, 3 . dQ ; : . . ' . " " 1

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BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: BVCA-301 — Analog Digital Communication

Time: Three hours

Maximum Marks: 70

Instructions: Attempt all questions. Questions of Section A, B and C Carry 15, 5 and 1 marks

Yr . BVCA - 302: Robotics with Arduino Note: Attempt all questions. Questions of Section A, B and C Carry 15, 5 and 1 marks each respectively.

SECTION — A

Q.1 Discuss different types of Robots. Elaborate their functionality. | OR . Discuss different types of Sensors along with their functionality. | Q.2 Draw block diagram of microcontroller and explain each component of it.

OR

What are the major steps to install Proteus simulation software? Also discuss the features available in it. 1 |

SECTION — B

Q.3 Discuss in detail about the functions and need of Industrial Robots. Q.4 What are the major components of Robot? Explain. Q.5 List some of the important reasons for using robots instead of human to perform a task. | Q.6 What are the advantages of Integrating Artificial Intelligence into Robotics? Q.7 Discuss different types of Robot Locomotion? Explain in detail. Q.8 What are the differences between voice and speech recognition? | Q4 What is a pressure sensor? List any two advantages of pressure sensor. | Qj What is servo motor? List any two application of servo motor. | Note: Answer in briefest possible way. | 1, The main objective(s) of Industrial robot is to:

- (a) To minimize the labor requirement |
- (b) To increase productivity
- (c) To enhance the life of production machines | : d) All of the above. — (2) |

Question 2. Which of the following branch process with sensor feedback in robotics?

- (a) Computer Engineering
- (b) Mechanical Engineering
- (c) Electrical Engineering
- (d) Electronic Engineering 3 Which of the following sensor work based on radio detecting and ranging?
- (e) Sonar
- (f) Radar

- (g) Intertial :
- (h) Biosensor 4 Why do the robot need sensor?
 - (i) To collect information from environment
 - (j) To map environment attribute to a quantitative measurement
- (k) Only option 1 is true
- (l) Both option 1 and 2 are true. . 5 Which sensor can be used In robots for measuring distance?
 - (m) Mpu6050
 - (n) LDR module
 - (o) Ultrasonic Sensor |
 - (p) None of above |

Question 6. A Robot is a:

- (a) Programmable
- (b) Multifunctional manipulator
- (c) Both a and b
- (d) None of the above

Question 7. Drives are also known as:

- (a) Actuators
- (b) Controller
- (c) Sensors |
- (d) Manipulator

Question 8. If a robot has k legs, then the number of possible events is:

- (a) $N=(2k-2)$
- (b) $N=(2k-2)!$
- (c) $N=(2*k-1)!$
- (d) $N=(2k-2)!$

Question 9. Which of the following is not an essential component for construction of robots?

- (a) Power Supply
- (b) Actuators

- (c) Sensors |
- (d) Energy

Question 10. Which of the following is not a type of Robot Locomotion? '

- (a) Legged
- (b) Wheeled
- (c) Tracked deslip
- (d) Tracked skid . eK

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2023-24

Paper: BVCA-301 — Analog Digital Communication

Time: Three hours

Maximum Marks: 70

Instructions: Answer the questions in each section as instructed.

BVCA-301 : Analog and Digital Communication Note: Answer the questions in each section as instructed. All questions carry marks as indicated. |

SECTION — A

Note: Long-Answer Questions each in 500 words approximately (Answer any 2 out of 4; Each question carries 12 marks). $2 \times 12 = 24$ |

Question 1. Define amplitude modulation and describe it in detail with suitable diagrams.
3

Question 2. Determine the following for DSBSC (i) Bandwidth (2Marks) (ii) Time domain representation . (5Marks) (iii) Frequency spectrum Waveform (3Marks) (iv) Advantages, Disadvantages and | Applications(2Marks)

Question 3. For a modulation coefficient $m = 0.2$ and an unmodulated carrier power $P_c = 1000W$, Examine the total sideband power, upper & lower side band power, modulated carrier power and total transmitted power. :

Question 4. State the principle of Angle Modulation. Derive phase deviation, modulation index, frequency deviation and percent modulation. :

SECTION — B

Question 1. Short-Answer Questions each in 200 words approximately. Answer any 6 out of 9; Each question carries 6 marks. $6 \times 6 = 36$

- (a) Define sampling and express Nyquist sampling rate.
- (b) What are the elementary signals define with graphs and mathematical functions?
- (c) Define System and how to check the linearity of a system? Give suitable example with proof.
- (d) What is Fourier Transform? Calculate the Fourier Transform of the following signals (i) $e^{-t}u(t)$ where $u(t)$ is the unit step signal (ii) $e^{at}u(-t)$. (€) If $x(t) = u(t)$ is an unit step signal then draw the following (i) $3x(t - 2)$ (ii) $5x(-2t + 1)$
- (e) Classify the different methods of Pulse modulation techniques.
- (f) Discuss the generation of DSB SC Waves. :

- (g) A 1000kHz carrier is simultaneously modulated with 300Hz, 800Hz and 2kHz audio sine waves. Select the frequencies present in the output.
- (h) Write the advantages of digital communication over analog communication.

SECTION — C

Note: Very Short-Answer Questions (Answer all 10; Each question carries 1 marks). 1x10=10
) Analog signals cannot be sent using digital techniques. (True (T)/False (F))

- (a) Digitizing a signal can reduce distortion. (True (T)/False (F))
- (b) Digitization removes noise and distortion from analog signals. (True (T)/False (F))
- (c) A digital signal can be changed from a 1 to a 0 by noise. (True (T)/False (F) : g 8)
- (d) Unlike analog, digital communications is not band-limited. (True (T)/False (F))
- (e) — The amount of digital data that can be sent is limited by the channel capacity (C).
(True (T)/False (F))
- (f) What are the causal and non-causal signals?
- (g) What is the bandwidth of an AM signal?
- (h) | Write the full form of DSB-FC, DSB-SC, and SSB-SC.
- (i) Write the Nyquist criterion for sampling.

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: BVCA-301b — Analog Digital Communication

Time: Three hours

Maximum Marks: 70

Instructions: Attempt all questions. Symbols have their usual meanings.

BVCA - 301: Analog and Digital Communication Candidates are required to write their answers in their own words as far as practicable.

SECTION — A .

Long Answer type questions to be answered in about 750 words each: 15x2=30 Marks

Question 1. Describe Analog and Digital Communication system. Discuss the fundamentals of Digital Modulation OR it Describe 8 QAM and 16 QAM with suitable Example.

Question 2. Describe the detection of Signals in AWGN. * OR Describe Error Correcting Code with reference to Convolutional Code.

SECTION — B

Short Answer type questions to be answered in about 550 words each Any Six: 5x6=30 Marks

Question 3. Describe the classification of Signals.

Question 4. Discuss PAM with Example.

Question 5. Describe Linear Block Code with example.

Question 6. What are Inter Symbol Interference?

Question 7. Describe different wireless standards.

Question 8. Discuss IEEE 802.11 a and n.

Question 9. Describe the classification of system.

Question 10. Discuss Hamming Code.

SECTION — C

Answer the following questions in a word/ Sentence: 10x1=10 Marks

Question 1. Full form of AWGN ,

Question 2. Define Parity

Question 3. Is 0000, 0101, 1010 and 1111 cyclic?

Question 4. Define Network communication.

Question 5. What is the function of Input transducer?

Question 6. Define BIBO stability criterion.

Question 7. Define Phase Modulation.

Question 8. Define Pulse time modulation.

Question 9. Define Phasor Diagram.

Question 10. Define Constellation Diagram. —_—0 ——— - - |

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: GEN-301 — Digital Logic and Processors

Time: Three hours

Maximum Marks: 70

Instructions: Attempt all questions. Symbols have their usual meanings.

COMPUTER APPLICATION GEN - 301 : Digital Logic and Processors Question Paper) > Answer All Sections. > Candidates are required to write their answers in their own words as far as practicable. . Section - A : (15x2=30) a Long Answer type questions to be answered in about 750 words each :

Question 1. Prove the following Boolean identities :

(a) $AC + ABC = AC$

(i) $A + AB = A + B$

(ii) $ABC + ABC + ABC = A(B + C)$

OR**Question 2.** Explain about the signal processing and types of it.**Question 3.** Convert the number system from one form to another. $G_{10} = (6)_{10} = ()_{16}$

(a) $(110010)_{10} = ()_{16}$

(b) $(A996)_{16} = ()_{10}$

OR**Question 4.** What are the basic gates in logic circuit. Give the brief explanation . about them with truth table.**Question 1.** em

(a)

SECTION — B : (5x6=30)

Attempt any six: Short answer type questions to be answered in 550 words if they are theoretical.

Question 1. What is combinational circuit. Design any combinational! circuit with two AND gates and one OR gate and draw the truth table.**Question 2.** Perform mathematical operation : |

(a) 1101 (i) $1111 + 1010 - 101$

Question 3. What will be the output for the given logic circuit if $X_1 = 1$ and $X_2 = 0$.

Question 4. Calculate the number of cells in K-map if we have 3 pts and draw the K-map as well as write the output in each cells.

Question 5. Demorganize the expression. $\overline{A+B}(C+D)$

Question 6. Convert the given logical expression in SOP. $ABC + ACD + ABCD$

Question 7. What are the latches and what are its types ? Explain about NAND latch.

Question 8. What is multiplexer ?

SECTION — C : (10x1=10)

Question 1. In K-map what is the abbreviation of K ?

Question 2. Give the one example of universal gate.

Question 3. What type of arithmetic operation performed by OR gate ?

Question 4. The binary variable can store the value of zero or one. This statement is true or false.

Question 5. Which device converts single input into the multiple output ?

Question 6. Does flip-flop store the 1-bit of information ?

Question 7. Does register use for storing the data ?

Question 8. Multiply (1001), with (10). Q.9 Write the full form of MSB.

Question 10. Write the 1's complement of (1000).

Question 3. Ke

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2023-24

Paper: BVCA-302 — Robotics Arduino

Time: Three hours

Maximum Marks: 70

Instructions: / 1. Attempt any two out of four questions in Section A.

- BGI INO. cia dhatjecssissosncoeditarcenrtpotterse BVCA-302 : Robotics with Arduino ee
ee | Instructions: | 1. Attempt any two out of four questions in Section A. (2*12=24 Marks)

Question 2. Attempt any six out of nine questions in Section B (6*6=36 Marks)

Question 3. Attempt all 10 questions in Section C (1*10=10 Marks)

SECTION — A

(Long Answer Type questions to be answered in about 500 words each)

Question 1. Explain the fundamental components and functions of both hydraulic and | pneumatic systems used in robotics. Provide a detailed comparison between the two systems, highlighting their similarities and differences. Additionally, discuss - the advantages and limitations of each system in the context of robotics applications. : | 2) Discuss the components and operation of a machine vision system used in | robotics. Explain the role of lighting, cameras, frame grabbers, and computer = ‘ processors in capturing and processing visual data for robot guidance and control. | Provide examples of machine vision applications in industrial automation and | robotics. ‘ 3) Explain the basic components of a hydraulic system and their functions. Provide examples of applications where hydraulic systems are commonly used. | 4) Describe the operation of a pneumatic system, including its basic components and their functions. Discuss how a compressor generates compressed air, and how pneumatic actuators convert this energy into mechanical power. | : SECTION-B | 5. Answer any six of the following in about 200 words each: : (a) Define the term "robotics" and explain its significance in modern industrial automation. Provide examples of industries where robotics is commonly utilized, highlighting its impact on productivity and quality improvement.

- (a) Explain the concept of degrees of freedom in industrial robots. Describe how degrees of freedom are related to the robot's ability to perform various tasks and navigate its work envelope efficiently.
- (b) Explain the historical development of robotics from its origins in science fiction to its modern applications in industries. Highlight key milestones and technological advancements that have shaped the field.
- (c) Discuss the advantages and disadvantages of different drive systems used in . industrial robots, including hydraulic drive, electric drive, and pneumatic drive. Provide examples of scenarios where each drive system is preferred based on performance requirements.
- (d) Define actuators and sensors in the context of robotics. Explain their respective roles in robotic systems and provide examples of each. Discuss how actuators and sensors

work together to enable robotic motion and perception in industrial and non-industrial applications.

- (e) Discuss the importance of precision in robot movement and its components: P.T. 0. spatial resolution, accuracy, and repeatability.
- (f) (c) Differentiate between fixed automation, programmable automation, and flexible automation. Provide examples of industries where each type of automation is commonly employed, and discuss their respective advantages and limitations. |
- (g) Explain the concept of ROS (Robot Operating System) and its significance in robotics development.
- (h) Describe the various types of motors used in robotics, including DC motors, AC motors, stepper motors, and servomotors.

SECTION — C

Question 6. Answer the following objective type questions:

- (a) In robot motion planning, what is the purpose of the "configuration space"?
- (b) The physical space occupied by the robot
- (c) The space of all possible joint configurations of the robot
- (d) The workspace where the robot performs tasks
- (e) The space-time continuum in which the robot operates
 - (i) What is the role of the "end effector" in a robot manipulator? ‘ a) To control the robot’s motion
- (f) To provide feedback to the operator
- (g) To act as the hand or tool of the robot
- (h) To process sensor data
 - (i) Which type of robot programming involves specifying the desired end-effector positions and orientations without detailing the intermediate joint motions?
- (i) Motion Planning __ b) Trajectory Planning
- (j) Inverse Kinematics
- (k) Forward Kinematics
 - (i) What does the term "ROS package" refer to in the Robot Operating System (ROS)?
- (l) A physical package containing robot components
- (m) A collection of software libraries, tools, and configuration files
- (n) The payload carried by a robot

- (o) A unit of measurement for robot speed : (Vv) Which concept in robotics involves a robot's ability to recover from errors or unexpected events without human intervention?
 - (p) Fault Tolerance
 - (q) Redundancy
 - (r) Closed-Loop Control
 - (s) Safety Interlock
 - (i) In the context of swarm robotics, what does the term "stigmergy" refer to?
 - (t) Communication through direct signalling
 - (u) Cooperation based on the use of pheromones or environmental cues
 - (v) Hierarchical control structure
 - (w) Swarm navigation using GPS | (vii) What is the origin of the term "robot"? . wine Count - - @ >)
 - (x) Greek
 - (y) Czechoslovakia
 - (z) German
 - (i) When did the modern technology of industrial robotics begin?
 - () 1920
 - () 1960
 - () 1980
 - () 2000
 - (i) How is robotics defined as industrial automation?
 - () A subset of automation
 - () Unrelated to automation
 - () Only involves computer programming
 - (i) What is another name for the Cartesian coordinate configurable robot?
 - () Spherical robot b) XYZ robot :
 - () Rectangular robot
 - () Hyperbolic robot = RRR EERE KKK KER EE
-
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BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: BVCA-303 — Programming In Linux

Time: Three hours

Maximum Marks: 70

*Instructions: Attempt all questions. Questions of Section A, B and C Carry 15,5 and**GEN - 302 : Introduction to IoT Note: Attempt all questions. Questions of Section A, B and C Carry 15,5 and 1 marks each respectively.***SECTION — A**

Q 1 What is the structure of Arduino sketch. Explain with suitable example a OR ' > What are the major components of Arduino Uno Microcontroller? Briefly explain. Q2 How you can develop IOT based project for the smart agriculture system?

. How you can develop IOT based project for the smart city?

OR**SECTION — B**

Q.3 What are the main components of the IOT architecture? Q.4 What are the primary hardware components that make up an embedded system? Q.5 What are some examples of sensors that can be used in Home automation? Q.6 What are some of the protocols used for IOT communication? Q.7 What are the difference between IPV4 and IPV6? Q.8 How could 5G cellular networks impact IOT? Q.9 What steps can an organization take to protect IOT systems and devices? Q.10 Discuss different types of Drown and their functionality. FT. O. (2) | 7

SECTION — C

Note: Answer in briefest possible way.

Question 1. Which of the following IOT network has a very short range:

- (a) Short Network
- (b) LPWAN
- (c) SigFox “
- (d) Short Range Wireless Network

Question 2. What is the main purpose of WoT (Web of Things) in the IoT?

- (a) Improve the usability and interoperability
- (b) Reduce the Security
- (c) Complex the development

- (d) Increase the cost 5 3 Which of the following is not an application of IOT |
- (e) Wearable's i
- (f) Smart Grid ce) Arduino ~ d) Smart City ' 4 What is the full form of the MQTT?
- (g) Multi-Queue Telemetry Things
- (h) Multiple Queue Telemetry Things
- (i) Message Queue Telemetry Things
- (j) Message Queue Telemetry Transport 5 Which of the following layers provides end to end communication in IOT?
- (k) Logical Layer
- (l) Data Link Layer
- (m) Transport layer |
- (n) Session Layer 6 Which of the following devices is used to measure the gases or liquid ?
- (o) Optical Sensor :
- (p) Gas Sensor
- (q) Smoke Sensor
- (r) Pressure Sensor 7 Which of the following topology is used for ZigBee Smart Energy?
- (s) Bus Topology
- (t) Ring Topology
- (u) Star Topology
- (v) Any Topology a @ " 8 Identify the protocol used to link all devices in IOT
- (w) TCP/IP
- (x) UDP
- (y) Network
- (z) HTTP

Question 9. In IOT as one of the key characteristics, devices have different hardware platforms and networks.

- (a) Sensors -
- (b) Heterogeneity
- (c) Security
- (d) Connectivity 10 Which of the following issues are considered in JOT?

- (e) Security Issue :
- (f) Reliability issue
- (g) Standard issue
- (h) All above RKEEK

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: BVCA-303 — Programming In Linux

Time: Three hours

Maximum Marks: 70

Instructions: Attempt all questions. Symbols have their usual meanings.

ROI No: srsancaccancsae BVCA - 303 : Programming in Linux ois Peo Answer All Sections. i Candidates are required to write their answers in their own words as far as practicable.

SECTION — A

Answer in about 750 words Each question carries 15 Marks ;[15x2=30] 1. (a) What is OS? Explain the Linux OS architecture and also list its features. (b) What is inode and dentry in context of Linux file-system? Describe the Linux file-system.

Question 2. a What is octal permission notation? Discuss Linux permission model with examples. Or.

(a) What is context switching? Explain the various states of Linux process in detail.

SECTION — B

Answer any SIX questions in about 550 words Each question carries 5 Marks [5x6=30]

Question 3. Write a shell program to calculate average and sum of given numbers.

Question 4. Write a shell program to get factorial of a given number.

Question 5. List and describe the modes of vi editor.

Question 6. Use example to explain the working of awk.

Question 7. Explain the background process.

Question 8. Discuss any five commands used in Linux with example.

Question 9. Write a brief note on Linux shell.

Question 10. Discuss briefly about the Dash, Activities and Grid GUI elements in Ubuntu Linux. Section-C Answer in one word Each question carries 1 Marks [1x10=10] 11, What is command used to display MAC address of a Linux Machine?

Question 12. Which command renames a file in Linux?

Question 13. Write the command used to print a file from terminal.

Question 14. In which file are passwords saved in Linux?

Question 15. What option of grep enables us to ignore the case of alphabetical characters?

Question 16. Name the Linux boot loader used in Ubuntu Linux?

Question 17. Which command is used to search forward in vi editor?

Question 18. Name the process that are run unattended. |

Question 19. What Key. combination will terminate a running command in terminal? |

Question 20. Name the software that runs GUI in Ubuntu Linux. > =

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2023-24

Paper: BVCA-303 — Programming In Linux

Time: Three hours

Maximum Marks: 70

Instructions: Answer the questions in each section as instructed. All questions carry marks as indicated.

TROMIN Ossian stettcattrskonesseossoeaseaplity BVCA-303 : Programming in Linux

SECTION — A

Note: Long-Answer Questions each in 500 words approximately (Answer any 2 out of 2x12=24 4; Each question carries 12 marks. !) Explain the architecture of Linux, emphasizing the roles of the kernel, shell, and file system.

Question 2. Describe the Linux file system hierarchy in detail, providing examples of directories at each level. Explain the purpose and usage of essential file system management commands such as ls, cp, mv, and rm. Illustrate their functionality with practical examples,

Question 3. Discuss the significance of Linux commands in performing various tasks efficiently. Explain how commands like touch, grep, wc, and cat contribute to file manipulation, searching, and text processing. Provide examples to demonstrate the practical utility of these commands in everyday computing tasks.

Question 4. Compare and contrast the advantages and disadvantages of using Linux as an operating system. Provide real-world examples to support your arguments.

SECTION — B

Question 5. Answer any six of the following in about 200 words each : 6x6=36 | (a) Write a shell script to print a pattern of asterisks (*) in the following manner: | * | * | Ke OK

- (a) Develop a shell script that accepts two numbers as input and performs addition, subtraction, multiplication, and division on them. (C) Discuss the significance of pipes in Linux. Provide an example scenario where pipes are used effectively with multiple commands. '
- (b) Discuss the purpose of the mkdir command in Linux. Provide an example demonstrating its usage in creating directories. (€) Define the tee command in Linux and explain its functionality.
- (c) What is the purpose of the cat command in Linux? How does it differ from other file viewing commands?
- (d) Explain the significance of the banner command in Linux. How can it be used to display text in a visually appealing manner?

- (e) Discuss the functionality of the grep command in Linux. __ (i) Describe the significance of the cal command in Linux. How does it differ from other calendar- related commands? Provide an example demonstrating its usage. P.T.0.

SECTION — C

Question 6. Answer the following questions in a word or sentence each: 1x10=10

- (a) Which command is used to display the current working directory in Linux? i , : (a) Is (b)PWD (c)cal (d)cp

(i) Which command is used to create a new directory in Linux?

- (b) mkdir (b)touch (c)rm (d)mv

(i) What does the man command do in Linux?

- (c) Displays the manual page of acommand (b) Moves files or directories

- (d) Creates a new file (d) Counts the number of words in a file

(i) What does the head command do in Linux?

- (e) Displays the first part of a file (b) Displays the last part of a file

- (f) Searches for a pattern in a file (d) Counts the number of lines in a file

(i) Which command is used to count lines in a file in Linux?

- (g) wc (b) sort (c) grep (d) banner |

(i) What is the purpose of the grep command in Linux?

- (h) Displays the calendar for the current month (b) Displays the current date and time

- (i) Counts the number of lines in a file (d) Searches for a pattern in a file

(i) Which command is used to count the number of words in a file in Linux?

- (j) wc (b) wc (c) ps (d) tee

(i) What is the function of the sort command in Linux?

- (k) Displays the sorted content of a file (b) Moves files or directories

- (l) Counts the number of lines in a file (d) Searches for a pattern in a file

(i) Which command is used to display the last part of a file in Linux?

- (m) head (b) tail (c) who (d) be

(i) What does the mkdir command do in Linux?

- (n) Removes a directory (b) Creates a new directory

- (o) Moves files or directories (d) Displays the manual page of a command ARK

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BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: BVCA-304 — Java Applet Programming

Time: Three hours

Maximum Marks: 70

Instructions: ANSWER ALL SECTIONS, CANDIDATES ARE REQUIRED TO WRITE THEIR ANSWERS IN THEIR

— REENO057601 COMPUTER APPLICATION BVCA - 304 : Java Applet Programming
NOTE: ANSWER ALL SECTIONS, CANDIDATES ARE REQUIRED TO WRITE THEIR ANSWERS IN THEIR OWN WORDS AS FAR AS PRACTICABLE. Section - A | Long Answer type questions to be answered in about 750 words each: 15x2=30 Marks

Question 1. What are Data types and Variables in Java. Differentiate between static, instance and local variables. Why | keywords cannot be used as variables?

Question 2. Discuss about if else Statement and Switch Statement. Write a program to display days of week for a number using Switch Statement. (Let us suppose days of week are assigned numbers as Sunday-0, Monday-1, | Tuesday-2 and so on. So, if user enters 2 value 4, the output should be Thursday)

Question 3. a) What are arrays? Explain its types and also discuss the ways to initialize them.

(a) Write a program to find the maximum element in an array of 20 integer values. | OR

Question 4. Explain about for, while and do-while loop structures in Java with example. What is the difference between entry-controlled loops and exit-controlled loops?

SECTION — B

Short Answer type questions to be answered in about 550 words each Any Six: 5x6=30 Marks

Question 1. How to read data using Scanner class in java? State at least three methods defined in the Scanner class to read data.

Question 2. What is concatenation of Strings? Write a program to concatenate two strings.

Question 3. Differentiate between break and continue statement with example.

Question 4. What are the features of Java Programming Language?

Question 5. Write a program to check whether the number is prime or not.

Question 6. Define JDK and JVM. Why Java is considered as platform independent language?

Question 7. What is a Constructor? Explain its types.

Question 8. What is an Object and how is it created? Create a Class Student with at least three attributes and create two objects of the class Student.

SECTION — C

Answer the following questions in a word: 10x1 Marks i

Question 1. 1) Who invented Java Programming?

(a) Guido van Rossum

(b) James Gosling

(c) Dennis Ritchie

(d) Bjarne Stroustrup PF: (2) 5,

(i) _____ is used to find and fix bugs in the Java programs. | :

(e) JVM .

(f) JRE

(g) JDK |

(h) JDB

(i) What does an Eclipse IDE allow? '

(i) Editing the program

(j) Compiling the program '

(k) Both Editing and Compiling the program |

(l) Editing, Building, Debugging and Compiling the program.

(i) SDK stands for_____

(m) Software Design Kit

(n) Serial Development Kit

(o) Software Development Kit

(p) Serial Design Kit

(i) Which of the following is not an OOPS concept in Java?

(q) Polymorphism

(r) Inheritance |

(s) Compilation i

(t) Encapsulation

(i) What will be the output of the following Java program? Classvariable_scope :

```
publicstaticvoidmain(String ares[]}) |
{
int x; i
x=5; |
{
int y = 6;
System.out.print(x +      +y);
}
System.out.println(x +      + y); |
}
}
```

(u) Compilation error

(v) Runtime error :

(w) 5656

(x) 565 amt) a rr SR , @)

(i) What will be the output of the following Java program? Class output

```
publicstaticvoidmain(String args[])
{
StringBuffer s1 = newStringBuffer( Quiz );
StringBuffer s2 = s1.reverse();
System.out.println(s2); ;
}
} j
```

(y) QuizziuQ

(z) ziuQQuiz

() Quiz

() ziuQ

(i) In which memory a String is stored, when we create a string using new operator?

() Stack

() String memory

() Heap memory

() Random storage space

(i) Which keyword is used for accessing the features of a package?

() package

() import

() extends

() export .

(i) Which of these values is returned by read method is end of file (EOF) is encountered?

() 0 |

() 1 : |

() -1 i

() Null ‘

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2023-24

Paper: BVCA-304 — Java Applet Programming

Time: Three Hours

Maximum Marks: 70

Instructions: Attempt all questions. Symbols have their usual meanings.

BVCA-304 : Java Applet Programming Answer All Sections. Candidates are required to write their answers in their own words as far as practicable.

SECTION — A

Long Answer type questions to be answered in about 500 words each: 12x2=24 Marks
Attempt any two:

Question 1. What do you understand by the concept of Object Oriented Language? Explain how Java is specified as an Object Oriented Language.

Question 2. Explain about for, while and do-while loop structures in Java with example. Why they are categorized as entry-controlled and exit controlled loops?

Question 3. What are operators ? Explain all the Operators supported by Java programming language. 4, Discuss Switch Statement. Write a program to calculate percentage of total marks, obtained in 5 subjects by a student and display the equivalent grade as S, A,B,C, D, or F. [91-100% as S grade; 81-90% as A grade; 71-80% as B grade and so on. F grade represents | Fail] , Section - B 5, Attempt any six (Answer in about 200 words each): 6 x 6 = 36 Marks

- (a) What do you understand by classes and objects? How objects are created from a class?
- (b) What are arrays? How will you implement one dimensional and two dimensional arrays?
- (c) Define JDK and JVM. Why Java is considered as platform independent language?
- (d) Differentiate between static, instance and local variables. :
- (e) Discuss break and continue statement in java. .
- (f) Write a program Count the digits of a number given input by a user.
- (g)) Write a java program find the area and circumference of a circle using the concept of objects and classes.
- (h) ' Discuss any four inbuilt methods for String manipulation given in java library.

SECTION — C

Question 6. Attempt all the questions: 10 x 1=10 Marks

- (a) What does an Eclipse IDE allow? g a) Editing the program
- (b) Compiling the program .
- (c) Both Editing and Compiling the program
- (d) Editing, Building, Debugging and Compiling the program.

Question 9. 7-0. Si abla s

- (a) Which keyword is used for accessing the features of a package?
 - (a) package
 - (b) import
 - (c) extends
 - (d) export
- (i) | Which of these values is returned by read() method is end of file (EOF) is encountered? ,
 - (e) 0
 - (f) 1
 - (g) -1
 - (h) Null
- (i) Which one of the following is not a Java feature?
 - (i) Object-oriented
 - (j) Use of pointers F c) Portable
 - (k) Dynamic and Extensible :
- (i) JRE stands for _____.
 - (l) Java run ecosystem
 - (m) JDK runtime Environment
 - (n) Java Runtime Environment
 - (o) None of these
- (i) Which class in Java is used to take input from the user?
 - (p) Scanner
 - (q) Input
 - (r) Applier
 - (s) None of these

(i) "Automatic type conversion is possible in which of the possible cases?"

(t) Byte to int

(u) Int to long

(v) Long to int

(w) Short to int

(i) Identify the output of the following program. String str = "abcde"; System.out.println(str.substring(1, 3));

(x) abc Coundi- G) @

(y) bc

(z) bed

() cd

(i) Find the output of the following code. int ++a = 100;

```
System.out.println(++a);
```

() 101

() Compile error as ++a is not valid identifier

() 100

(i) None i (X) _ Find the output of the following code. Public class Solution

```
Public static void main(String args[]){ .  
Int i;  
for(i = 1; i < 6; i++){  
| if(i > 3) continue;  
| }  
| System.out.println(i);  
| }
```

() 3

() 4 5

() 6 oS

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: BVCA-305 — Network Config Management

Time: Three hours

Maximum Marks: 70

Instructions: ANSWER-ALL SECTIONS. CANDIDATES ARE REQUIRED TO WRITE THEIR ANSWERS IN THEIR

COMPUTER APPLICATION BVCA - 305 : Network Configuration and Management
NOTE: ANSWER-ALL SECTIONS. CANDIDATES ARE REQUIRED TO WRITE THEIR ANSWERS IN THEIR OWN WORDS AS FAR AS PRACTICABLE. my Section - A Long Answer type questions to be answered in about 750 words each: 15x2=30 Marks Question. 1Describe OSI Model with suitable drawing. Compare OSI and TCP/IP Models. : : OR Question. 2Explain the process of inter-VLANs network communication. What do you understand by Layer 3 switch? . Question. 3What is a 802.11 Protocol. Explain the 802.11—infrastructure network and components. Lok OR > Question. 4Discuss any three Internet Applications with illustrative diagrams. eA Section - B Short Answer type questions to be answered in about 550 words each Any Six: 5x6=30 Marks Question. '1What are various Network Topologies?

Question 2. Describe Repeater, Hub and Switch. Compare Switch with Bridge? Question. 3Discuss the 802.3 Protocol and its categories? Question. 4what do you understand by Distribution System? Question. 5What is Access Point? What are its various deployment methods? Question. 6Explain BGP Protocol. Discuss its two types? Question. 7What is DNS server? Explain with suitable diagram? Question. 8How Wireshark is helpful in network troubleshooting? aaaa en Bg Section - C Answer the following questions in a word: 10x1 Marks

Question 2. UTP Stands for ‘

Question 2. In network models cables are considered in layer. Question. 3In network models routers are considered in layer. Question. 4The WLC and APs use a protocol to communication. Question. 5WPA stands for Question. 6The broadcasting address of a network is 192.168.1. P Question. 7Subnetting is the short form of . Question. 8UTM is a firewall or not. Question. 9° ":Which one is more reliable-TCP or UDP.

Question 10. > The tool used for trace the routing path of the packet is . nes

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BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2023-24

Paper: BVCA-305 — Network Config Management

Time: Three hours

Maximum Marks: 70

Instructions: Answer all sections. Candidates are required to write their answers in their own words as far as

| ! re] | RIL NO ..n.0essncssecssessrsesecesnessnssnnse | BVCA-305 : Network Configuration and Management Ss eI er SS Se | Note: Answer all sections. Candidates are required to write their answers in their own words as far as ae practicable. | SECTION-A | Note: Long Answer type questions to be answered in about 500 words each: 12x2=24

Question 1. What is Layer 3 switch? Explain the process of inter- VLANs network communication with a suitable diagram.

Question 2. Describe various layers of OSI Model with suitable drawing.

Question 3. What is WLAN? Explain components of WLAN.

Question 4. Consider an IP address 8.1.4.5/16. Find the number of Host Bits, number of Subnet bits. Number of subnets and number of hosts per subnet. Assuming only one mask is used throughout the network.

SECTION — B

Note: Short Answer type questions to be answered in about 200 words each: 6x6=36

Question 5. Answer any six of the following:

- (a) What are various Network Devices, explain with their working?
- (b) Describe Hub, Bridge and Switch.
- (c) Discuss the Ethernet CSMA/CD Protocol.
- (d) What do you understand by CAPWAP?
- (e) What is WLC? What are its various deployment methods?
- (f) Explain SFTP Protocol.
- (g) What is MAC addressing?
- (h) How is PRTG helpful in network monitoring?
- (i) Describe Router and compare it with switch.

SECTION — C

Question 6. Answer the following questions in a words each:

- (a) SNMP stands for ,
 - (i) In OSI models cables are considered in layer.
 - (ii) In Network models routers are considered in layer.
 - (iii) The WLC and APs use a protocol to communication.
 - (iv) New IP addressing schema is known as ‘
 - (v) /16 is a CIDR of a network, its dotted decimal equivalent is 5
 - (vi) The equivalent of Subnetting in layer 2 is 2
 - (vii) UTM is a firewall or not.
 - (viii) Which one is more reliable — TCP or UDP.
 - (ix) The tool used for trace the routing path of the packet is KRKEKK

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2023-24

Paper: BVCA-306 — GEN

Time: Three hours

Maximum Marks: 70

Instructions: 1, Attempt any two out of four questions in Section A.

GEN - 302 : Introduction to IoT Instructions: 1, Attempt any two out of four questions in Section A. (2*12=24 Marks)

Question 2. Attempt any six out of nine questions in Section B (6*6=36 Marks)

Question 3. Attempt all 10 questions in Section C (1*10=10 Marks)

SECTION — A

Question 1. Explain 3 and 5-layer IoT architecture in detail.

Question 2. Differentiate between microprocessors and microcontrollers. What are the applications areas of Microprocessors and Microcontrollers?

Question 3. What is the purpose of sensors in IoT? What are the required criteria to choose a suitable sensor to measure the desired physical parameter?

Question 4. What do you understand by the term IoT Ecosystem? What are the components of an IoT system? Describe all.

SECTION — B :

1, What do you mean by IoT? What are the advantages and disadvantages of IoT applications?

Question 2. What is cloud? What are the basic characteristics of cloud computing?

Question 3. Differentiate RISC and CISC microprocessors.

Question 4. What is the difference between transducers, sensors and actuators?

Question 5. What do you mean by Embedded System? Explain its characteristics.

Question 6. Explain the components of the Microcontroller.

Question 7. Define the Term: a) Data Processing b) Automation c) Communication Protocols

Question 8. Explain any 6 applications of IoT.

Question 9. Discuss the principles of operation for each type of sensor and their applications in IoT devices.

SECTION — C

Question 1. Which protocol provide Full-Duplex communication?

- (a) WebSocket
- (b) HTTP
- (c) MQTT
- (d) CoAP

Question 2. The RFID technology consists of:

- (a) Antenna
- (b) Transponder
- (c) Transceiver -d. All of these Fr LsO: Q)

Question 3. Connected car is :

- (a) BAN B.LAN C. WAN D. None Of These

Question 4. The full form of LoRa is: :

- (a) label range radio |
- (b) long radio range ,
- (c) none of these _5. The different wireless networks used in cellular network:
- (d) GSM : | B, CDMA C. LTE i D. All Of The Above

Question 6. Which of the following is Lightweight protocol used in IoT Applications?

- (a) MQTT : b. WebSocket
- (b) HTTP , d All of the above 3

Question 7. Which of the following wireless technologies provide mesh network architecture in easiest way? : |

- (a) Bluetooth |
- (b) ZigBee
- (c) Wi-Fi
- (d) None of these

Question 8. What is the full form of HTTP?

- (a) Hyphenation Text Test Program

- (b) Hypertext Transfer Protocol
- (c) Hypertext Transfer Package
- (d) None of the above

Question 9. An entity in IoT:

- (a) Businesses
- (b) Governments
- (c) Consumers
- (d) All of the above

Question 10. ZigBee is based on which of the following standard:

- (a) IEEE 802.15.4 B. IEEE 802.5 C. IEEE 802.4.13 D. Both a and b

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2023-24

Paper: CS-102 — Digital Logics and Circuits

Time: Three Hours

Maximum Marks: 70

*Instructions: Attempt **five** questions including Question No. 1, which is compulsory. Write your Roll No. at the top immediately on receipt of this question paper.*

Question 1.

- (a) How does changing the base of a number system impact the representation and arithmetic operations performed on numbers? [3]
- (b) What is an unweighted and weighted code? Provide an example of each. [3]
- (c) What is the primary difference between combinational and sequential circuits? [3]
- (d) Explain the concept of a multiplexer and its applications. Design a 4-to-1 multiplexer using basic logic gates. [3]
- (e) How many 1's are present in the binary representation of $4 \times 256 + 8 \times 16 + 12$? [2]

Question 2.

- (a) Compute: $(B1.21)_{16} + (211)_4 + (131)_8 = (?)_8$ [4]
- (b) How is a negative number represented in sign-magnitude form? Calculate the sum of -36 and $+52$ using signed number addition. [5]
- (c) Express the following functions in Sum of Minterms and Product of Maxterms and implement using a universal gate of your choice: [5]
 - (i) $F(A, B, C, D) = A'D + C'D' + ABD$
 - (ii) $F(A, B, C) = (ABC + BC)(A'B'C + BC)$

Question 3.

- (a) Minimize the expression with don't-care conditions $F(A, B, C, D) = \sum(0, 1, 5, 13, 14, 15) + \sum d(3, 4, 7, 10, 11)$ using K-map. Draw the logic diagram of the minimized circuit. Why are reflected codes used in K-Map? [7]
- (b) Determine the Prime-Implicants and essential Prime-Implicants of the following Boolean function using the Tabulation method: [7]

$$F(A, B, C, D, E, F, G) = \sum(20, 28, 30, 52, 60)$$

Question 4.

- (a) What are Half and Full-Adder circuits? Explain in detail with a logic diagram and truth table. For the addition of two 8-bit numbers, what is the required number of half adders and full adders? Explain with a logic diagram. [5]
- (b) Design a circuit that compares two 4-bit numbers X and Y . The outcome of the comparison is specified by three binary variables indicating whether $X = Y$, $X > Y$, or $X < Y$. [5]
- (c) Implement a Full-Subtractor circuit using a multiplexer. [4]

Question 5.

- (a) What is ROM? Design a combinational circuit using a ROM that accepts a 2-bit number and generates an output binary number equal to the cube of the input number. [5]
- (b) What is a code converter, and why is it important in digital systems? Design a BCD to Excess-3 code converter. [5]
- (c) What is a Programmable Logic Array (PLA)? Explain its basic structure and functionality. [4]

Question 6.

- (a) Explain the various types of flip-flops used in digital circuits: SR, JK, D, and T flip-flops. Explain how each can be implemented using basic logic gates. For each type, describe its truth table and characteristic equation. [6]
- (b) What is an excitation table for SR, D, T, and JK flip-flops, and how does it differ from a truth table? [4]
- (c) What is a Register? Design a 4-bit Register. [4]

Question 7.

- (a) What are counters and their types? Design a 3-bit synchronous up-counter using a T flip-flop. Explain with a timing diagram. [7]
- (b) What is the primary purpose of state reduction and state assignment in sequential circuit design? Explain with an example. [7]

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2023-24

Paper: CS-102 — Object Oriented Programming

Time: Three hours

Maximum Marks: 70

Instructions: Attempt five questions including Question No. 1, which is compulsory. All undefined symbols have their usual meanings.

Question 1. Attempt all questions: [2 × 9 = 18]

- (a) Differentiate between procedural programming and object-oriented programming.
- (b) What is dynamic binding in C++?
- (c) Explain the purpose of a destructor.
- (d) What is a reference variable? Show its usage with an example.
- (e) What is a copy constructor? Write its prototype.
- (f) Can a virtual function be declared inline? Why or why not?
- (g) Find the output of the following code segment:

```
#include <iostream>
using namespace std;
class Test {
    static int count;
public:
    Test() { count++; }
    static int getCount() { return count; }
};
int Test::count = 0;
int main() {
    Test t1, t2, t3;
    cout << "Count=" << Test::getCount() << endl;
    return 0;
}
```

- (h) Identify and explain the compile-time error in the following code block:

```
#include <iostream>
using namespace std;
class Box {
    double width;
public:
    Box(double w) : width(w) {}
};
void printWidth(Box b) {
    cout << b.width << endl;
}
```

- (i) What is the use of `this` pointer in C++?

Question 2.

- (a) What is encapsulation? How is it implemented in C++? Compare private, protected, and public access specifiers. [6]
- (b) Explain the concept of inline functions. What are the rules and situations where inline function expansion is ignored by the compiler? [4]
- (c) Discuss function overloading with a suitable code example. [4]

Question 3.

- (a) Explain the dynamic memory allocation operators `new` and `delete` in C++ with syntax and examples. Differentiate between `delete` and `delete[]`. [6]
- (b) What are friend functions and friend classes? Discuss their advantages and disadvantages with a suitable example. [8]

Question 4.

- (a) What is constructor overloading? Write a C++ class representing a `Complex` number with default, parameterized, and copy constructors. [8]
- (b) Write the rules for overloading operators in C++. How does overloading a binary operator using a friend function differ from overloading it using a member function? [6]

Question 5.

- (a) Write a C++ program to overload the binary addition operator `+` to add two `Matrix` objects. [8]
- (b) Explain unary operator overloading with an example of overloading pre-increment `++` and post-increment `++` operators. [6]

Question 6.

- (a) What are the different types of inheritance supported in C++? Explain the concept of virtual base class in multiple inheritance with a diagram and code snippet to show how ambiguity is resolved. [8]
- (b) Discuss the difference between compile-time polymorphism and runtime polymorphism. How is runtime polymorphism achieved in C++? [6]

Question 7.

- (a) What is a pure virtual function and an abstract class? Write a program demonstrating the use of virtual destructors. [7]
- (b) What is exception handling? Explain the use of `try`, `catch`, and `throw` blocks in C++ with a sample program. [7]

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BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2022-23

Paper: CS-103 — Numerical Computing

Time: Three hours

Maximum Marks: 70

Instructions: ATTEMPT ANY FIVE QUESTIONS. THE FIGURES IN THE RIGHT-HAND MARGIN INDICATE

Question 1.

- (a) Express the following decimal numbers in binary, octal and hexadecimal: [6]
(i) 22.623
(ii) -10.215
- (b) What are the three different types of errors encountered in numerical calculations? Explain each of them briefly. [3]
- (c) Determine the roots of the following equation using Newton-Raphson method correct to three decimal places: [5]

$$x^3 - 5x + 3 = 0$$

Question 2.

- (a) Derive the Regula-Falsi method for calculation of roots of non-linear equation. Write the steps of calculation of roots using this method. Discuss the scenario where this method would give large errors. [8]
- (b) Calculate the roots of the following equation using Regula-Falsi method, correct to three decimal places: [6]

$$x^3 + x^2 + x + 7 = 0$$

Question 3.

- (a) Describe the LU decomposition method for solving system of linear equations. [2]
- (b) Use the LU decomposition method to solve the following system of linear equations: [7]

$$\begin{aligned} 5x - 2y + z &= 4 \\ 7x + y - 5z &= 8 \\ 3x + 7y + 4z &= 10 \end{aligned}$$

- (c) Describe the method of diagonalization of a real symmetric matrix using Jacobi method. [5]

Question 4. For the matrix A given below, calculate:

$$A = \begin{bmatrix} 4 & 3 & 2 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{bmatrix}$$

(a) The largest Eigen value and the corresponding Eigen vector. [7]

(b) Inverse of the matrix. [7]

Question 5.

(a) Derive Simpson's 1/3rd Rule for numerical integration. [4]

(b) Integrate the following function using:

$$I = \int_0^1 \frac{1}{1+x} dx$$

(i) Trapezoidal Rule

(ii) Simpson's 1/3rd Rule

using $h = 0.125$. Also compare the results obtained in each case with the actual value of the integral. [10]

Question 6.

(a) Derive the Newton-Gregory forward interpolation formula. [4]

(b) Use Newton's Divided difference method to calculate $f(0.25)$ and $f(0.27)$ for the following tabulated function: [8]

x	0.20	0.22	0.24	0.26	0.28	0.30
$f(x)$	1.6596	1.6698	1.6804	1.6912	1.7024	1.7139

(c) State two applications of interpolation. [2]

Question 7.

(a) Use Picard's method to solve the differential equation:

$$\frac{dy}{dx} = 1 + xy, \quad y(0) = 1$$

Find the value of y for $x = 0.5$ correct to 3 decimal places. [7]

(b) Use Predictor-Corrector method to solve the following differential equation:

$$\frac{dy}{dx} = y(1 + x^2), \quad y(0) = 1$$

Use $h = 0.05$ and compute $y(0.3)$ correct to 3 decimal places. [7]

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2023-24

Paper: CS-103 — Numerical Computing

Time: Three hours

Maximum Marks: 70

Instructions: Attempt ANY FIVE questions.

ROUGIN OS pocssesn so sccteocsseebevenas CS-103 : Numerical Computing

Question 1. a. Diagonalize the given matrix: : [7] 52; O22

Question 0. 3220 t & -

- (a) Solve the following equation using bisection method to find a real root between 2 and 3 (correct to 3 decimal places) [7] xlogix —1.2=0 , .

Question 2. For the following system of linear equations: $9x_1 + 2x_2 + 4x_3 = 20$, $x_1 + 10x_2 + 4x_3 = 6$, $2x_1 - 4x_2 + 10x_3 = -15$ a. Check the convergence of the iterative procedure. [4] se _b. Use the Gauss-Siedel iterative procedure to solve the above (correct to 3 significant digits). [10]

Question 3. a. Derive the Lagrange interpolation formula. What are the advantages and dis-advantages of using it? [7] _

- (a) Fit 3-degree Lagrange polynomial to find $f(x)$ at $x = 3.8$ for the following function: [7] ASHE Saad ce ER Se $f(x)$ 6 Chae TE 2 Saee

Question 4. .a. Use the Newton's divided difference formula to Gerive; : [7] i, Trapezoidal Rule

- (a) Simpson's 1/3 Rule :

- (a) Use Gauss-Legendre 2-point formula to calculate: [7] 1 Je — $10x + 6$ dx

Question 4.

Question 5. Solve the following equations (correct to 4 significant digits) using: [14] d $2x$, $y(0)=1$, $0 < x < 0.5$, $h=0.1$ dx : i, Euler's Method ii, Picard's Method Pe. Te [0]

Question 6. For the given matrix calculate: [14]

- (a) Inverse matrix using Gauss Elimination method.
(i) Largest Eigen value and corresponding Eigen vector using iterative method, $1 < 3$:
4 S152

Question 2. 40 | . 7. Compute the root between | and 2 of the equation given below using: [14] $x^3 - 2x^2 - 437x - 5 = 0$

- (a) Regula-Falsi method :

(i) Newton Raphson method OO | ee ee eee ey Se

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BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2021-22

Paper: GEN-201 — Website Designing

Time: Three hours

Maximum Marks: 70

Instructions: THERE ARE THREE SECTIONS. ALL SECTIONS ARE COMPULSORY.

GEN - 201 : Website Designing NOTE : THERE ARE THREE SECTIONS. ALL SECTIONS ARE COMPULSORY.

SECTION — A: Multiple Choice Questions[2 × 10 = 20]

Question 1. Answer the following multiple choice questions:

(a) The correct sequence of HTML tags for starting a webpage is:

- (i) <html>, <head>, <title>, <body>
- (ii) <head>, <title>, <html>, <body>
- (iii) <html>, <body>, <title>, <head>
- (iv) <html>, <head>, <body>, <title>

(b) What will happen if the following JavaScript code is executed?

```
var count = 0;
while (count < 10) {
    console.log(count);
    count++;
}
```

- (i) An error is displayed.
- (ii) An exception is thrown.
- (iii) The values of count variable are stored in a particular location.
- (iv) The values of count from 0 to 9 are displayed in the console.

(c) How to create a hyperlink in HTML?

- (i) javatpoint.com
- (ii) javatpoint.com
- (iii) javatpoint.com
- (iv) <a>www.javatpoint.com

(d) How to create a checkbox in HTML?

- (i) `<input type="checkbox">`
 - (ii) `<input type="button">`
 - (iii) `<checkbox>`
 - (iv) `<input type="check">`
- (e) Which of the following is the correct way to start an ordered list with a numeric value of 4?
- (i) `<ol type="1" start="4">`
 - (ii) `<ol type="1" initial="4">`
 - (iii) `<ol type="1" begin="4">`
 - (iv) `<ol type="1" num="4">`
- (f) Which of the following is the correct syntax to make the background-color of all paragraph elements yellow?
- (i) `p {background-color: yellow;}`
 - (ii) `p {background-color: #yellow;}`
 - (iii) `all {background-color: yellow;}`
 - (iv) `all p {background-color: #yellow;}`
- (g) The CSS property used to specify the transparency of an element is:
- (i) `opacity`
 - (ii) `filter`
 - (iii) `visibility`
 - (iv) `overlay`
- (h) Which of the following syntax is correct in CSS to make each word of a sentence start with a capital letter?
- (i) `text-transform: capitalize;`
 - (ii) `text-style: capital;`
 - (iii) `transform: capitalize;`
 - (iv) `text-transform: capital;`
- (i) In JavaScript, the `x === y` statement implies that:
- (i) Both x and y are equal in value and data type.
 - (ii) Both x and y are equal in value, type, and reference address.
 - (iii) Both x and y are equal in value only.
 - (iv) Both are not the same.

SECTION — B: Short Answer Questions

[5 × 6 = 30]

Question 2. What is the difference between DIV and SPAN in HTML? Explain with examples.

OR

What is the difference between class selectors and id selectors? Explain with examples.

Question 3. What is the purpose of the z-index and how is it used?

What is HTML? How is a website created with the help of HTML & JavaScript?

Question 4. What is the difference between CSS2 and CSS3?

What are Event Bubbling and Event Capturing?

Question 5. In how many ways can you create an array in JS?

State the difference between the operators “==” and “===” with examples.

SECTION — C: Long Answer Questions

[10 × 3 = 30]

Instructions: Answer any three questions.

Question 6. Explain the step-by-step process of creating a responsive website.

Question 7. Name some new features which were not present in HTML but are added to HTML5.

Question 8. What is JavaScript and what are the scopes of a variable in JavaScript?

Question 9. What is CSS? Also explain the different ways CSS can be included in HTML.

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2021-22

Paper: GEN-202 — Programming In Python

Time: Three Hours

Maximum Marks: 70

Instructions: Attempt all the Parts. The figures in right

: Paper : GEN-202 Programming in Python Note: Attempt all the Parts. The figures in right margin indicate the marks.

PART — A

Note: Multiple Choice Type Question and Short type Questions 10x2=20 .

Question 1. What will be the output of: the following code snippet ?
`a=3, b=1
print(a, b) a, b=b, a
print(a, b)`

(a) 3 1 1 3

(i) 3 1 1 1

(ii) 1 3 3 1
2. What will be the result of below command in python language :
`round(0.4) - round(-0.5)`

(b) 0

(i) 1

(ii) 2 / 3

Question 3. What is the output of the below program in python :
`print(0.2 + 0.4 == 0.6)`

(a) True

(i) False

(ii) Error

(iii) Depends on machine

Question 4. Which type of programming does Python support ?

(a) object-oriented programming

(i) "structured programming

(ii) functional programming

(iii) all of the mentioned

Question 5. All keywords in Python are in

- (a) Capitalized
 - (i) Lower case | ' (iii) UPPER CASE
 - (ii) None of the mentioned

Question 6. What will be the output of the following Python code ?
`i=1 ; while True : if i%3 ==0: , break — print (i) . i+=1 : | fi 123`

- (a) error
- (b) 12
- (c) none of the mentioned | 3 PTO

Question 7. What will be the output of the following Python function ?
`min(max(False,-3,-4), 2, 7)`

- (a) 4
 - (i) -3
 - (ii) 2 :
 - (iii) False

Question 8. What arithmetic operators cannot be used with strings in Python ?

- (a) * (iit) -
 - (i) +
 - (ii) All of the mentioned

Question 9. What are the two main types of functions in Python ?

- (a) System function (it) Custom function
 - (i) Built-in function & User defined function
 - (ii) User function

Question 10. What is output of `print (math. pow (3, 2))` ?

- (a) . 9.0
 - (i) None (iit) 9 .
 - (ii) None of the mentioned 4 . Part - B Note: Internal Choice-Either or Type Questions :- 5*4=20

Question 11. Write a python program to accept three integer numbers in same line and find their sum.

OR

Write a python program to find sum of two . numbers using command line arguments.

Question 12. Write Python Program to display even numbers between m and n. OR - Write a python program that display stars in right angled triangular form using nested loops.

Question 13. Write a python program to retrieve the elements from 2D array and display them using for loop.. : OR Write a function that return the result of | addition, subtraction, multiplication and division. |

Question 14. Write the python program to calculate factorial | values using. recursion. : | OR | . Write a python program to read all the strings from the text file and display them. .

Question 15. Write a Python program to check given character is vowel or consonant. OR - , Write a Python program to convert Fahrenheit into Celsius.. , : Part - C | Note : Open choice - 3 out 4 questions) 3x 10=30

Question 16. Write a Python program to find L.C.M. of two : numbers. , 6 L

Question 17. How to create a dictionary and explain its different operations. Elaborate how to iterate | over a dictionary.

Question 18. Consider a rectangle class. Write Python code to check the area of the first rectangle is greater than second by overloading '>' Operator.

Question 19. Discuss the technical strengths of python. What is the difference between list and tuples in Python ? - KKK 7 300 wn % ict 2 i FB ' p . ' : ' a ° ' , '

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2013-14

Paper: BCS-401 — Computer Org Arch

Time: Three hours

Maximum Marks: 70

Instructions: ANSWER ALL QUESTIONS.

ROU NOs esesesnssaseneacenoereass NOTE : ANSWER ALL QUESTIONS.

Question 1. Answer in brief any four parts from the following : 4x3=14

- (a) What are the functions of program counter/ instruction pointer and stack pointer in any CPU? Explain with suitable examples.
- (b) What are meant by static and dynamic RAM and what are the advantages of one over the other?
- (c) What is meant by DMA and how is it realized by means of the DMA interface and CPU?
- (d) What is meant by a procedure (subroutine)? Differentiate between (i) subroutine and interrupt subroutine (ii) NEAR and FAR procedures.
- (e) What is the difference between instructions and directives? Explain with example the functions performed by the ASSUME and DW directives.

Question 2. a) What is meant by the addressing modes of any CPU? Explain with suitable examples the addressing modes of 8086 CPU. 5

OR

What are the functions of ALU in any CPU? Explain the construction and working of one bit (stage) ALU. How will you design the ALU of 8 bit CPU using one bit ALU's? 5

- (a) What is meant by flag (status) register in any CPU? Differentiate between the control and conditional flags in the flag register. Find the status of different conditional flags of 8086 after the performance of the addition of hexadecimal numbers 5439H and 456AH by the ALU of 8086. 5
- (b) Mention the difference between hardwired and microprogrammed Control Unit. 4

Question 3. a) What is meant by memory hierarchy and what is its need in a large computer system? Explain in brief the cache memory organization and its advantages. 3

- (a) What is meant by virtual memory and what is its need? Differentiate between the logical memory address and physical memory address. Explain the method of physical memory address calculation in protected virtual addressing mode of a CPU with 20 address lines and 16 bit segment registers. 5
- (b) Differentiate between mask ROM, PROM, EPROM, EEPROM and flash memory, Mention in brief different applications of ROM, 4 2

Question 4. a) What is meant by interrupt? Name and explain in brief the different types of interrupts that are possible in any computer system. What is meant by priority of interrupts?

Name the different methods of assigning priority to interrupts and explain any one of these methods. 5

OR

What is the need of I/O interface in computer organization? Write in brief about synchronous and asynchronous serial data communication. Explain the need of MODEM for computer data communication to long distances. 5

- (a) Differentiate between simple I/O, strobed I/O and handshaking modes of I/O data transfer. Explain in brief the different modes of operation of programmable peripheral interface 8255. Find the control word that must be written in the control register of 8255 for using it in simple I/O mode with Port A as input Port B as output Port C upper as output and Port C lower as input. 5
- (c) What is meant by I/O processor? Explain in brief about its use in computer organization. 4

Question 5. a) Explain the operations performed on executing the following instructions : 5

- (a) MOV AL, [BX]
- (i) IDIVCX
- (ii) TEST DX, [BP] [D1]
- (iii) ROR AX, CL and
- (iv) JBE NEXT where NEXT is a label and [CL] = 04
- (b) Write an assembly language program to multiply the unsigned hexadecimal numbers 2A and 3C stored as data somewhere in memory. The product is also to be stored in memory. Comments should follow each statement to explain the program. 5
- (c) | What is meant by string manipulation instructions and where are these used? Explain the functions performed by the instructions CMPSW and SCASB. 4

OR

What is meant by a MACRO and how is it defined and called? Explain with suitable example the meaning of MACRO expansion. 4

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2013-14

Paper: BCS-403 — Fundamentals Of Computing

Time: Three hours

Maximum Marks: 70

*Instructions: Answer any five questions, including question no. 1, which is compulsory.***Question 1.** Write the full name and function of the following terms: [2 × 7 = 14]

- (a) UNIVAC I
- (b) ULSI
- (c) CD-ROM
- (d) ARPANET
- (e) FTP
- (f) URL
- (g) HTTP

Question 2.

- (a) What is computer and what are its characteristics? Explain each. [7]
- (b) Differentiate among computer program, software and hardware. Explain the system software and application software with appropriate examples. [2+5]

Question 3.

- (a) Divide $(1010110)_2$ by $(110)_2$. [3]
- (b) Multiply $(10101)_2$ with $(101)_2$. [2]
- (c) Add these binary numbers: 11111, 11111, 10111, 1011, 10111. [3]
- (d) Convert the binary number 11011101.0111_2 to decimal number system. [3]
- (e) Convert the decimal number 352_{10} to binary number system. [3]

Question 4.

- (a) What do you mean by software development steps? Explain each step. [7]
- (b) What is programming language? Describe the different types of programming language with merits and demerits of each. [7]

Question 5.

- (a) What is Internet? How is it different from the World Wide Web? Write the various applications of Internet. [2+2+3]
- (b) What are the steps involved in solving problems using computers? Explain each. [7]

Question 6.

- (a) Write the characteristics of an algorithm. Draw a flowchart to check if a given year is a leap year or not. [3+4]
- (b) What is pseudocode? Write a pseudocode to find the sum of digits of a given number. [2+5]

Question 7. Write short notes on the following:

- (a) Evolution of Computers [4]
- (b) Structured Programming and Logic/Control Structures [5]
- (c) Flowchart with its merits and demerits. [5]

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2013-14

Paper: BSC-044 — Intro To Computer Programming C

Time: Three Hours

Maximum Marks: 70

Instructions: Answer any five questions, including Question No. 1, which is compulsory. All undefined symbols have their usual meanings.

Question 1. Answer any six questions:

[3 × 6 = 18]

- (a) What are the steps involved in solving problems using computer?
- (b) What are logical operators? List them with their meaning.
- (c) What is the difference between `scanf()` and `printf()`?
- (d) What is bottom up approach of program designing?
- (e) Define data type. Write the names of any three data types.
- (f) What is a header file? Write the names of any three header files.
- (g) What is the difference between local and global variables?

Question 2.

- (a) What is an algorithm? Draw the flow chart for comparing three given numbers and display the largest number. [5]
- (b) What are the advantages of High level languages over Machine level languages? Write the names of any three high level languages. [3]
- (c) What are the differences between Compiler and Interpreter? [3]

Question 3.

- (a) Explain the various decision making statements in C language with an example for each. [6]
- (b) Explain the difference between a 'while' loop and a 'do-while' loop with a suitable example. [6]

Question 4.

- (a) Discuss about the precedence of operators in C. What will be the output of the following program: [4]

```
void main()
{
    int x;
    x = -3 + 4 - 7 * 8 / 5 % 10;
```

```
printf("x=%d\n", x);  
}
```

- (b) What is an array? How do you declare and initialize a one dimensional array? Illustrate with an example. [5]
- (c) Write a 'C' program to convert the given temperature in Fahrenheit to Celsius. [4]

Question 5.

- (a) Define pointer. What are the advantages of using pointers? [4]
- (b) What is the difference between a String and Array of characters? [3]
- (c) Explain any three String handling library functions in 'C' with an example for each. [6]

Question 6.

- (a) Define Function in C language. What is the difference between a user defined function and library function? [4]
- (b) Explain the parameter passing techniques of functions through an example. [6]
- (c) What do you mean by function prototype? [3]

Question 7.

- (a) Define storage classes in C Programming. List various storage classes used in 'C' with their properties. [6]
- (b) What is a structure? How do you declare a structure variable and access its members? Explain with an example. [7]

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2023-24

Paper: BVCA-201 — Introduction To Databases

Time: Three Hours

Maximum Marks: 70

Instructions: Please attempt all Sections. Question of Section — A, B and C carry 12, 6 and 1

Reh-No. 76932 , : TT | Subject: Computer Application Note: Please attempt all Sections. Question of Section — A, B and C carry 12, 6 and 1 marks each respectively SECTION — A Note: Please answer in approximately 500 words each. Attempt any two from the following question. Each question carries 12 Marks ‘ Question 1. Describe the three-schema architecture. Why do we need mappings between schema levels? How do different schema definition languages support this architecture?

Question 2. Discuss the different types of user-friendly interfaces and the types of users who typically use each.

Question 3. Discuss the various reasons that lead to the occurrence of NULL values in relations

Question 4. Discuss the role of a high-level data model in the database design process.

SECTION — B

Note: Please answer approximately in 200 words each. Each question carries 6 marks each. | Attempt any six from following: | Question 5. A. What is the difference between logical data independence and physical data independence? Which one is harder to achieve? Why? B. Discuss the different data types of SQL. C. Discuss the entity integrity and referential integrity constraints. Why each is considered important? D. Discuss Functional dependency with suitable example. E. What is a relationship type? Explain the differences among a relationship instance, a relationship type, and a relationship set. F, What is a participation role? When is it necessary to use role names in the description of relationship types? G. What is meant by a recursive relationship type? Give some examples of recursive relationship types. H. Why are duplicate tuples not allowed in a relation? I. What is the difference between a key and a superkey? EEG)

SECTION — C

Note: Please answer in briefest possible way. Each question carries 1 mark each.

Question 6. Define the following terms

- (a) Data model
- (b) Database schema
- (c) Client/server architecture

- (d) Owner entity type :
- (e) Weak entity type i
- (f) Entity g. Attribute Value
- (g) Relation state | i. Degree of a relation .
- (h) Normalization —_—_— Oo -

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2023-24

Paper: BVCA-401 — Cloud Computing

Time: Three hours

Maximum Marks: 70

Instructions: Answer the questions in each section as instructed. All questions carry marks

z : : BVCA-401: Cloud Computing Rg Instructions: Answer the questions in each section as instructed. All questions carry marks as indicated. : Section-A Very Short Answer type questions: [10x1=10]

Question 1. Which of the following is a characteristic of PaaS? - a) Supports only one programming language - b) Does not integrate with web services - c) Provides an ability to "Auto-scale" - d) Requires manual hardware updates 2, PaaS is primarily used for: ~ a) Networking hardware : - b) Developing, testing, running, and managing applications - c) Providing email services - d) Hosting web pages

Question 3. Which model involves the infrastructure residing at the customer's premises? - a) Public cloud ' - b) Private cloud - c) Hybrid cloud - d) Community cloud

Question 4. Which of these is an example of an IaaS provider? - a) Google App Engine : - b) Amazon Web Services (AWS) - c) Salesforce - d) Slack

Question 5. SaaS applications are: - a) Hosted locally on user devices - b) Accessible over the internet PT. 0. sees a - c) Only for enterprise users - d) Managed by end-users : | 6. Which service model allows users to avoid the cost and complexity of purchasing and managing physical servers? - a) SaaS : ' | - b) JaaS | -c) PaaS - d) NaaS :

Question 7. In which model do users have control over operating systems, storage, and deployed applications? | - a) SaaS - b) PaaS | - c) IaaS - d) NaaS | 8. SaaS applications often integrate with: — esa). On-premise-software-only - : | - b) Other cloud-based services | - c) Local hardware solutions | - d) None of the above | 9. Which deployment model offers resources over the internet to anyone willing to pay? : - a) Public cloud : - b) Private cloud : - c) Hybrid cloud - d) Community cloud

Question 10. PaaS is advantageous for integration because it: i - a) Requires less manual coding - b) Provides pre-configured development environments - c) Eliminates the need for middleware - d) Reduces application complexity) PP plexity aie Q) 2?

SECTION — B

Short Answer type questions: (Attempt any 6)

[6X6=36]

Question 1. What are the advantages of using cloud computing?

Question 2. What are the characteristics of cloud architecture that separate it from a traditional one? !

Question 3. What does infrastructure-as-a-service refer to?

Question 4. What does the acronym SaaS mean? How does it relate to cloud computing?

Question 5. What is difference between block level and file level storage?

Question 6. Explain different types of cloud computing services.

Question 7. What is difference between different types of servers?

Question 8. What are basic requirements of a secure cloud software?

Question 9. Discuss the use of hypervisor in cloud computing

SECTION — C

Long Answer type questions: (Attempt any 2)

[12x2=24]

Question 1. What is Virtualization in Cloud Computing? Explain its types and benefits.

Question 2. What is RAID? How it is useful and explain its types?

Question 3. What are the different types of Clouds in Cloud Computing? Explain with examples.

Question 4. Explain the role of cloud technologies for social networking. Discuss the operational and economic benefits of SaaS.

—x—

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2022-23

Paper: BVCA-401 — Web Technology

Time: ThreeHours

Maximum Marks: 70

Instructions: Attempt all questions. Symbols have their usual meanings.

| REFNO 065162 | Paper : BVCA-401 | Cloud Computing SECTION — A | Very Short Answer Type Questions | 10x1=10 ° | 1. Who are the Cloud service providers in a cloud | ecosystem ? | 2. Who are the Cloud Consumers in a cloud . | ' ecosystem ? (c) | 3. Who are the Direct customers in a cloud ecosystem ? P.7T.0.

Question 4. What is cloud ?**Question 5.** Name the cloud services model.)**Question 6.** Who is the father of cloud computing. | :**Question 7.** Give few examples of hybrid cloud. . | 8 Can you name some open-source cloud! | computing platform databases ?**Question 9.** Give few examples of public cloud. |**Question 10.** Give few examples of private cloud.**SECTION — B : | _____**

_ Short Answer Type Questions (Do any 6) = > 6x6=36**Question 1.** What are the advantages of using cloud | computing? - 5**Question 2.** What are the characteristics of cloud, architecture that separate it from a traditional . one?**Question 3.** What does infrastructure-as-a-service refer to ? | :, Io.**Question 4.** What does the acronym SaaS mean ? How does | it relate to cloud computing ?) 5. What do you know about the Multi-cloud | strategy ? | 6. Explain different types of cloud computing | services.**Question 7.** What is AWS ? What types of services does it | provide ?**Question 8.** What are basic requirements of a secure cloud | software ? ‘**Question 9.** Discuss the use of hypervisor in cloud computing. :**SECTION — C _____**

. Long Answer Type Questions (Do any 2) . 4 12x2=24

Question 1. What is Virtualization in Cloud Computing ? | Explain its types and benefits.

Question 2. What are some of the popularly used cloud computing services ? 3 P.T.0.

Question 3. What are the different types of Clouds in Cloud - Computing ? Explain with examples.

Question 4. Explain the role of cloud technologies for social networking. Discuss the operational and, . economic benefits of Saas. , kik | ‘ |

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2022-23

Paper: BVCA-402 — Database Management

Time: Three Hours

Maximum Marks: 70

Instructions: Attempt any two questions from section-A,

Probability and Statistics | Note: Attempt any two questions from section-A, any six questions from section-B and all questions from section-C. The marks for each question are given against the respective questions.

SECTION — A 2x 12=24**Question 1.** (a) State and prove Markov's inequality.

(a) Derive Chebyshev's inequality using Markov's inequality.

(b) Let $f(x)$ be the uniform distribution on $0 \leq x \leq 5$ and 0 everywhere else. Give a bound using Chebyshev's for $P(1 \leq X \leq 4)$ and also calculate the actual probability. How do they compare? $3+3+6=12$

Question 2. (a) Give the axiomatic definition of probability.

(a) Define convergence in distribution.

(b) Consider the joint probability mass function p_{ij} of two discrete random variable X and Y as follows with support $(x, y) = (1, 1), (1, 2), (2, 2)$.

$\{(1, 2), (2, 2)\}$. Construct a joint probability distribution table of X and Y . Are X and Y independent? Find $E(X)$ and $E(Y)$. $2+2+8=12$

Question 3. (a) Define simple and composite hypothesis with examples. (b) What is the Type-I and Type-II error in hypothesis testing? (c) Urban storm water can be contaminated by many sources, including discarded batteries. When ruptured, these batteries release metals of environmental significance. A sample of 49 Panasonic AAA batteries gave a sample mean zinc mass of 2.06g and a sample standard deviation of 0.140g. Does this data provide compelling evidence for concluding that the population mean zinc mass exceeds 2.0g? Construct a 95% confidence interval for population mean zinc mass. Choose the appropriate critical value ($z_{0.05} = 1.64$, $t_{0.05, 48} = 1.68$, $z_{0.01} = 2.33$, $z_{0.995} = 1.96$). (d) Let X be a random variable with PDF given by $f(x) = \frac{1}{2}x^2(2x+c)$, $0 < x < 1$, 0, otherwise. Find the value of c . Also compute mean, variance and CDF of random variable X . If $Y = \frac{1}{3}X^2$ be any function of random variable X , then find $E(Y)$ and $Var(Y)$. $2+2+8=12$

SECTION — B 6x6=36

Question 5. (a) If the random variables X and Y independently follow exponential distribution with parameters λ and 4 , respectively, then find the expectation and variance of $Z = X + Y$.

(a) What do you mean by standardization of a variable ? $4+2=6$ REFNO 065195 : | f

Question 6. (a) Show that the Pearson's correlation coefficient between the variable X and Y is independent of change of scale and origin. .

(a) Assume neither A nor B is an impossible - event. If A and B are mutually exclusive, will they be independent ? $4+2=6$

Question 7. (a) Show that Poisson distribution is a limiting case of Binomial distribution.

(a) What is p-value in hypothesis testing ? $4+2=6$ |

Question 8. A person uses his car 30% of the time, walks 30% of the time and rides the bus 40% of the time as he goes to work. He is late 10% of the time when he walks; he is late 3% of the time when he drives; and he is late 7% of the time he takes the bus. What is the probability he took the bus if he was late ? What is the probability he walked if he is on time ? What is the probability that he'll be on time ? 6 4 , | REFNO 065195

Question 9. Vehicles pass through a junction on a busy road at an average rate of 240 per hour. Find the probability that none passes in a given minute. What is the expected number passing — in three minutes ? Find the probability that this expected number actually pass through in a given three-minute period. $G = 2.72 \sim 3$, é |

Question 10. (a) State and prove the law of total probability. (b) Give an example where two events are mutually exclusive but not exhaustive. Also give an example where two events are not mutually exclusive but exhaustive. $(4+2)$

Question 11. Define Normal distribution. Also discuss its important properties. - $(2+4)$

Question 12. (a) Suppose the average CGPA of students is normally distributed with population mean

Question 8. 7 and standard deviation 0.2. At 10% level of significance, what is the minimum number of sample size needed to obtain a margin of error of 0.013, ' (b) What is cumulative distribution function? What are the criteria that a valid CDF need to satisfy ? 4

Question 13. The results of a national survey showed that on average, adults sleep 6.9 hours per night. Suppose that the standard deviation is 1.2 hours. Use Chebyshev's theorem to calculate the minimum percentage of individuals who sleep between 4.5 and 9.3 hours. Use Chebyshev's theorem to calculate the minimum percentage of individuals who sleep between 3.9 and 9.9 hours. Assume that the number of hours of sleep follows a normal distribution. Use the empirical rule to calculate the percentage of individuals who sleep between 4.5 and 9.3 hours per day. How does this result compare to the value that you obtained using Chebyshev's theorem in the first part of this question ? 6

SECTION — C

Question 14. Answer all the questions in a Sentence each : $10 \times 1 = 10$

- (a) What is the sampling distribution of sample variance ?
- (b) What is the range of Pearson's correlation coefficient ?
- (c) When will central limit theorem fail, even if the sample size is large ? : (d) What is the relation between power of the test and Type-II error ? | 6 , | P . | REFNO 065195 a

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| (e) Construct a sample space for the event }  
7 tossing a coin till a first head appear.  
: (f) Let  $P(B) > 0$  so that  $P(A/B)$  is defined. if  
| - ACB, what is  $P(A/B)$  ?  
: (g) Define probability mass function with  
: suitable example.  
: (h) How can we get the binomial distribution  
from Bernoulli random variables ? .  
: (i) If mean = median = mode, what can we  
| say about that distribution ?  
| (j) What is the formula of sample covariance  
| kee  
7 200
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BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2023-24

Paper: BVCA-402 — Database Management

Time: Three Hours

Maximum Marks: 70

Instructions: Answer the questions in each section as instructed.

| 76992 | (Probability and Statistics) | Note: Answer the questions in each section as instructed.
 | All questions carry marks as indicated. | SECTION - A | Note: Long Answer Questions.
 Answer any 2 out of 4, | Each question carries 12 marks. 2x12=24

Question 1. Derive mean and variance of Normal distribution i.e., | 2. What is simple linear regression model ? Obtain the | expression of f_j and $f_{\cdot}$. | 3. What is distribution function ? Explain properties of the distribution function with its proof. P. F.20.

Question 4. What is likelihood function. How can we obtain the estimate of the parameter of any distribution function | (give its theory). In random sampling from Normal | distribution ie, $N(y, \sigma^2)$, find the maximum | likelihood estimator (MLE) for (i) for σ^2 when y is | known (ii) for σ^2 when y is known. |

SECTION — B

Note : Short Answer Questions. Answer any 6 out of 9, Each | question carries 6 marks).
 6x6=36 |

Question 5. (a) Calculate the mean and standard deviation if the) $f(x)$ has form : $32x$, for $2 < x < 4$ FO see “ 0; otherwise

- (a) From a city population, the probability of | selecting (i) a male or a smoker is $7/10$, (ii) a male smoker is $2/5$ and (iii) a male, if a smoker is already selected is $2/3$. Find the probability of selecting (a) a non-smoker, (b) a male, and (c) a | smoker, if a male is first selected.
- (b) The contents of urns, I, II or III] are as follows : Urn I: 1 white, 2 black and 3 red balls, Urn II : 2 white, 1 black and 1 red balls, | Urn III : 4 white, 5 black and 3 red balls. | (2) : One urn is chosen at random and two balls drawn from it. They happen to be white and red. What is the probability that they come from urns I, II or III.
- (c) Define chi square test statistic. What are the conditions for the validity of chi-square test ? (ec) What are the non-parametric distributions. Define Kolmogorov-Smirnov statistic.
- (d) For any three events A, B and C, prove the following :
- (e) $P(A \cup B | C) = P(A | C) + P(B | C) - P(AB | C)$.
 (i) $P(C | A) = P(B | A)$ provided that $B \subset C$ and $P(A) > 0$, $P(B | A) < P(C | A)$.
- (f) Discuss statistical hypothesis. Define null and alternative hypothesis with an example.
- (g) Define Chebyshev's inequality and Markov inequality.

- (h) Define Statistics. According to you, where it can be used ? Give at least three examples.

SECTION — C

Note: Very Short Answer Questions Answer all 10, Each question carries 1 Mark. 10x 1=10

Question 6. (i) In an experiment a coin is thrown five times. Write down the sample space.

- (a) Write the additive property of Poisson distribution. (3) P.T.O.
- (b) Define weak law of large numbers.
- (c) Define type IJ error.
- (d) Who discovered the normal distribution ?
- (e) Write the pdf of standard normal variate.
- (f) Give two examples of Poisson distribution. (viii) For any three events A, B and C: $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$
- (g) Define exhaustive events.
- (h) Write the mean and variance of standard normal variate. (4) 200

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2022-23

Paper: BVCA-403 — Cloud Computing

Time: Three Hours

Maximum Marks: 70

Instructions: This question paper carries three sections :

. REFNO 065163 | Paper : BVCA-403 | Critical Design Thinking in social Science | Note: This question paper carries three sections : | Section-A, B and C. All questions are : compulsory. | Section -A' t | Note: Answer any two of the following each in | 500 words approximately. Each question. : carries 12 marks. $12 \times 2 = 24$ | 1. What do you mean by critical thinking ? Describe the different barriers to critical thinking. P.T.O, REFNO 065163 |

Question 2. Explain the meaning of thinking. Differentiate between critical thinker and uncritical thinker.

Question 3. What is design thinking ? Describe the phases of design thinking. /

Question 4. What are essential design thinking skills? Discuss any two design thinking skills. *

SECTION — B

Note: Answer any six of the following questions. in approximately 200 words. Each question carries 6 marks : $6 \times 6 = 36$

Question 5. (a) Describe the benefits of critical thinking!

- (a) Highlight the importance of critical _ thinking in life. |
- (b) How critical thinking helps employees at 4 workplace ? Discuss. * (dj) Discuss how critical thinking makes personal life better.
- (c) Why design thinking skills matter ? Discuss. : REFNO 065163
- (d) Describe how one can learn design thinking.
- (e) Differentiate between hard skill and soft skills.
- (f) Describe the functions of communication.
- (g) Suggest strategies to improve communications skills.

SECTION — C

Note: Answer each of the following questions in a word or a sentence. Each question carries 1 mark. $10 \times 1 = 10$

Question 6. (a) Who introduced the term critical thinking ?- , | (b) Explain consistency as critical thinking standard. | c) Differentiate between argument and | explanation. . (ad) Why people with critical thinking skills are | in demand by organizations ?

(a) Why should we study logic ? REFNO 065163

(b) What is meant by reflective thinking ?

(c) What is not an argument ? : .

(d) Give the meaning of argument.

(e) What is meant by analytical skills ? (j How one can practice active listening ? kik

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2023-24

Paper: BVCA-403 — Cloud Computing

Time: Three Hours

Maximum Marks: 70

Instructions: This question paper carries three sections : Section /

| 76993 : Paper : BVCA-403 | (Critical Design Thinking in Social Science) | A, Band C. All questions are compulsory. | SECTION -A | Note: Answer any two of the following questions in | 500 words approximately. Each question carries | 12 marks.

Question 1. Explain the concept of critical thinking. Describe the benefits of critical thinking.) : 2. Who is a critical thinker ? Discuss the characteristics of a critical thinker. | 3. Why design thinking skills matter ? How can we learn design thinking ? Discuss. Pie

Question 76993.**Question 4.** What is the significance of studying logic ? What is not an argument ? Explain.**SECTION — B |**

Note: Answer any six of the following questions in about 200 words approximately. Each question carries 6 marks.

Question 5. (a) Explain the concept of sociocentrism.

(a) What is the importance of critical thinking at workplace ? Discuss.

(b) Describe the phases of design thinking.

(c) Explain fairness as critical thinking standard. (ec) What are the design thinking skills ? Describe team building as design thinking skill.

(d) Explain the functions of communication.

(e) Describe analytical skill as essential skill for design thinking.

(f) Differentiate between arguments and explanations.

(g) How can we identify premises and explanations ? Explain. SECTION — C Note: Answer each of the following questions. Each question carries 1 mark.

Question 6. (a) What do you mean by thinking ?

(a) Explain how one can improve communication skills. | 76993

(b) How logical correctness can be practiced ?

(c) What are the levels of communication ? | (e) List some of the most common barriers to critical | thinking.

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2022-23

Paper: BVCA-403b — Cloud Computing

Time: Three Hours

Maximum Marks: 70

Instructions: . This question paper carries three sections :

| ‘ a | . REFNO 065163 : _ Paper: BVCA-403_ | Critical Design Thinking in social Science
| Note: . This question paper carries three sections : | Section-A, B and C. All questions
are . compulsory. | Section-A | Note: Answer any two of the following each in | 500 words
approximately. Each question | carries 12 marks. 12x2=24 —_ 1. What do you mean by
critical thinking ?

(a) Describe the different barriers to critical , thinking. REFNO 065163

Question 2. Explain the meaning of thinking. Differentiate between critical thinker and uncritical thinker. : |

Question 3. What is design thinking ? Describe the phases of design thinking.

Question 4. What are essential design thinking skills ? Discuss any two design thinking skills.

SECTION — B

Note: Answer any six of the following questions in approximately 200 words. Each question, carries 6 marks : 6x6=36

Question 5. (a) Describe the benefits of critical thinking. | _

- (a) Highlight the importance of critical thinking in life.
- (b) How critical thinking helps employees at workplace ? Discuss.
- (c) Discuss how critical thinking makes * personal life better. | (e) Why design thinking skills matter ? Discuss. 2 REFNO 065163
- (d) Describe how one can learn design ° thinking.
- (e) Differentiate between hard skill and soft skills.
- (f) Describe the functions of communication.
- (g) Suggest strategies to improve communications skills.

SECTION — C

Note: Answer each of the following questions in a word or a sentence. Each question carries 1 mark, ‘ 10x1=10 . 6. (a) Who introduced the term critical thinking ?

- (a) Explain consistency as critical thinking standard.
- (b) Differentiate between argument and explanation.
- (c) Why people with critical thinking skills are in demand by organizations ? e) Why should we study logic ? [= | REFNO 065163
- (d) What is meant by reflective thinking 2
- (e) What is not an argument ? | | _ (h) Give the meaning of argument.
- (f) What is meant by analytical skills ? |
- (g) How one can practice active listening ? | seek * 4 | : - 4 200 :

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2022-23

Paper: BVCA-404 — Android Development

Time: Three Hours

Maximum Marks: 70

Instructions: Answer the questions in each section as

Paper : BVCA-404 an Microprocessor Note: Answer the questions in each section as instructed.
. All question carry marks as indicated. - - Section - A . Note: Long Answer Questions
each in 500 words approximately (Answer any 2 out of 4. Each question carries 12 marks)
2x12=24 ‘

Question 1. Explain the architecture of the 8085 -microprocessor, describing each component
and how they interact with each other. P.O,

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BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2023-24

Paper: BVCA-404 — Android Development

Time: Three Hours

Maximum Marks: 70

Instructions: Answer the questions in each section as

REFNO 076994 Microprocessor Note : Answer the questions in each section as instructed. All questions carry marks as indicated.

SECTION — A

Note: Long Answer Questions each in 500 words approximately (Answer any 2 out of 4; Each question carries 12 marks). 2x 12=24

Question 1. Explain pin diagram of 8085 microprocessor in details.

Question 2. Describe internal diagram of microprocessor

Question 8085. PTO. REFNO 076994

Question 3. What are interrupts ? Differentiate between software interrupts and hardware interrupts?

Question 4. How many types of addressing mode does 8085 microprocessor have? Explain any two of them.

SECTION — B

Note: Short Answer Questions each in 200 words approximately (Answer any 6 out of 9; Each question carries 06 marks). 6x6=36

Question 5. (a) Explain three step execution cycle with details.

(a) What is universal gates?

(b) Differentiate between microprocessor, | microcontroller and Microcomputer.

(c) What is assembly language?

(d) Explain the memory hierarchy diagram in detail. |

(e) Explain all three buses: address bus, data | bus control bus. |

(f) Describe 8085 microprocessor instructions and their types.) ! ass tii a | REFNO 076994

(g) Draw a block diagram which represents the microprocessor - based system and explain the basic functionality.

(h) Explain 8085 microprocessor architecture.

SECTION — C

Note: Very Short Answer Questions (Answer all 10; Each question carries 01 mark). 1x10=10

Question 6. (i) What is microprocessor?

- (a) Write the list of logical operations.
- (b) What are the three main units of a microprocessor made of?
- (c) Convert "101101110" into decimal number.
- (d) Draw a logic gate which represents a NAND gate.
- (e) Find out the complement of "001001110".
- (f) What is machine language?
- (g) Pin diagram is external diagram or internal diagram. (3 r+). REFNO 076994
- (h) Write the functionality of RD and WR in short.
- (i) What are maskable interrupts? (4 200)

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2022-23

Paper: BVCA-405 — Information Security

Time: Three Hours

Maximum Marks: 70

Instructions: Section-A: Attempt any 2 out of 4 questions and

| “ | REFNO 065196 | . Paper : BVCA-405 | Android Programming .-receiptofthis question-
paper . ww,) Note: Section-A: Attempt any 2 out of 4 questions and each question is of 12
marks., h ion i k

SECTION — B: Attempt any 6 out of 9 questions and

: each question is of 6 marks.

SECTION — C: All questions are compulsory and

are of 1 mark each.

SECTION — A ;

Note: Answer the questions in approximately 500 _ “ words each : 2x12=24 . 1. Discuss the
Android Architecture. Explain the i importance and functions of different layers. | F PTs

Question 2. a) Write a Flutter code Snippet to create a simple Application with a single
page containing a centered "Hello, Flutter!" Text.

- (a) Explain the concept of state management in Flutter. Discuss the different state manage-
ment approaches in Flutter and provide an examples of when to use each.,

Question 3. Write the flutter code to build an Application with following specifications :

- (a) The first page of the Application should have _ an App Bar with the text "Flutter App".
(b) The first page should have a body with text | "First Page" and an image found in the
assets | folder with the name “flutter.jpg”. |
(c) The Application should have a floating button | at the bottom of the screen. which
takes you to the second page.
(d) The second page should contain an App bar with the text "Second Page" and a back
button which takes you back to the homepage of the . application. . (e The second
page should have a body with text | "Second Page" and an image found in the , ‘ assets
folder with the name "second.jpg". |

Question 4. What is ART (Android Run Time] ? Discuss the features and working of ART.

SECTION — B]

Note: Answer the questions in about 200 words each : 6x6=36 ' REFNO 065196

Question 5. What is AndroidManifest.xml file ? Write purpose and name of its key components.

Question 6. What is an APK file ? How can you build an apk file in flutter ? . -%, Discuss the directory structure of an Android project. .

Question 8. What is the use of ListView.builder widget in Flutter ? Explain the different components of the ListView.builder widget using an example. (c) 9, What is the difference between Dalvik Virtual Machine (DVM) and Android Runtime (ART) ?

Question 10. What is Flutter ? Write about some disadvantages of using Flutter. , Loe sone

Question 11. What is the purpose of the 'pubspec.yaml' file in a Flutter project ? Explain the process of adding external dependencies (packages) to a Flutter | project using the 'pubspec.yaml' file. Provide an : example of adding a popular Flutter package and | its usage.

Question 12. Discuss the importance of using 'async' and 'await' in Flutter when working with asynchronous operations. Provide an example of using 'async' __ and 'await' in Flutter code. = - 13: Write a Flutter Dart code snippet to create a simple | Flutter app that displays a list of items with images | and text. Include the necessary widget tree, data | source and any relevant Flutter packages. | | 3 PT.O. " Section - C. Note: Answer in one word of one sentence each : : g , Re 10x1=10

Question 14. Which programming language is primarily used for Flutter app development ?

Question 15. Which programming language is primarily used 3 :... for Android app development ?- :

Question 16. Which Flutter command is used to create a new : Flutter project ?~ |

Question 17. Which Flutter widget is commonly used for | ' creating scrollable lists of items P- | _ 18. Which file format is commonly used for storing persistent data in Android ? 19: What is the purpose of the 'setState' method in Flutter ? extasciitilde . : - 20. What is the primary advantage of using Flutter for cross-platform app development ?

Question 21. Which Flutter widget is used for navigation between different screens or pages in an app? .. -.

Question 22. What is the primary role of the 'main.dart' file in a Flutter project. .

Question 23. What was the name of the first Android Smartphone ? : 4 200

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2023-24

Paper: BVCA-405 — Information Security

Time: Three Hours

Maximum Marks: 70

Instructions: Answer any two questions in about 500

| REFNO 076995 Android Programming

SECTION — A: Attempt any 2 out of 4 questions and _____

each question in of 12 marks.

SECTION — B: Attempt any 6 out of 9 questions and _____

each question in of 06 marks.

SECTION — C: All questions are compulsory and are _____

of 1 mark each.

SECTION — A _____

Note: Answer any two questions in about 500 words each. 2x 1224

Question 1. Discuss the Android Architecture. Explain the importance and functions of different layers. PF. REFNO 076995**Question 2.** Discuss the directory structure of an Android project.**Question 3.** What are the features of Android OS? Discuss the advantages and disadvantages of Android | in detail. Also write about its versions and | | major breakthroughs. 4 |**Question 4.** What is ART (Android Run Time)? Discuss the | features and working of ART. |**SECTION — B** | _____

Note: Answer any six questions in about 200 words each. 6x6=36

Question 5. (a) What is Android Manifest.XML file? Write purpose and name of its key components.

(a) What is an APK file? Discuss in detail. How | can you build an APK file in flutter?

(b) What is the difference between List view and List View. builder ? Explain with examples.

(c) What is the difference between Dalvik | Virtual Machine (DVM) and Android Runtime (ART)? | . (e) (i) What are the different types of Widget in Flutter?

- (i) What is the Purpose of the Scaffold widget in Flutter? - 2 REFNO 076995 (f Name the three common image file formats. Explain the characteristics of each one of them and write the scenarios in which they are used.
- (d) What is the difference between a programming language and Framework? How is flutter different from Kotlin (for android) and Swift (for iOS)?
- (e) Discuss the architecture of Flutter Framework. Explain the inner working of components of Flutter architecture.
- (f) Briefly explain the use of the following :
- (g) Material App ()
- (h) set State ()

SECTION — C

Question 6. Answer in one word or one sentence each : 10x1=10

- (a) Which file format is commonly used for storing persistent data in Android?
 - (i) What is the primary advantage of using Flutter for cross- platform app development?
 - (ii) Which widget is used to render the elements side by side in flutter?
 - (iii) What language is used for developing native Android Apps? REFNO 076995
 - (iv) Which database is most commonly used in Mobile applications? | (vi) Which widget is used to give margin | between the edge of screen and the | element or between two elements? | (vii) How can you get the screen size of the device in Flutter? (Write the name of the widget used). | (viii) What is the API level of the latest Android | version?
 - (v) Which programming language is used in the Flutter framework?
 - (vi) Which file contains the application permissions? KeE 4 200

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2023-24

Paper: CS-104 — Computer Org Arch

Time: Three hours

Maximum Marks: 70

Instructions: Answer five questions in all including Question No. 1, which is compulsory. Answers should be brief

PRGA N AIR Op ocececescroksso vazscstosisudbvsnineice CS-104: Computer Organization and Architecture Note: Answer five questions in all including Question No. 1, which is compulsory. Answers should be brief and to the point and may be supplemented with neat Sketches, wherever required. Terms and abbreviations have their standard meaning. Figures on the right-hand side margin indicate Max. Mark for each question,

Question 1. (a). A cache memory is with 8 cache lines (0-7). The memory block requests are in 7 the order- 4, 3, 25, 8, 19, 6, 25, 8, 16, 35, 45, 22, 8, 3, 16, 25, 7 (1) Which cache line will have memory block 7 if fully associative mapping with LRU policy is used. Also, calculate the hit ratio and miss ratio. (2) Which cache line will have memory block 7 if direct mapping is used. Also, calculate the hit ratio and miss ratio.

(a) What are the basic differences among a branch instruction, a call sub-routine instruction and a program interrupt?

Question 2. (a) Design a 4-bit Binary Adder-Subtractor and a Binary Incrementor. Explain their working.

(a) Why register —transfer— language is preferred for describing the internal organization of digital computers?

(b) What is an instruction cycle, and write the phases of the Instruction cycle?

Question 3. (a) Design a bus system using three-state buffers. The number of registers is 6, and the size of each register is 4 bits.

(a) What is the difference between isolated I/O and memory-mapped I/O?

(b) Write the difference between Static RAM and Dynamic RAM.

Question 4. (a) A computer uses a memory unit with 1 Giga words of 16 bits each. The computer supports three addressing modes (register mode, register indirect mode, direct mode) and 35 unique micro-operations. The instruction has four parts: an addressing mode part, an operation code, a register code part to specify one of 180 registers and an address part.

(a) Find the minimum number of words required for storing one instruction.

(b) Draw the instruction word format and indicate the number of bits in each part.

(c) Explain any four addressing modes with suitable examples.

(d) Explain any four applications of the stack in computer organization. 4

Question 5. (a) What is virtual memory? Explain Virtual address Mapping using Pages with 7

Question 2. 0. necessary examples. Sige

(a) What is the locality of reference in cache memory? Explain its major types and 4 how they are achieved?

(b) Explain word addressable and byte addressable memory. 3

Question 6. (a) We have to provide a memory capacity of 4096 bytes using 128 x 8 Ram chips. 5

(a) How many chips are required? (ii) How many lines of the address bus must be used to access 4096 bytes of memory? (iii) How many of lines in (ii) will be common to all chips? (iv) How many lines must be decoded for chip select? Specify the size of decoders.

(b) What is an interrupt? Explain its major types. 5

(c) List out the importance of the I/O interface. 4

Question 7. (a) Explain the working of DMA (Direct Memory Access) with the block diagram. 8

(a) What is the disadvantage of the strobe method? Explain how the handshake 6 method solves the problem? 3

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2022–23

Paper: CS-104 — Computer Organization & Architecture

Time: Three Hours

Maximum Marks: 70

*Instructions: Attempt any **five** questions including Question 1 which is compulsory.***Question 1.**

[5 + 5 = 10]

- (a) State True or False and justify briefly:
- (i) All high-level languages are compiled.
 - (ii) All low-level languages are compiled.
 - (iii) Storage is not a part of data processing.
 - (iv) Concept of jitter is related to audio-video data.
 - (v) Port address is used for source to destination delivery.
- (b) Briefly define the following:
- (i) Malware
 - (ii) Browser Hijacker
 - (iii) Macro Virus
 - (iv) FTP

Question 2.

[10 + 5 = 15]

- (a) What is data processing? What are its different types? Explain each of them briefly.
- (b) Explain the concept of stored program computers. Why is it considered better than hard-wired computers?

Question 3.

[10 + 5 = 15]

- (a) What do you understand by a web crawler? Explain the working of a web crawler. What type of pages cannot be accessed by a web crawler?
- (b) What is personalization of search results? State some of the parameters used for personalization of search results.

Question 4.

[10 + 5 = 15]

- (a) Explain the different types of network topologies with examples. Also give their advantages and disadvantages.
- (b) Explain the different components of a communication system.

Question 5.

[10 + 5 = 15]

- (a) What is the role of Internet Service Providers (ISPs) in the Internet? Explain their different types.
- (b) Differentiate between protocol and standards. What are the different types of standards used in communication? Name any one standard organization.

Question 6.

[10 + 5 = 15]

- (a) Differentiate between a client and a server. Explain the roles of an e-mail client and e-mail server. How does message transfer take place in an e-mail communication system? Explain.
- (b) Explain the applications of ICT in Education.

Question 7.

[10 + 5 = 15]

- (a) Explain the memory hierarchy of a computer system.
- (b) Differentiate between a CPU and a GPU. List some tasks where a GPU performs better than a CPU and explain why.

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2023-24

Paper: CS-104b — Computer Org Arch

Time: Three hours

Maximum Marks: 70

Instructions: Answer five questions in all including Question No. 1, which is compulsory. Answers should be brief

CS-104: Computer Organization and Architecture a Note: Answer five questions in all including Question No. 1, which is compulsory. Answers should be brief and to the point and may be supplemented with neat Sketches, wherever required. Terms and abbreviations have their standard meaning. Figures on the right-hand side margin indicate Max. Mark for each question.

Question 1. (a). A cache memory is with 8 cache lines (0-7). The memory block requests are in 7 the order- 4, 3, 25, 8, 19, 6; 25, 8, 16, 35, 45, 22, 8, 3, 16, 25, 7 (1) Which cache line will have memory block 7 if fully associative mapping with LRU policy is used. Also, calculate the hit ratio and miss ratio. (2) Which cache line will have memory block 7 if direct mapping is used. Also, calculate the hit ratio and miss ratio.

(a) What are the basic differences among a branch instruction, a call sub-routine 7 instruction and a program interrupt?

Question 2. (a) Design a 4-bit Binary Adder-Subtractor and a Binary Incrementor. Explain their 6 working.

(a) Why register transfer language is preferred for describing the internal 4 organization of digital computers?

(b) What is an instruction cycle, and write the phases of the Instruction cycle? 4

Question 3. (a) Design a bus system using three-state buffers. The number of registers is 6, and 6 the size of each register is 4 bits.

(a) What is the difference between isolated I/O and memory-mapped I/O? 4

(b) Write the difference between Static RAM and Dynamic RAM. : 4

Question 4. (a) A computer uses a memory unit with 1Giga words of 16 bits each. The computer 6 supports three addressing modes (register mode, register indirect mode, direct mode) and 35 unique micro-operations. The instruction has four parts: an addressing mode part, an operation code, a register code part to specify one of * 180 registers and an address part. :

(a) Find the minimum number of words required for storing one instruction.

(b) Draw the instruction word format and indicate the number of bits in each part.

(c) Explain any four addressing modes with suitable examples. 4

(d) Explain any four applications of the stack in computer organization. 4

Question 5. (a) What is virtual memory? Explain Virtual address Mapping using Pages with 7 necessary examples. os

(a) What is the locality of reference in cache memory? Explain its major types and 4 how they are achieved?

(b) Explain word addressable and byte addressable memory. 3

Question 6. (a) We have to provide a memory capacity. of 4096 bytes using 128 x 8 Ram chips. 5

(a) How many chips are required? (ii) How many lines of the address bus must be | : used to access 4096 bytes of memory? (iii) How many of lines in (ii) will be | common to all chips? (iv) How many lines must be decoded for chip select? Specify the size of decoders.

(b) What is an interrupt? Explain its major types. 5

(c) List out the importance of the I/O interface. 4

Question 7. (a) Explain the working of DMA (Direct Memory Access) with the block diagram. 8

(a) What is the disadvantage of the strobe method? Explain how the handshake 6 method solves the problem? 2

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2022-23

Paper: CS-105 — Computer Organization Architecture

Time: Three hours

Maximum Marks: 70

Instructions: Answer five questions in all including Question No. 1, which is compulsory. Answers should be brief

CS-104/CS-105 : Computer Organization and Architecture | Note: Answer five questions in all including Question No. 1, which is compulsory. Answers should be brief and to the point and may be supplemented with neat Sketches, wherever required. Terms and abbreviations have their standard meaning. Figures on the right-hand side margin indicate Max. Mark for each question,

Question 1. (a) A 4-way set-associative cache memory unit with a capacity of 32 KB is built 7 using a block size of 16 words. The word length is 32 bits. The size of the physical address space is 8 GB. Find the number of bits required for the TAG field with explanation.

- (a) Explain the stack and its operations with suitable example. Explain its relevance 7 in computer organization.

Question 2. (a) Draw a block diagram of an 8-bit ALU with 4-bit status register and explain its 6 working.

- (a) Draw a diagram of a bus system for 4 registers of 4 bits each using multiplexer. 4 at (c) Design a 4-bit adder-subtractor and explain its working. 4

Question 3. (a) An instruction is stored in location 300 with its address field at location 301. The address field has the value 400. A processor register R1 contains the number 200. Evaluate the effective address if the addressing mode of the instruction is: [6]

- (a) Direct
(b) Immediate
(c) Relative
(d) Register indirect

- (a) What is the difference between isolated I/O and memory mapped I/O? [4]

- (b) Write short notes on Strobe control and Handshaking. [4]

Question 4. (a) How to evaluate the arithmetic statement $X = A + B + C * (D + E + F)$ 6

- (a) Using a general register computer with three address instruction.

- (i) | Using a general register computer with two address instruction. (iti) | Using a general register computer with one address instruction.

- (b) Why I/O interface is required? Give any three reasons. 4
- (c) What is memory hierarchy and why do we need it? Explain. 4

Question 5. (a) Explain the working of DMA (Direct Memory Access) with the block diagram. 8

- (a) Differentiate between Horizontal and Vertical Microinstruction format for 6 a microprogrammed control unit. i P.T- 0 da

Question 6. (a) What are the basic differences among a branch instruction, a call sub-routine 5 instruction and a program interrupt?

- (a) What is an interrupt? Explain its major types. 5 :
- (b) A computer has 32-bit instructions and 10-bit addresses. If there are 158 two- 4 . address instructions, how many one address instructions can be formulated? (c)

Question 7. (a) We have to provide a memory capacity of 2048 bytes using 256 x 8 Ram chips. 8

- (a) How many chips are required? (ii) How. many lines of the address bus must be used to access 2048 bytes of memory? (iii) How many of lines in (ii) will be common to all chips? (iv) How many lines must be decoded for chip select? Specify the size of decoders.
- (b) Explain shift micro operations with suitable block diagrams and examples. Also 6 mention how to detect overflow condition in the arithmetic shift micro operation.
- — — — — ee | oe [2]

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2023-24

Paper: CS-105 — Discrete Mathematics

Time: Three hours

Maximum Marks: 70

Instructions: Attempt any 5 questions including Question No. 1 which is compulsory.

ROLE NOs: ssesiscsssscncvsnesoascstseennette Note: Attempt any 5 questions including Question No. 1 which is compulsory. : 1. Answer in one word/sentence: 110

- (a) . Write the full form of IP,
- (b) Name the first internet.
- (c) Name one protocol used for file transfer.
- (d) Name the protocol used for transfer over World Wide Web. (. The malware that can spread automatically over a network is called:
- (e) The malware that locks victim's file and demands money for unlocking is called:
- (f) Give example of a device that can act both as an input and an output device:
- (g) Can a computer virus spread automatically over the network?
- (h) Processor speed is measured in:
- (i) Network speed is measured in:

Question 2. a. Differentiate between data and information with the help of an example. [5]

- (a) Explain in detail what is data processing along with all the steps that are followed for data processing. : [10]

Question 3. a. What are the different types of computer software? Explain with example. [5]

- (a) What are the different types of digital computers? Explain each of them in detail. Also give examples of each of them. [10]

Question 4. a. Differentiate between compiled and interpreted languages with the help of example. [5]

- (a) What is an Electronic Mail (E-Mail)? What are its advantages and disadvantages. Give example of an email client. Explain the basic functionalities of an E-mail Client. [10]

Question 5. a. Explain the use of ICT in Health Care Sector. [5] : b. Explain any 5 different types of computer viruses in detail. [10]

Question 6. a. What is Indexing in the context of a search engine? Explain. [5]

- (a) Explain the IP addresses used for addressing over the Internet. Also differentiate between the IPv4 and IPv6 addressing schemes. Why are two different schemes used? [10]

Question 7. a. What are the different criteria of judging the quality of service provided by a network? Explain each of them briefly. [5]

- (a) Explain the structure of World Wide Web in detail. [10]

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2023-24

Paper: CS-403 — Programming For Problem Solving

Time: Three hours

Maximum Marks: 70

Instructions: Question 1 is compulsory, attempt any four questions from the remaining six questions.

Question 100. |B (CS - 403 : Programming for Problem Solving | Note: Question 1 is compulsory, attempt any four questions from the remaining six questions.

Question 1. (a) Explain 5 major types of programming languages with examples. [5]

(a) What is the difference between the top-down and bottom-up approaches of programming? [5]

(b) What are the steps involved in solving a problem? Explain your answer with an example. [4]

Question 2. (a) Write an algorithm to find the greatest between 3 numbers. [4]

(a) Design an algorithm and flowchart to convert a decimal number to a binary number. [6]

(b) Compare and contrast compilers from interpreters. [4]

Question 3. (a) Describe the order of precedence with regards to operators in C language. [5] ? (b) Explain different data types used in C language? [5]

(a) Write the steps involved in converting a C source code to executable code. [4]

Question 4. (a) Write C Program to find the factorial of a number using recursion. [6]

(a) What will be the outcome of the following conditional statement if the value of variable s is 10? `s >= 10 && S < 25 && S != 12`. [4]

(b) Write a call-by-value function to swap two numbers. [4]

Question 5. (a) Explain with example pre-increment(`++i`) and post-increment (`i++`). [4]

(a) Difference between formal argument and actual argument? [3]

(b) What are variables and in what way is it different from constants? [4]

(c) Explain call-by-value and call-by-reference. [3]

Question 6. (a) Write the difference between insertion sort and selection sort. [4]

(a) Write the pseudo of the quick sort algorithm. [5]

(b) What will be the output of the following C code? [5] `2 | #include<stdio.h>`

```
2 int main()
3 { .
4 int a=1,b=2,c=3,d=4,e;
5 e=ctd=b*a; is
3 printf("%d, \td\n", e, d); ;
7 t
```

Question 7. (a) How Insertion Sort works? Write the pseudo of insertion sort. Explain step-by-step to sort an array A[3, 8, 5, 4, 1, 9, -2] in descending order. [7]

- (a) What is Searching Algorithm? Explain different types of search operations.
- (b) Explain step-by-step to search element '4' in array A[3, 8, 5, 4, 1, 9, -2] using binary search algorithm. 7]

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2014

Paper: CSB-401 — Numerical Computing

Time: Three hours

Maximum Marks: 70

Instructions: Answer five questions including Question No. 1 which is compulsory. Use of calculator is allowed. All undefined symbols have their usual meanings.

Question 1.

- (a) Find the maximum absolute error and maximum relative error in $3x^2y/z$ when $\delta x = \delta y = \delta z = 0.001$ at $x = y = z = 1$. [3]
- (b) Define Normalized floating point number. Represent 54.75×10^7 in normalized floating point mode. [3]
- (c) Prove or disprove the relations: [4]
- (i) $\Delta^n \equiv \delta^n E^{n/2}$
- (ii) $\Delta + \nabla \equiv \frac{\Delta}{\nabla} - \frac{\nabla}{\Delta}$
- (d) Find the missing value of the following data: [3]

x	1	2	3	4	5
y	7	–	13	21	31

- (e) Using Jacobi method, find all the eigenvalues and corresponding eigenvectors of the matrix: [3]

$$A = \begin{bmatrix} 3 & \sqrt{3} \\ \sqrt{3} & 3 \end{bmatrix}$$

- (f) Using forward differences, obtain the formula for the second derivative of $y(x)$ at $x = x_0$. [3]
- (g) Use Runge-Kutta second order method to approximate y when $x = 0.1$, given that $\frac{dy}{dx} = x + y$, $y(0) = 1$. [3]

Question 2.

- (a) Solve the equation $x \tan x + 1 = 0$ by Regula-Falsi method starting with $x_0 = 2.5$ and $x_1 = 3$ correct to three decimal places. [6]
- (b) Perform four iterations of Gauss-Seidel method to solve the equations: [6]

$$x + 2y + 10z = 61$$

$$10x + y + 2z = 44$$

$$2x + 10y + z = 51$$

by taking the initial approximation as $[0, 0, 0]$.

Question 3.

(a) Obtain the order of convergence of the secant method in finding the root of the equation $f(x) = 0$. [6]

(b) From the following table of values of x and y , obtain $\frac{d^2y}{dx^2}$ at $x = 2.2$: [6]

x	1.0	1.2	1.4	1.6	1.8	2.0	2.2
y	2.7183	3.3201	4.0552	4.9530	6.0496	7.3891	9.0250

Question 4.

(a) Using the iterative method, find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 10 & 1 \\ 2 & 0 & 1 \\ 3 & 3 & 2 \end{bmatrix}$$

by assuming the initial inverse of the matrix to be

$$\begin{bmatrix} 0.4 & 2.4 & -1.4 \\ 0.14 & 0.14 & -0.14 \\ -0.85 & -3.8 & 2.8 \end{bmatrix}$$

[6]

(b) By means of Newton's divided difference formula, find a polynomial and hence find the value of $f(8)$ from the following data: [6]

x	4	5	7	10	11	13
y	48	100	294	900	1210	2028

Question 5.

(a) Prove the identity: [6]

$$y_x - \Delta^2 y_x + \Delta^3 y_x - \Delta^5 y_x + \Delta^6 y_x - \Delta^8 y_x + \dots = y_x - \Delta^2 y_{x-1} + \Delta^4 y_{x-2} - \Delta^6 y_{x-3} + \Delta^8 y_{x-4} - \dots$$

(b) Solve the following differential equation using Runge-Kutta method of order four: [6]

$$\frac{dy}{dx} = xy + y^2, \quad y(0) = 1,$$

at $x = 0.1$ and 0.2 .

Question 6.

(a) Derive Newton-Cotes quadrature formula. Hence obtain Simpson's one-third rule of integration. [6]

(b) Apply Gauss-Jordan method to solve the system of linear equations: [6]

$$2x + 3y - z = 5$$

$$4x + 4y - 3z = 3$$

$$2x - 3y + 2z = 2$$

Question 7.

- (a) Using Milne's method, find $y(0.8)$ if $y(x)$ is the solution of $\frac{dy}{dx} = 1 - y^2$, given that $y(0) = 0, y(0.2) = 0.2027, y(0.4) = 0.4228, y(0.6) = 0.6844$. [6]
- (b) Derive two-point Gauss-Legendre formula for integration. [6]

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2022–23

Paper: GEN-401 — Real Analysis

Time: Three Hours

Maximum Marks: 70

Instructions: This question paper carries three sections: Section A, B, and C. All questions are compulsory.

Section A — Long Answer Questions

[12 × 2 = 24]

Attempt any two questions. Each question carries 12 marks.

Question 1. Calculate the divergence and curl of the vector field $\mathbf{F} = x^2 \hat{i} + xy \hat{j} + yz \hat{k}$. Interpret the physical meaning of these operators.

Question 2. Prove that $f(x) = 2x + 3$ is continuous at $x = 1$.

Question 3. Find the first and second order partial derivatives of the function $f(x, y) = 3x^2 + 4xy - 5y^2$.

Question 4. Calculate the gradient of $f(x, y, z) = x^2 + xy + z^2$ and evaluate it at the point $(1, 1, 1)$.

Section B — Short Answer Questions

[6 × 6 = 36]

Answer any six out of the following nine questions. Each question carries 6 marks.

Question 5. Compute the curl of the vector field $\mathbf{G} = \sin(x) \hat{i} + \cos(y) \hat{j}$.

Question 6. Prove that $f(x) = x^2$ is continuous but not uniformly continuous on $\mathbb{R}$.

Question 7. Calculate the directional derivative of $f(x, y) = x^2 + y^2$ at the point $(1, 1)$ in the direction of $\hat{i} + \hat{j}$.

Question 8. Prove that the function $f(x, y) = x^2$ is differentiable at $x = 0$.

Question 9. Use the limit definition of a derivative to find $f'(x)$ for $f(x) = x^2$.

Question 10. Compute $\frac{\partial^2}{\partial x \partial y}(3x^2 + 4xy - 5y^2)$.

Question 11. Calculate the unit vector perpendicular to both $\mathbf{A} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\mathbf{B} = \hat{i} + \hat{j} - \hat{k}$.

Question 12. Compute the limit $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$.

Question 13. Show that the function $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.

Section C — Very Short Answer Questions

[1 × 10 = 10]

Answer all ten questions. Each question carries 1 mark.

Question 14. What is the divergence of $\mathbf{F} = x\hat{i} + y\hat{j} + z\hat{k}$?

Question 15. True or False: Every continuous function is differentiable.

Question 16. What is the range of the function $f(x) = x^2$?

Question 17. True or False: $f(x) = \sin(x)$ is uniformly continuous on $\mathbb{R}$.

Question 18. What is the value of $\lim_{x \rightarrow 0} \frac{\sin x}{x}$?

Question 19. What is the cross product of $\mathbf{A} = \hat{i} + \hat{j}$ and $\mathbf{B} = \hat{j} - \hat{k}$?

Question 20. True or False: If f and g are continuous functions, then f/g is also continuous wherever $g \neq 0$.

Question 21. What is the curl of $\mathbf{F} = x^2\hat{i} + xy\hat{j} - z\hat{k}$?

Question 22. True or False: If a function is differentiable, it is continuous.

Question 23. True or False: The dot product of two vectors is a scalar.

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2023-24

Paper: GEN-401 — Real Analysis

Time: Three Hours

Maximum Marks: 70

Instructions: Answer the questions in each section as instructed. All questions carry marks as indicated.

(Math | : Real Analysis)

SECTION — A 10x1=10

(Very Short Answer Type Questions) Note: Answer all questions.

Question 1. Define Nabla operator.**Question 2.** State Sandwich Theorem.**Question 3.** What is null vector ?**Question 4.** What is dot product in $\mathbb{R}^n$?**Question 5.** Find curl of $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.**Question 6.** Find the divergence of $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.**Question 7.** Find the gradient of $9 = xyz$:**Question 8.** Define the irrotational vector.**Question 9.** Define scalar triple product of orthogonal system.**Question 10.** Define uniform continuity of a real valued function.**SECTION — B 6x6 =36**

(Short Answer Type Questions) Note: Answer any six questions.

Question 1. State Cauchy criterion for series of function.**Question 2.** . State and prove Weirstrass M-test .**Question 3.** Define piecewise continuous function and give an example.**Question 4.** Show that: $V(\mathbb{R}) = 7$ **Question 5.** Define curl of a vector field. And find out curl of $\mathbf{V}$ where $\mathbf{V} = (xyz, xyz, xyz)$.
Same a

Question 6. Define uniform convergence of series of function and discuss the convergence of series $\sum x^n$ in $[0, 1]$.

Question 7. Define the continuity of real valued function. Show that $f(x) = x$ is continuous by ϵ - δ definition.

Question 8. Show that If the function f is continuous on $[a, b]$ and f is differentiable in (a, b) and $f'(c) = 0$ for all $c \in (a, b)$ then f is constant.

Question 9. Determine which of the following functions are even and which are odd ?

(a) $f(x) = \sin x$

(b) $f(x) = x^\circ \cos x$

(c) $f(x) = 1$

(d) $f(x) = x \tan x$ | SECTION -C 12x2=24 | (Long Answer Type Questions) Note: Answer any two questions. | 1. (a) Define gradient of scalar function @ defined on $\mathbb{R}^n$ and also define directional derivative of the | function 9.

(e) Find the directional derivative of $g(x, y, z) = x^4 y + z^2$ in the direction of $A = i + j + k$.

Question 2. (a) Define the Fourier series.

(a) Define Square wave function and find out the Fourier series of square wave function.

Question 3. (a) State and prove mean value theorem.

(a) State Rolle's theorem.

Question 4. (a) Find the unit normal vector to the surface equation of ellipsoid $(x/a)^2 + (y/b)^2 + (z/c)^2 = 1$ at the point $(a/3, b/3, c/3)$.

(a) Define a periodic function. Check whether $f(x) = \sin x$ is periodic or not ?

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2022-23

Paper: GEN-402 — Data Structures Algorithms

Time: Three hours

Maximum Marks: 70

Instructions: Attempt all questions. Symbols have their usual meanings.

GEN-402 : Data Structure and Algorithms Fe einen a Answer All Sections. ‘ Candidates are required to write their answers in their own words as far as practicable.

SECTION — A

: Long Answer type questions to be answered in about 500 words each: 12x2=24 Marks :
Attempt any two: .

Question 1. Define Linked Lists. What are the basic operations performed on Linked List? Write methods for creation of Linked list and insertion of any three elements at the beginning of the list. . : .

Question 2. What is a Binary tree 2j Explain Preorder, Inorder and Postorder traversal of binary tree , with suitable example. =

Question 3. What is a graph data structure? Explain Breadth First Search Algorithm with suitable - example. Differentiate between Breadth First Search and Depth First Search.

Question 4. Explain Stack Data Structure. Write a program to implement stack using array (Show creation, insertion, deletion and display of elements). . s — . * . Section-B mo eee
<7 Attempt any six (Answer in about 200 words each): : 6x6=36

Question 1. Explain Binary Search with. suitable example. What is the prerequisite condition to implement binary search? . .

Question 2. Explain Prim’s algorithm to find minimum spanning tree. Find the cost of minimum spanning tree of the given graph:)="—@)—).

Question 3. Explain Bubble Sort with suitable example.

Question 4. What is recursion? Write a program to find factorial of a given number using recursion.

Question 5. Explain Insertion Sort using suitable example. * 6, Give the algorithm to evaluate postfix expression using stack. Solve the postfix expression 524 *+7- using the algorithm and give the result. - . .

Question 7. What is an AVL tree? Explain LL and LR rotations. . ”

Question 8. Define a queue data structure. Explain the operations enqueue and dequeue in a queue. 6

Question 9. Write a java program to find the minimum and maximum element in an array of about 20 elements. (The elements must be taken input by the user).

SECTION — C

Attempt all the questions: 10 x 1=10

Question 1. Binary Search Tree (BST) has the property that the value of each node in the left subtree is , the value of the root node, and the value of each node in the right subtree is the value of the root node.

- (a) Greater, Greater 'ij F
- (b) Greater, Less
- (c) Less, Less
- (d) Less, Greater .

Question 2. In a stack, which operation is used to remove an element from the stack? a, Pop' :

- (a) Push , :
- (b) Peek . .
- (c) Empty .

Question 3. Which of the following data structures is not a linear data structure?

- (a) Stack
- (b) Queue '
- (c) linked List. .
- (d) Binary Tree (c)

Question 4. Which of the following data structures finds its use-in-recursion? +. CE cr

- (a) Stack J" b. Arrays ,
- (b) LinkedList z!
- (c) Queues Be

Question 5. Which of the following is a Divide and Conquer algorithm?

- (a) BubbleSort
- (b) Selection Sort :
- (c) Heap Sort . , 1°
- (d) Merge Sort

Question 6. Which data structure is needed to convert infix notation to postfix notation?

- (a) Tree .
- (b) Branch
- (c) Stack
- (d) Queue : :

Question 7. Evaluate the postfix expression $ab + cd/-$ where $a=5$, $b=4$, $c=9$, $d=3$.

- (a) 23
- (b) 15
- (c) 6 .
- (d) 10 . :

Question 8. The optimal data structure used to solve TowerofHanoi is__ . . (2) |

- (a) Tree -
- (b) Heap
- (c) Priority queue” ∴.
- (d) Stack

Question 9. What is a linear data structure with FIFO behavior? (Answer in one word) :

Question 10. What is the time complexity of Merge Sort? (Answer in one word). x x y 1
anemone see ee ee ee iS - eee rs aye Pe Ne eT F 1 1

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper: BCS-501 — Net Centric Computing

Time: Three hours

Maximum Marks: 70

Instructions: ANSWER ANY FIVE QUESTIONS INCLUDING QUESTION NO. 01, WHICH !5

Question 1.

- (a) Mention the difference between two types of transmission technology: broadcast links and point-to-point links. [3]
- (b) What are the basic services provided by third generation (3G) mobile phones? [3]
- (c) Mention the advantage of DHCP protocol over BOOTP protocol. [3]
- (d) What are the four primary quality of service (QoS) parameters? [3]
- (e) What is domain name space? [2]

Question 2.

- (a) Explain the functionalities of various layers in brief of the TCP/IP reference model. [7]
- (b) Explain three main multiplexing techniques: FDM, TDM, and WDM. Also, distinguish between synchronous and statistical TDM. [7]

Question 3.

- (a) Five-bit messages are transmitted using a Hamming code. How many check bits are needed to ensure that the receiver can detect and correct single bit errors? Show the bit pattern transmitted for the message 11010. Assume that even parity is used in the Hamming code. [7]
- (b) Explain the reason for moving from Stop-and-Wait ARQ protocol to the Go-Back-N ARQ protocol. [4]
- (c) What is Point-to-Point Protocol (PPP)? [3]

Question 4.

- (a) Datagram subnets route each packet as a separate unit independent of all others. Virtual Circuit subnets do not have to do this, since each data packet follows a predetermined route. Does this observation mean that virtual circuit subnets do not need the capability to route isolated packets from an arbitrary source to an arbitrary destination? Explain your answer. [7]
- (b) Draw the IPv4 datagram header format and explain the fields in brief. [4]

- (c) Explain any static routing algorithm. [3]

Question 5.

- (a) What are the Byte numbers and sequence numbers in TCP protocol? Explain the three-way handshaking method of connection establishment of a TCP connection. [7]
- (b) What is the difference between open-loop congestion control and closed-loop congestion control? Explain any two methods of congestion control in datagram subnets. [7]

Question 6.

- (a) Explain the role of Message Transfer Agent (MTA) in Email architecture. [5]
- (b) What are cookies? [4]
- (c) What is the main objective of 'Simple Network Management Protocol' (SNMP)? Explain how it works. [5]

Question 7.

- (a) What is the difference between symmetric and asymmetric key cryptography? Explain RSA algorithm for asymmetric key cryptography. [7]
- (b) What is a digital signature? How does it help in network security? [7]

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper: BCS-502 — Operating System Concepts

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all, including Question No.1, which is compulsory. Figures or 8 Pets

Saal Av Paper: BCS-502 Operating System Concepts: Time: Three Hours te bya Note: Answer five questions in all, including Question No.1, which is compulsory. Figures or 8 Pets

Question 1. Answer any six of the followings: a

- (a) Define the term Operating System and distinguish between internal and external views of an operating system.
- (b) What are the main design goals of an operating system? Explain them in brief.
- (c) What is starvation problem? Mention which scheduling algorithms suffer from this problem.
- (d) Distinguish between a process and a thread. State the major advantages of multithreading.
- (e) What is System Resource-Allocation Graph? Describe.
- (f) When Dynamic Loading and Dynamic Linking are required?
- (g) Briefly discuss about any three free-space management schemes for disk.

Question 2. (a) Briefly explain the different process states and transition among the states with a transition diagram.

- (a) Explain the three-level scheduling, which may coexist in a complex operating system.
- (b) Consider the following five processes, with the length of the CPU burst given in milliseconds:
P1: 10, P2: 5, P3: 8, P4: 4, P5: 6
Burst Time | P1 P2 P3 P4 P5 |
|-----|
| 10 5 8 4 6 |
|-----|
Illustrate the First Come First Serve (FCFS), Non-Preemptive Shortest Job First (SJF) and Round Robin (RR) (Time Quantum=10 milliseconds) scheduling algorithms using Gantt Chart. Which algorithm will give the minimum average time?

Question 3. (a) What do you mean by memory fragmentation? Differentiate between internal and external memory fragmentation.

- (a) Define the notion of Logical and Physical Addresses. How is logical address translated to a physical address in Paging? Explain with a neat diagram.

Question 4. (a) What is Demand Paging? Explain the steps involved in Page fault handling.

- (a) What is Belady's anomaly? Illustrate this anomalous behavior in FCFS with a suitable example.
- (b) Explain the Working Set Model. Explain how it is used for enhancing effective memory bandwidth in practice.

Question 5. (a) Describe the physical and logical views of a disk storage system. (a)

- (a) Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order is 86, 1470, 913, 1774, 948, 1509, 1622, 1756, 130. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms
- (b) FCFS
 - (i) SSTF
 - (ii) SCAN

Question 6. (a) Describe the deadlock detection algorithm for a system with multiple instances of 6 Resource Types.

- (a) Consider the following snapshot of a system: (1+4+2 Allocation Max Available ABCD
ABCD ABCD PO 0012 001322 1520 P1 1000 1750 PA LS 8 Be GS P3 0632 0652 P4.0014
0656 Answer the following questions using the banker's algorithm:
- (b) What is the content of the matrix Need?
- (c) Is the system in a safe state?
- (d) If a request from process P1 arrives for (0, 4, 2, 0), can the request be granted immediately?
7 Write notes on the followings: 3+4+3+3
- (e) Dining Philosopher Problem.
- (f) Critical Section and Semaphores.
- (g) Indexed File Allocation Method.
- (h) Directory Structure.

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper: BCS-503 — Discrete Mathematical Structures

Time: Three Hours

Full Marks: 70

*Instructions: Answer any five questions. Terms/Symbols used have their standard meanings.***Question 1.** Define the following terminologies and explain with appropriate examples: $[2 \times 7 = 14]$

- (a) Poset
- (b) Regular Graph
- (c) Clique
- (d) Fundamental Circuit
- (e) Bipartite Graph
- (f) Minimal Covering
- (g) De Morgan's Law

Question 2.

- (a) Explain partition of a set and equivalence classes with a suitable example. [3]

- (b) What is a Hasse Diagram? Draw the Hasse Diagram for the relation R on set $A = \{a, b, c, d, e\}$ where R is given as:

$$R = \{\langle a, a \rangle, \langle a, c \rangle, \langle a, d \rangle, \langle a, e \rangle, \langle b, b \rangle, \langle b, c \rangle, \langle b, d \rangle, \langle b, e \rangle, \langle c, c \rangle, \langle c, d \rangle, \langle c, e \rangle, \langle d, d \rangle, \langle e, e \rangle\}$$

[4]

- (c) Find the inverse of the function $f(x) = 4e^{x+2}$. [3]

- (d) A mobile manufacturing company assembled 30 mobiles. The features available are Bluetooth, Wi-Fi and 4G network. It is known that 15 of the mobiles have Bluetooth, 8 of them have Wi-Fi, 6 of them have 4G network, and 3 of them have all three features. Find at least how many mobiles do not have any features at all. [4]

Question 3.

- (a) For an algebraic system $(A, \vee, \wedge)$ defined by a lattice $(A, \leq)$, show that: [5]

(i) $a \leq a \vee b, a \wedge b \leq a$

(ii) $a \vee (a \wedge b) = a, a \wedge (a \vee b) = a$ where $a, b \in A$.

- (b) Define Boolean lattice and Boolean algebra. Prove that an element in a Boolean lattice has a unique complement. [5]

- (c) Given a function $f : \{0, 1\}^3 \rightarrow \{0, 1\}$ defined as:

$$\begin{array}{ll} f(0, 0, 0) = 1, & f(0, 0, 1) = 0, \\ f(0, 1, 0) = 1, & f(0, 1, 1) = 0, \\ f(1, 0, 0) = 0, & f(1, 0, 1) = 0, \\ f(1, 1, 0) = 0, & f(1, 1, 1) = 1. \end{array}$$

Express the function in disjunctive normal form and conjunctive normal form of Boolean expression. [4]

Question 4.

- (a) Determine whether each of the following compound propositions is a tautology or not: [3]

(i) $(\neg p \wedge \neg q) \rightarrow (p \rightarrow q)$

(ii) $(p \rightarrow q) \rightarrow (p \wedge \neg q)$

(iii) $[p \wedge (p \rightarrow \neg q)] \rightarrow q$

- (b) Show that $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$ and $\neg \forall x(P(x) \rightarrow Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x))$ without truth table. [4]

- (c) Solve the following recurrence relations: [7]

(i) $4a_r - 20a_{r-1} + 17a_{r-2} - 4a_{r-3} = 0$

(ii) $a_r + 5a_{r-1} + 6a_{r-2} = 3^r - 2r + 1$

Question 5.

- (a) Prove that the number of vertices of odd degree in a graph is always even. [3]

- (b) Draw the graphs of the following types: [5]

(i) Graph is both Hamiltonian and Eulerian.

(ii) Graph is Eulerian but not Hamiltonian.

(iii) Graph is Hamiltonian but not Eulerian.

- (c) What is meant by center of a graph? Prove that a tree may have one or two centers. [2]

- (d) What will be the minimum and maximum possible height of a binary tree with n vertices? [2]

- (e) What are the Kuratowski's two graphs? Also draw these two graphs. [2]

Question 6.

- (a) How to prove that a graph is non-planar? Show with an example. [4]

- (b) What is geometric dual of a graph? If r and μ denote the rank and nullity of a graph G , and if r^* and μ^* denote the same for the dual graph G^* , then prove that $r^* = \mu$ and $\mu^* = r$. [4]

- (c) Let $G = (V, E)$ be a connected weighted and undirected graph, where $V = \{a, b, c, d, e, f, g\}$ is the set of vertices and

$$E = \{\langle(a, b), 3\rangle, \langle(a, c), 2\rangle, \langle(b, d), 1\rangle, \langle(c, d), 5\rangle, \langle(c, f), 7\rangle, \langle(d, e), 3\rangle, \langle(d, f), 6\rangle, \langle(e, g), 8\rangle, \langle(f, g), 4\rangle\}$$

is the set of edges along with weights. Find the minimal spanning tree of G using Kruskal's and Prim's algorithms with demonstration of steps in both methods. [6]

Question 7.

- (a) What is chromatic partitioning? Explain with an example. [3]
- (b) What is chromatic polynomial? Find the chromatic polynomial of a complete graph of n vertices and a tree of n vertices. [5]
- (c) Explain the reduced incidence matrix A_f , fundamental circuit matrix B_f , and fundamental cutset matrix C_f of a connected graph. Use a common graph and common spanning tree to explain these matrices. What is the relationship among them? [6]

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BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper: BCS-504A — System Analysis Design

Time: Three hours

Maximum Marks: 70

Instructions: ANSWER ANY FIVE QUESTIONS AND THE FIGURES IN THE RIGHT HAND MARGIN INDICATE:

Question 1.

- (a) Write the IEEE definitions of the following terms in connection with software: [6]
- (i) Error
 - (ii) Fault
 - (iii) Failure
- (b) Design an ER diagram for a Library Management System. Specify all constraints that should hold on the database. Make sure that it has at least three entity types, two relationship types, a weak entity type, a superclass/subclass relationship, and an n -ary ($n > 2$) relationship type. [8]

Question 2.

- (a) Describe software crisis and the goals of software engineering. [7]
- (b) Write four important attributes of software products and differentiate them from hardware products. [7]

Question 3.

- (a) Compare prototyping and iterative software development process models by listing their strengths and weaknesses. [7]
- (b) Describe at least five important characteristics of an SRS. [7]

Question 4.

- (a) What is the role of problem analysis in software development? Explain the use of Data Flow Diagrams in problem analysis with an example. [7]
- (b) What are use cases? Where are these use cases used during the software development process? Explain through an example. [7]

Question 5.

- (a) What do you understand by a module? What are the modularization criteria used for function-oriented design to produce modular designs? [7]

- (b) Explain all four major steps of structured design methodology using one example. [7]

Question 6.

- (a) Write the purpose of the following diagrams of UML used in object-oriented design: [6]

- (i) Class diagram
- (ii) Interaction diagram
- (iii) Activity diagram

- (b) What do you understand by test cases and test criteria? Write the differences between black-box and white-box testing. [8]

Question 7. Write short notes on the following: [14]

- (a) Test oracle
- (b) Software maintenance
- (c) Data dictionary
- (d) Open-closed principle

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper: CS-105 — Discrete Mathematics

Time: Three hours

Maximum Marks: 70

Instructions: ATTEMPT ANY FIVE QUESTIONS. THE FIGURES IN THE RIGHT-HAND MARGIN INDICATE

Question 1.

- (a) Define a set. Give another representation of the following sets: [4]
- (i) $\{x \mid x \text{ is an integer and } 5 \leq x \leq 12\}$
 - (ii) $\{2, 4, 8, \dots\}$
 - (iii) All the countries of the world.
 - (iv) $\{x \mid x \text{ is an integer and } x^2 = 2\}$
- (b) Define number theoretic, total, and partial function. Give one example of each. [6]
- (c) Let a relation be defined over a finite set. Answer the following considering the matrix representation of the relation: [4]
- (i) How to check that the relation is a function.
 - (ii) How to check that the relation is a one-to-one function.
 - (iii) How to check that the relation is an onto function.
 - (iv) How to check that the relation is a one-to-one and onto function.

Question 2.

- (a) Define partial order relation, POSET, and Lattice. [4]
- (b) Let $X = \{1, 2, 3, 4, 6, 8, 12, 24\}$ and the relation $\leq$ be such that $x \leq y$ if x divides y . Answer the following questions for the POSET $(X, \leq)$: [10]
- (i) List all ordered pairs of relation $\leq$.
 - (ii) Give relation matrix of the relation $\leq$.
 - (iii) Find the matrix of inverse relation of $\leq$ from matrix of relation $\leq$.
 - (iv) Draw the Hasse diagram of $(X, \leq)$.
 - (v) Find the maximal and minimal element, if it exists.
 - (vi) Find the greatest and least element, if they exist.
 - (vii) Find all upper bounds of $\{2, 6\}$.

- (viii) Find the least upper bound of $\{2, 6\}$, if it exists.
- (ix) Find all lower bounds of $\{4, 6\}$.
- (x) Find the greatest lower bound of $\{4, 6\}$, if it exists.

Question 3.

- (a) Show that the conclusion “If I do not finish writing the program, then I will wake up feeling refreshed”, logically follows from the premises: “If you send E-mail message, then I will finish writing the program,” “If you do not send me an E-mail message, then I will go to sleep early,” and “If I will go to sleep early, then I will wake up feeling refreshed.”
[4]
- (b) Write the negation of the premises and conclusion given in 3(a). [4]
- (c) Describe the procedure proof by contradiction. Using the procedure proof by contradiction, show that the conclusion Q logically follows from the premises P and $P \rightarrow Q$.
[6]

Question 4.

- (a) Indicate the variables that are free and bound in the following statements. Also show the scope of the quantifiers: [5]
 - (i) $(\forall x)P(x, y)$
 - (ii) $(\forall x)(P(x) \rightarrow Q(x))$
 - (iii) $(\forall x)(P(x) \rightarrow (\exists y)R(x, y))$
 - (iv) $(\forall x)(P(x) \rightarrow R(x)) \vee (\forall x)(P(x) \rightarrow Q(x))$
 - (v) $(\exists x)P(x) \wedge Q(x)$
- (b) Discuss the limitation of propositional calculus, and how it is overcome in predicate calculus. [4]
- (c) Suppose the domain of the propositional function $P(x)$ consists of the finite set $\{x_1, x_2, x_3, \dots, x_n\}$. Express the following statements without using quantifiers: [5]

$$\exists x P(x), \quad \forall x P(x), \quad \neg \exists x P(x), \quad \exists x \neg P(x), \quad \forall x \neg P(x)$$

Question 5.

- (a) Define an Euler circuit in a graph. Give necessary and sufficient condition that a connected graph have an Euler circuit. [3]
- (b) Does the Hasse diagram given in question 2(b)(iv) have an Euler circuit? If yes, give at least one Euler circuit present in this graph. [2]
- (c) Define Hamiltonian circuit in a graph. If they exist, give necessary and sufficient conditions (or necessary or sufficient condition) that a connected graph have a Hamiltonian circuit. [3]
- (d) Prove that in a graph the number of nodes with odd degree must be even. [6]

Question 6.

- (a) Can five houses be connected to two utilities without connection crossings? Justify your answer with proper reasoning. [4]
- (b) For which values of n the following graphs are planar: [4]
- (i) K_n
 - (ii) W_n
 - (iii) $K_{m,n}$
 - (iv) Q_n
- (c) For which value of n are these graphs bipartite? And also find their chromatic number: [4]
- (i) K_n
 - (ii) C_n
 - (iii) $K_{m,n}$
 - (iv) Q_n
- (d) Find the chromatic number of the Hasse diagram of the POSET given in question 2(a). [2]

Question 7.

- (a) Define phrase-structured grammar and describe the Chomsky classification of phrase-structured grammar. [5]
- (b) Let $G = (V, T, P, S)$ be the phrase-structured grammar with $V = \{0, 1, A, B, S\}$, $T = \{0, 1\}$, and set of productions $P = \{S \rightarrow 0A, S \rightarrow 1A, A \rightarrow 0B, B \rightarrow 1A, B \rightarrow 1\}$. Answer the following: [9]
- (i) Show that 10101 belongs to the language generated by G .
 - (ii) Show that 10110 does not belong to the language generated by G .
 - (iii) What is the language generated by G ?

Question 8.

- (a) Give formal definition of a finite state machine which produces an output. Model a two-input logical AND gate using any finite state machine. [6]
- (b) Show that the total number of one-to-one functions from A to B is $n(n-1)(n-2)\dots(n-m+1)$, where $|A| = m$ and $|B| = n$. [3]
- (c) Model any problem of your choice using a recurrence relation and also give the solution of the modelled problem. [5]

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BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022–23

Paper: CS-106 — Operating System Concepts

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks.***Question 1.**

[4 + 3 + 3 + 4 = 14]

- (a) Explain the different states of a process with the help of a state diagram.
- (b) Name any two differences between logical and physical addresses.
- (c) What is meant by pre-paging? Is it better than demand paging? Justify.
- (d) What will be the output of the following code? Draw the execution tree explaining all process creations.

```
1 #include <stdio.h>
2 #include <unistd.h>
3 int main() {
4     if (fork() || fork())
5         fork();
6     printf("1 ");
7     return 0;
8 }
```

Question 2.

[2 + 3 + 3 + 3 + 3 = 14]

- (a) List out the benefits and challenges of thread handling.
- (b) What is thrashing? How can this problem be resolved?
- (c) Explain the principle of locality.
- (d) Explain the working set model.
- (e) Explain Compaction with a suitable example.

Question 3.

[7 + 7 = 14]

- (a) Why are Translation Look-aside Buffers (TLBs) important? Explain the details stored in a TLB table entry.
- (b) Explain the concept of demand paging in detail with neat diagrams.

Question 4.

[4 + 10 = 14]

- (a) What are the different principles which must be considered while selecting a CPU scheduling algorithm? Support your answer with an example.

- (b) Consider the following set of processes with CPU burst times (in milliseconds):

Process	Arrival Time	Burst Time	Priority
P1	0	3	3
P2	1	5	1
P3	3	2	3
P4	9	5	2
P5	12	5	1

Draw a Gantt chart and find the average waiting time and average turnaround time for:

- (i) FCFS
- (ii) SJF (Preemptive)
- (iii) SJF (Non-Preemptive)
- (iv) Priority Scheduling (Preemptive; smaller number = higher priority)
- (v) Round Robin (Time Quantum = 1 ms)

Question 5.

[6 + 8 = 14]

- (a) Compare the following memory management techniques on their strengths and weaknesses: (i) Fixed Partition, (ii) Dynamic Partition, (iii) Simple Paging.
- (b) Given the following page reference string:
 1, 2, 3, 2, 5, 6, 3, 4, 6, 3, 7, 3, 1, 6, 3, 4, 5, 3, 2, 4, 3, 4, 5, 1
 Number of page frames = 4. Show the page trace and calculate the number of page faults for the following page replacement policies: (i) LRU, (ii) Optimal, (iii) FIFO. Also explain Belady's anomaly.

Question 6.

[8 + 6 = 14]

- (a) What is busy waiting with respect to process synchronization? Explain how a semaphore reduces the severity of this problem. Also define and distinguish between:
- (i) General (counting) semaphores
 - (ii) Binary semaphores
 - (iii) Strong semaphores
 - (iv) Weak semaphores
- (b) The following program consists of 3 concurrent processes (P0, P1, P2) and 3 binary semaphores initialized as S0=1, S1=0, S2=0:

<pre> /* P0 */ while(true){ wait(S0); print("0"); signal(S1); signal(S2); } </pre>	<pre> /* P1 */ while(true){ wait(S1); print("1"); signal(S0); } </pre>	<pre> /* P2 */ while(true){ wait(S2); print("2"); signal(S0); } </pre>
--	--	--

How many times will process P0 print '0'? Give step-by-step explanation.

Question 7. Write the Banker's Algorithm. Consider the following snapshot of a system at time T_0 with 5 processes (P0–P4) and 4 resource types (A, B, C, D): $3 + 4 + 3 + 4 = 14$

Process	Allocation				Maximum				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

Answer the following:

- What is the content of the NEED matrix?
- Is the system in a safe state? If yes, justify with step-wise calculations.
- Which processes may cause deadlock if the system is not safe?
- If a request from process P1 arrives for (0, 4, 3, 1) — can the request be granted immediately? Justify.

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper: CS-107 — Database Management Systems

Time: Three hours

Maximum Marks: 70

Instructions: Answer any Five questions. Each question carrying 14 marks.

Question 1. A. Describe the three-schema architecture. Why do we need mappings between schema levels? [6]

- (a) Discuss the main characteristics of the database approach and how it differs from traditional file systems.

Question 2. A. Explain the differences among an entity, an entity type, and an entity set. [6]

- (a) Consider the following set of requirements for a university database that is used to keep track of students' transcripts. [8]

“The university keeps track of each student's name, student number, social security number, current address and phone, permanent address and phones, birthdate, sex, class (freshman, sophomore, . . . , graduate), major department, minor department (if any), and degree program (B.A., B.S., . . . , Ph.D.). Some user applications need to refer to the city, state, and zip code of the student's permanent address and to the student's last name. Both social security number and student number have unique values for each student.

- Each department is described by a name, department code, office number, office phone, and college. Both name and code have unique values for each department.
- Each course has a course name, description, course number, number of semester hours, level, and offering department. The value of course number is unique for each course.
- Each section has an instructor, semester, year, course, and section number. The section number distinguishes sections of the same course that are taught during the same semester/year; its values are 1, 2, 3, . . . , up to the number of sections taught during each semester.
- A grade report has a student, section, letter grade, and numeric grade (0, 1, 2, 3, or 4).

Design an ER schema for this application and draw an ER diagram for that schema. Specify key attributes of each entity type and structural constraints on each relationship type. Note any unspecified requirements and make appropriate assumptions to make the specification complete.”

Question 3. A. Discuss the entity integrity and referential integrity constraints. Why is each considered important? [6]

- (a) Consider the following six relations for an order processing database application in a company: [8]

CUSTOMER (Cust#, Cname, City)
 ORDER (Order#, Odate, Cust#, Ord_Amt)
 ORDER_ITEM (Order#, Item#, Qty)
 ITEM (Item#, Unit_price)
 SHIPMENT (Order#, Warehouse#, Ship_date)
 WAREHOUSE (Warehouse#, City)

Here, Ord_Amt refers to the total dollar amount of an order; Odate is the date the order was placed; Ship_date is the date an order is shipped from the warehouse. Assume that an order can be shipped from several warehouses. Specify the foreign keys for the above schema, stating any assumptions you make. Then specify the following queries in relational algebra:

- (i) List the Order# and Ship_date for all orders shipped from Warehouse number 'W2'.
- (ii) List the Warehouse information from which the Customer named 'Jose Lopez' was supplied his orders. Produce a listing: Order#, Warehouse#.
- (iii) Produce a listing: CUSTNAME, #OFORDERS, AVG_ORDER_AMT, where the middle column is the total number of orders by the customer and the last column is the average order amount for that customer.
- (iv) List the orders that were not shipped within 30 days of ordering.

Question 4. A. What is Join operation in SQL? Discuss each type of Join operation with an example. [6]

- (a) Consider the following employee database, where the primary keys are underlined. Give an expression in SQL for each of the following queries. [8]

Employee (employee-name, street, city)
 Works (employee-name, company-name, salary)
 Company (company-name, city)
 Manages (employee-name, manager-name)

- (i) Find the names of all employees who work for First Bank Corporation.
- (ii) Find the names and cities of residence of all employees who work for First Bank Corporation.
- (iii) Find the names, street addresses, and cities of residence of all employees who work for First Bank Corporation and earn more than \$10,000.
- (iv) Find all employees in the database who live in the same cities as the companies for which they work.

Question 5. A. Compute the closure of the set of functional dependencies $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$ for relation schema $R = (A, B, C, D, E)$. List the candidate keys for R . [6]

- (a) Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies $F = \{AB \rightarrow C, AD \rightarrow E, B \rightarrow F, F \rightarrow GH, D \rightarrow H\}$. What is the key for R ? Decompose R into 2NF, then 3NF relations. [8]

Question 6. A. Discuss the ACID properties of the transaction in detail. [6]

- (a) Discuss the different measures of transaction equivalence. What is the difference between conflict equivalence and view equivalence? [8]

*** END ***

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper: CS-108 — Data Structures Algorithms

Time: Three hours

Maximum Marks: 70

Instructions: Attempt any Five Questions. 4

CS - 108 : Data Structure and Algorithms Note: Attempt any Five Questions. 4

Question 1. (a) Explain primitive and non-primitive data structures with suitable examples.

(a) Distinguish array and linked list with suitable illustrations.

Question 2. (a) What is Binary Search Tree? How BST is different from B Tree? | Explain with suitable example.

(a) Discuss tree traversal with suitable example. ::. .

Question 3. (a) Define Quick sort. Sort the following using Quick sort algorithms. 12, 98, 01, 55, 03, 77, 00, 10 Se

(a) Discuss the order of complexity of Quick sort in all the cases.

Question 4. (a) Explain Greedy and Dynamic programming approach. Take your own problem a i | and solve the problem using both the approach. __ a**Question 5.** (a) Define asymptotic notations with suitable examples.(a) Find upper, lower and tight bound of running time of quadratic function | $f(n) = 99n^2 + 28n + 4$. a .**Question 6.** (a) Define back tracking, Discuss through an example. 7

(a) Define Graph as a data structure. Explain adjacency matrix and adjacency list with suitable example, . a

Question 7. (a) Define shortest path problem with illustrations. an

(a) Define Hashing and Collision. What are the methods to handle the collisions? Explain with an example. ns oe —__

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BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper: CS-109 — Data Communication

Time: Three hours

Maximum Marks: 70

Instructions: ANSWER ANY FIVE QUESTIONS, ALL QUESTIONS CARRY EQUAL MARKS.

CS - 109 : Data Communication

Question 1. Write a short note on any four out of six techniques/protocols: $3.5 \times 4 = 14$

Question 2. Amplitude Modulation(AM) ,

Question 5. Frequency-Division Multiplexing (FDM) : : ‘

- (a) Delta Modulation (DM)
- (b) Cyclic Redundancy Check (CRC) .
- (c) High Level Data Link Control (HDLC) .
- (d) Direct Sequence Spread Spectrum (DSSS)

Question 2. _ a) What is wireless (unguided) transmission media? What are the three major types of wireless - transmission media, describe each in detail. 7 Cee 6 “

- (a) What are circuit-switched and packet-switched networks? Discuss differences among them. 8

Question 3. .

- (a) What is the result of scrambling the sequence 11100000000000 using one of the following scrambling techniques? Assume that the last non-zero signal level has been positive. 6
- (b) B8ZS (ii) HDB3 (The number of nonzero pulses is odd after the last substitution)
- (c) What is the purpose of Pulse Code Modulation (PCM) technique? Describe the components of a PCM encoder with an example. c 8:

Question 4. . .

- (a) Describe the following mechanisms for modulating digital data into an analog signal with examples: Amplitude shift keying (ASK), Frequency shift keying (FSK), Phase shift keying (PSK), and Quadrature amplitude modulation (QAM). 6
- (b) Station 4 needs to send a message consisting of 9 packets to Station B using asliding window (window size 3) and go-back-n error control strategy. All packets are ready and immediately available for transmission. If every Sth packet that A transmits gets lost

(but no acks from B ever get lost), then what is the total number of packets that A will transmit for sending the message to B? Explain through diagram. 8

Question 5.

- (a) Discuss Frequency Hopping Spread Spectrum (FHSS) with a suitable diagram. 6
- (b) A sender needs to send the four data items given in hexadecimal: 3456, ABCC, 02BC, and EEEE. Answer the following questions: 8 : i. Calculate the checksum at the sender site. ii, Calculate the checksum at the receiver site if the second data item is changed to ABCE. ~ iil. 'Find the checksum at the receiver site if the second data item is changed to ABCE and the third data item is changed to 02BA.

Question 6.

- (a) What is data link control? Describe the key components of data link control. 6
- (b) Why we need flow and error control? Discuss the sfop-and-wait flow control technique with an example. 8 Seek END eee eRe |

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper: CS-109 — Data Communication

Time: Three hours

Maximum Marks: 70

Instructions: Attempt any five questions. Figures on the right-hand side margin indicate max

Question 1. (a) Define a System. What are the key elements of a system? Distinguish between Conceptual, Physical and Logical System Models. [7]

(a) Explain through a neat diagram the hierarchy of information levels and information systems in an organization. [7]

Question 2. (a) Explain the Software Development Life Cycle (SDLC) model along with its major advantages and disadvantages. [7]

(a) Explain the commonly used Fact Finding Tools for Preliminary Investigation. [7]

Question 3. (a) What is Feasibility Study? Explain the various types of feasibility analysis carried out [7] during system analysis.

(a) What is the purpose of designing SRS document? Explain the typical structuring [7] elements of SRS document.

Question 4. (a) Distinguish between Coupling and Cohesion as design principles. What is desirable [7] and why?

(a) Describe the following terms in reference to System testing: [7] Error, Fault, Failure, Reliability

Question 5. (a) Distinguish between verification & Validation. What are the various levels of [7] software system testing?

(a) What is the major difference between Blackbox and Whitebox testing? Explain the [7] Various criteria for testcase generation in the two approaches.

Question 6. (a) Describe the commonly used methods for System Conversion. [7]

(a) Differentiate between Corrective, Adaptive and Perfective maintenance of a system. [7]

Question 7. Write short notes on any two of the following: [14]

(a) Tangible and Intangible Costs and Benefits

(b) Structured Analysis

(c) Role of a System Analyst

(d) Software Cost Estimation

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BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper: CSB-501 — Object Oriented Programming Using C++

Time: Three Hours

Maximum Marks: 70

Instructions: Answer any FIVE questions including question no. / which is compulsory. The figure in

BS] SemZ/ 242 Paper No. CSB-501: Object Oriented programming Using C++ . the right hand side indicates marks. eee ee

Question 1. (a) State whether the following statements are TRUE or FALSE: 5X1=5

(a) Polymorphism is extensively used in implementing inheritance.

(i) in C++, class is a user defined data type.

(ii) The scope resolution operator can be used to uncover a hidden variable. iv The default visibility mode in C++ is private (v In C++ both virtual constructors and destructors are allowed. b Fill in the blanks:- 5X1=5

(b) The operator « is called the operator.

(i) A takes a reference to an object of the same class as itself as an argument. (iti) An is one that is not used to create objects. iv C++ uses a unique keyword called to represent an object that invokes a member function.

(ii) C++ supports a mechanism known as ___ to achieve runtime polymorphism. (c Find errors, if any, in the following C++ statements:- AXi=4

(c) `int *p = new;`

(i) `enum (green, yellow, red);`

(ii) `for (i= 1; int i <10; i=+} cout <<i<<"\n";`

(iv) `char name{3} = USA ;`

: 2. (a) What are the unique advantages of an object-oriented programming paradigm? 5

(d) Write output of following program:- 5

```
#include <iostream.h>
int a = 20;
main()
{
  int a= 40; .
  {
    int k= a;
    int a= 50;
    cout<< INNER block\ n ; .
    cout<<"k="<<k<<"\n";
    cout<<"a="<< a<<"\n";
    COULS<"_hae" <<rac< \ n :
```

```

}
cout<<    OUTER BLOCK\ n ;
cout<<"as"<<a<< \ n ;
cout<<"::gae"<<im<\n";
}

```

(e) What is a reference variable? What is the major use of this variable? 4 2 3, (a) What do you mean by overloading of a function? When do we use this concept? Explain with example. 6 ‘

(f) When do we declare a member of a Class static? 3 c) Write a program in C++ that shows how to use a common friend function to exchange the private values of two classes. 5

Question 4. (a)What is a constructor and destructors? Also write various types of constructors available in C++. 5

(a) Write any five rules for overloading operators. 5 (c) How many arguments are required in the definition of an overloaded unary operator. Explain it with an example. 4

Question 5. a Explain the effect of inheritance on the visibility of members by diagram. 5 (b) Write a program in C++ that shows multilevel inheritance. 5 (c Write any four rules for virtual functions in C++. 4

Question 6. a Both cin and getline() functions can be used for reading a string. Comment. 4

(a) What is file mode? Describe the various file mode options available in C++? 5 (c) Write a program in C++ that reads two files simultaneously. 5

Question 7. Write short notes on any two of the following:- [7X2=14] (2) OOAD b) Class Templates

(a) Exception handling

(b) Class Wizard —K—

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper: CSB-502 — Net Centric Computing

Time: Three hours

Maximum Marks: 70

Instructions: ANSWER FIVE QUESTIONS INCLUDING QUESTION NO. 1 WHICH IS COMPULSORY.

Question 1. a) What is meant by flow control? 2

- (a) What are the fields of UDP header? 3
- (b) Distinguish between bus topology and ring topology. 3
- (c) What is the count to infinity problem? 3
- (d) What are the disadvantages of DES algorithm? 3

Question 2. a) Explain the four basic Network topologies, and cite an advantage of each type. 5

- (a) What are the responsibilities of the network layer in the Internet model? 5
- (b) Describe the multiplexing techniques used to combine digital signals. 4

Question 3. a) Explain in detail, with example, the various error detecting methods. 7

- (a) Suppose that we model “A Simplex Protocol for a Noisy Channel” using the finite state machine model. How many states exist for the communication channel? How many states exist for the complete system? 7

Question 4. a) What is internetworking? Discuss the various global addressing schemes and the issues in forwarding the IP (IP packets). ,

- (a) Explain why routing is very important in Networking. Illustrate any one of the routing algorithms used in Network. 7

Question 5. a) Differentiate connectionless and connection oriented services at Transport Layer. Draw the header format of TCP and explain the purpose of each field. 7

- (a) What is the difference between open loop congestion control and closed loop congestion control? Name the policies that can prevent congestion. 7

Question 6. a) What is the difference between a primary server and a secondary server? What are the two main categories of DNS messages? 4

- (a) Write short notes on: 6
- (b) WWW ii) browser iii) URL

(c) | What is the mechanism of cookies? 4

Question 7. a) Describe Simple Network Management Protocol (SNMP) and explain the role of MIB (Management of Information Base). 7

(a) In asymmetric key cryptography, how do you think two persons can establish two pairs of keys between themselves? Explain with an example. 7 RRR

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper: CSB-503 — Discrete Mathematical Structure

Time: Three hours

Maximum Marks: 70

*Instructions: Attempt any FIVE questions, The figures in right margin indicate the marks.***Question 1.**

- (a) Define successor of a set. Give two sets A and B such that $A \in B$ and $A \subset B$. [4]
- (b) Define difference and symmetric difference of sets. Describe in general how these two set operations are achieved when sets are represented using bit strings. [5]
- (c) Define partition and covering of a set. Generate universal set E using $A_1 = \{3, 4, 5\}$, $A_2 = \{1, 2\}$, and then give the partition of generated universal set using sets A_1, A_2 . [5]

Question 2.

- (a) Consider the following relation R on set $A = \{1, 2, 3, 4\}$:

$$R = \{(a, b) \mid a \text{ divides } b\}$$

- (i) List all ordered pairs of relation R . [1]
- (ii) Represent relation R using digraph. [1]
- (iii) Give relation matrix of the relation R . [1]
- (iv) Find the matrix of R^{-1} from matrix of relation R . [1]
- (b) (i) Define partial order relation and poset. [2]
- (ii) Using only matrix operations show that the relation R defined in Question 2(a) is a partial order relation and hence (A, R) is a poset. [3]
- (iii) Draw the Hasse diagram of the above partial order relation. [1]
- (iv) Find the maximal and the minimal elements, if they exist, of the above poset (A, R) . [2]
- (v) Find the greatest and the least elements, if they exist, of the above poset (A, R) . [2]

Question 3.

- (a) Let A be a set having n elements. Show the following:
- (i) Total possible reflexive relations on A is $2^{n(n-1)}$. [2]
- (ii) Total possible symmetric relations on A is $2^{n(n+1)/2}$. [2]
- (iii) Total possible symmetric and reflexive relations on A is $2^{n(n-1)/2}$. [2]

- (iv) Total possible anti-symmetric relations on A is $2^n \cdot 3^{n(n-1)/2}$. [2]
- (b) Define number theoretic function, partial function, and primitive recursive function and give at least one example of each. [4]
- (c) Show that the total number of one-to-one functions from A to B is $n(n-1)(n-2) \cdots (n-m+1)$, where $|A| = m$ and $|B| = n$. [2]

Question 4.

- (a) Prove the following without constructing truth table:
 - (i) $\neg p \wedge (p \vee q) \rightarrow q$ [3]
 - (ii) $[p \wedge (p \rightarrow q)] \rightarrow q$ [3]
- (b) Represent the following facts using propositions. Also find what rules of inference are used in each of these arguments: [6]
 - (i) Alice is a mathematics major. Therefore, Alice is either a mathematics major or a computer science major.
 - (ii) Jerry is a mathematics major and a computer science major. Therefore, Jerry is a mathematics major.
 - (iii) If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will get sunburned. Therefore, if I go swimming, then I will get sunburned.
- (c) What is the limitation of statement calculus and how is it removed in predicate calculus? [2]

Question 5.

- (a)
 - (i) Prove that in a group $(G, \circ)$, if every element is its own inverse, i.e., $a^2 = e$ for all $a \in G$, then G is an abelian group. [5]
 - (ii) Let $(\mathbb{R}, +)$ denote the additive group of real numbers and $(\mathbb{R}^*, \cdot)$ denote the multiplicative group of positive reals. Define a function $f : \mathbb{R} \rightarrow \mathbb{R}^*$ as $f(x) = a^x$, $a > 0$. Show that f is a homomorphism. [5]
- (b) Let f be a homomorphism from $(\mathbb{Z}, +)$ to $(\mathbb{Q}^*, \cdot)$, $\mathbb{Q}^* = \mathbb{Q} - \{0\}$. Suppose $f(2) = 4$. Find out $f(6)$ and $f(-8)$. [4]

Question 6.

- (a) Model the round-robin tournament using a graph. [3]
- (b) What do the in-degree and out-degree of a vertex in the graph represent? [3]
- (c) Draw the graph of the following: [4]
 - (i) K_5
 - (ii) C_5
 - (iii) W_5

- (iv) Q_3
- (d) For which values of n are these graphs bipartite? Also find their chromatic number: [4]
 - (i) K_n
 - (ii) C_n
 - (iii) W_n
 - (iv) Q_n

Question 7.

- (a) Can five houses be connected to two utilities without connections crossing? [2]
- (b) A connected planar graph has 30 edges. If a planar representation of this graph divides the plane into 20 regions, how many vertices does this graph have? [4]
- (c) Find the complements of every element, if they exist, of (S_n, D) for $n = 75$, where S_n is the set of divisors of n , and D is the division relation. [5]
- (d) Determine whether the following statements are true, false, or not valid: [3]
 - (i) $2 + 2 = 4$ iff $1 + 1 = 2$.
 - (ii) If $1 + 1 = 3$, then unicorns exist.
 - (iii) $x + y = 2$.

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper: CSB-504 — Operating System Concepts

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all, including Question No.1, which is compulsory. Figures on the right-hand side

Paper: CSB-504 (Operating System Concepts) Note: Answer five questions in all, including Question No.1, which is compulsory. Figures on the right-hand side

Question 1. Answer all the parts

- (a) Briefly explain the main design goals of an operating System. [3]
- (b) What are the differences a process and a thread? Describe some benefits [4] using threads.
- (c) Explain the difference between internal and external fragmentation. Why [4] should they be avoided?
- (d) List the four conditions for the occurrence of a deadlock and describe the [5] resource-allocation graph.
- (e) Briefly discuss the terms: Dynamic Loading, Busy form of waiting, and [6] Monitor.

Question 2. (a) Briefly explain the different process states and transition among the states [2] with transition diagram.

- (a) Explain the terms Process Contro! Block and Context Switching. [4]
- (b) There are five processes A to E to run. Their arrival times are 0,1,3,9 and 12 (6 seconds, respectively. Their processing times are 3, 5, 2, 5 and 5 seconds, respectively. What is average turn-around times using First-Come-First- Served, Shortest-Job-First and Round-Robin (with 1 second Time Quantum) Scheduling?

Question 3. (a) Discuss Paging Memory Management Scheme with an illustrative example 8) and explain the hardware support required for Paging.

- (a) Consider a paging system with the page table stored in memory. [4]
- (b) tfa memory reference takes 200 nanoseconds, how long does a paged memory reference take?
 - (i) if we add TLBs and 75 percent of all page-table references are found in TLBs, what is the effective memory reference time? assume that finding a page-table entry in TLBs is 10 times faster than that of the memory)

Question 4. (a) What is Demand Paging? Explain the steps involved in Page fault handling. [4]

- (a) Briefly describe Belady's anomaly. | [2]
- (b) A small computer has 4 frames. A process makes the following list of page [6] references 1, 2, 3, 4, 1, 5, 2, 3, 1, 2. How many page faults occur using FIFO, Second Chance and Least-Recently Used page replacement algorithms?

Question 5. (a) Describe the physical and logical views of a disk storage system. [3]

- (a) Consider the following sequence of disk track requests: 27, 129, 110, 186, [9] 147, 41, 10, 64 and 120. Assume that initially the head is at track 30 and is moving in the direction of decreasing track number. Compute the number of PTO, tracks the head traverses using FIFO, SSF and Elevator algorithms.

Question 6. (a) Explain the contiguous and linked file allocation methods with major and the [6] FAT scheme for linked allocation.

- (a) Explain the concepts involved in maintaining the file system security 2)
- (b) Consider a swapping system in which memory consist of the following hole [4] sizes in memory order: 10K, 4K, 20K, 18K, 7K, 12K, and 15K. What hole is taken for successive segment requests of
- (c) 12K
- (i) 10K
- (ii) 9K Using First-fit, best-fit, worst-fit and next-fit strategies. 7 Write notes on any three of the followings 3x4=12
- (d) Disk free space management schemes .
- (e) Types of Operating System
- (f) Semaphore
- (g) Deadlock Recovery
- (h)

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013–14

Paper: CSB-601 — Database Management System

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions including Question 1 which is compulsory. Terms and symbols used have their standard meaning.*

Question 1. Fill in the blanks: [10]

- (a) The number of attributes in a relation is called the _____ of the relation.
- (b) Immunity of the external schemas to changes in the internal schema is referred to as _____.
- (c) When one column of a table refers to the values in another column of the same table, it introduces a _____ integrity constraint.
- (d) The virtual page table never changes during the execution of a transaction — this property is called _____.
- (e) Along with the **GROUP BY** clause, one can use the _____ clause to define a condition on the group of tuples.
- (f) _____ and _____ are two unary operations in relational algebra.
- (g) 4NF is based on the concept of _____ dependency.
- (h) A schedule S is not conflict serializable if the precedence graph contains _____.
- (i) _____ and _____ are the two desirable properties of relational schema decomposition.
- (j) _____ is an abstraction through which relationships are treated as higher-level entities.

Question 2. [3 + 4 + 3 + 5 = 15]

- (a) Explain the difference between primary key and superkey.
- (b) What are multi-valued attributes, and how can they be handled within the database design?
- (c) What is meant by a recursive relationship type? Give an example.
- (d) Define cardinality ratio. Draw an ER diagram to illustrate different cases of cardinality ratio.

Question 3. [6 + 6 + 3 = 15]

- (a) What is the lossless-join property of decomposition? Suppose that we decompose the schema $R = (A, B, C, D, E)$ into $R_1 = (A, B, C)$ and $R_2 = (A, D, E)$. Check whether this decomposition is lossy or lossless if the following set F of functional dependencies holds: $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$.
- (b) Define full, partial, and transitive functional dependencies. Explain the normal form based on transitive functional dependency with a suitable example.
- (c) Define multi-valued functional dependency.

Question 4.

[6 + 6 + 3 = 15]

- (a) Discuss the immediate update technique of recovery. How is it different from the deferred update technique? Why is it called the UNDO/REDO method?
- (b) What is the two-phase locking protocol? What are some variations of the two-phase locking protocol? Why is strict two-phase locking often preferred?
- (c) What is a recoverable schedule? Why is recoverability of schedules desirable?

Question 5.

[5 + 10 = 15]

- (a) With a suitable example, explain the difference between natural join, equi-join, and Cartesian product operations.
- (b) Consider the following relational database. Answer each of the queries in SQL:

Branch(bname, bcity, assets)
Customer(cname, street, ccity)
Depositor(cname, account#)
Account(bname, account#, balance)

- (i) Find account numbers of accounts with balances between Rs. 70,000 and Rs. 1,00,000.
- (ii) Find all customers whose street includes the substring "main".
- (iii) Find branches and their average balances where the average balance is more than Rs. 2,000.
- (iv) Delete the records of all accounts with balances below the average.
- (v) Increase all balances by 10% whose balance is greater than the average.

Question 6.

[6 + 3 + 6 + 5 + 4 + 6 = 30]

- (a) Describe the ANSI/SPARC three-level architecture. How are these three levels related to the concept of physical and logical data independence?
- (b) Describe the steps of query processing with a flow diagram.
- (c) Explain any three heuristic rules for query optimization through examples.
- (d) Define weak entity type. Explain how it is handled in relational database designing.
- (e) Explain the Select and Project operators in Relational Algebra through an example.

- (f) What are the database security issues? What is the difference between discretionary and mandatory access control?

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013–14

Paper: CSB-602 — Data and File Structures

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer **five** questions in all, including Question 1 which is compulsory. Figures on the right-hand side indicate marks.*

Question 1. Answer **all** parts:

[3 × 6 = 18]

- (a) Define the terms: Abstract Data Types (ADT) and Data Structure.
- (b) State the advantages of using postfix notation over infix notation.
- (c) Define Double-ended Queue (Deque).
- (d) What is a Binary Heap? Distinguish between Max-Heap and Min-Heap.
- (e) List the basic operations carried out on a linked list.
- (f) Briefly discuss the following terms: Tree, Degree of a node, and Height of a tree.
- (g) What do you mean by collision in hashing? Briefly explain any two collision resolution methods.

Question 2.

[2 + 5 + 6 = 13]

- (a) Define ADT Array.
- (b) What is a sparse matrix? Give a structure for its efficient representation.
- (c) Give an algorithm for addition of two sparse matrices stored in the representation you proposed above.

Question 3.

[2 + 6 + 5 = 13]

- (a) What are the two important criteria to be satisfied by any recursive function?
- (b) Explain the implementation of a stack using an array with algorithms for all operations on a stack.
- (c) Write an algorithm to convert an infix expression to a postfix expression, and explain it through an example.

Question 4.

[3 + 10 = 13]

- (a) What do you mean by order of growth in reference to time complexity of an algorithm?
- (b) Explain the quick sort algorithm with an example. Also mention any three strategies for choosing a pivot element.

Question 5.

[4 + 9 = 13]

- (a) Define full binary tree and complete binary tree with an example for each.
- (b) Explain binary tree traversals (in-order, pre-order, post-order) with recursive algorithms.

Question 6.

[7 + 6 = 13]

- (a) Explain Breadth First Search (BFS) traversal on a graph with an example.
- (b) Write an algorithm for adding two polynomials using a linked list.

Question 7. Write notes on any **two** of the following:

[6.5 × 2 = 13]

- (a) Hashing
- (b) Binary Search Algorithm (Iterative and Recursive)
- (c) Sequential and Index-Sequential file organization techniques
- (d) External sorting methods with an example

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013–14

Paper: CSB-603A — System Analysis and Design

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer all parts:

[3 + 3 + 4 + 4 = 14]

- (a) What is the difference between a program and software? Explain.
- (b) Write the purpose of CASE tools.
- (c) What is system testing? Explain.
- (d) What are the myths associated with software?

Question 2.

[7 + 7 = 14]

- (a) Compare the Waterfall model with the Spiral model of software development.
- (b) What are fact-finding techniques and where are they used?

Question 3.

[10 + 4 = 14]

- (a) An airline reservation is an association between a passenger, a flight, and a seat. Select a few pertinent attributes for each of these entity types and represent a reservation in an E-R diagram.
- (b) What is the difference between a decision tree and a decision table?

Question 4.

[6 + 8 = 14]

- (a) Differentiate between the terms: Data Flow Diagram (DFD), Flowchart, Data Dictionary, and E-R diagram.
- (b) Define module coupling and explain the different types of coupling.

Question 5.

[4 + 10 = 14]

- (a) What is the difference between testing and debugging? Explain.
- (b) Write in detail about all types of software testing methods.

Question 6.

[2 + 12 = 14]

- (a) What is the difference between verification and validation?

(b) Write short notes on the following:

- (i) Modular top-down analysis
- (ii) Maintenance process
- (iii) User interface design

Question 7.

[5 + 4 + 5 = 14]

- (a) What is Object-Oriented Analysis and modelling?
- (b) What are the characteristics of a Software Requirements Specification (SRS)?
- (c) Write about the significance of the coding phase in SDLC.

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2013–14

Paper: CSB-604C — Operations Research

(Major Elective Course – I)

Time: Three Hours

Maximum Marks: 70

*Instructions: Attempt **five** questions including Question 1 which is compulsory. Use of calculator is allowed. All undefined symbols have their usual meanings.*

Question 1. Answer any **four** of the following: [4 × 4 = 16]

- (a) Find the graphical solution of the following LPP:

$$\begin{aligned} &\text{Maximize } Z = 2x_1 + 2x_2 \\ &\text{Subject to: } x_1 + x_2 \leq 7 \\ &\quad 0.5x_1 + x_2 \leq 2 \\ &\quad x_1, x_2 \geq 0 \end{aligned}$$

Is it a well-defined LPP?

- (b) How do you calculate the mean and variance of the earliest occurrence time T_E of event i in a project arrow diagram, given a_i , b_i , and m_i as the optimistic, pessimistic, and most likely times respectively?
- (c) What are slack, surplus, and artificial variables? Explain their use for finding the initial basic feasible solution of a LPP.
- (d) What is a non-linear programming problem? Define local and global minima and maxima of a single-variable non-linear function $f(x)$.
- (e) Define the following with an example of each:
- (i) A directed and an undirected cycle
 - (ii) A connected network
 - (iii) An undirected path and an undirected network

Question 2. [6 + 8 = 14]

- (a) Explain the steps of the Simplex method for solving a LPP.
- (b) Solve the following LPP using the Simplex method:

$$\begin{aligned} &\text{Maximize } Z = 3x_1 + 5x_2 + x_3 \\ &\text{Subject to: } 2x_1 + 3x_2 \leq 3 \\ &\quad x_2 + x_3 \leq 8 \\ &\quad 3x_1 + 2x_2 + 4x_3 \leq 15 \\ &\quad x_1, x_2, x_3 \geq 0 \end{aligned}$$

Question 3. If D_{ij} is the duration of an activity (i, j) , write the conditions when activity (i, j) lies on a critical path. Define total float and free float of activity (i, j) . For the following project network, compute the total float, free float, and identify the critical path: $[6 + 8 = 14]$

Activity	Duration (days)	Activity	Duration (days)
1-2	3	3-5	4
1-3	2	4-6	5
2-4	4	5-6	7
2-5	6	5-7	3
3-4	5	6-7	2

What are the uses of total and free floats?

Question 4.

$[6 + 8 = 14]$

- Give the algorithm for processing N jobs through 3 machines, and state the assumptions which must be satisfied for the algorithm to be valid.
- Determine the optimal sequence of 5 jobs on 3 machines A, B, C with no passing between machines. Processing times (in hours) are:

Machine	Job 1	Job 2	Job 3	Job 4	Job 5
A	4	9	8	6	5
B	5	6	2	3	4
C	8	10	6	7	5

Question 5.

$[6 + 4 + 4 = 14]$

- What do you understand by an equipment replacement problem? Name the replacement policies for equipment that fails completely.
- Explain how to find the dual of the general form of a primal LPP.
- With the help of a table, explain how the optimal solutions of a general primal and its dual are related.

Question 6.

$[4 + 10 = 14]$

- What do you understand by “network analysis”? List the problems that can be addressed by network analysis.
- The following network gives links and associated lengths of roads in Seervada Park. Location O is the entrance and T is the end station. Park management wants to install telephone lines to establish phone connections among every pair of stations from O to T, with minimal total miles of line.

Link	Nodes	Length (miles)
1	O – A	2
2	O – B	4
3	O – C	3
4	A – B	1
5	A – D	4
6	B – C	6
7	B – D	2
8	C – E	7
9	D – E	3
10	D – T	4
11	E – T	5

Can this problem be solved by the Minimum Spanning Tree Problem (MSTP) technique?
Give the algorithm of MSTP and find a solution if possible.

Question 7.

$$[4 + 3 + 7 = 14]$$

- (a) What is a convex programming problem?
- (b) What is the direct substitution method of solving a non-linear programming problem?
- (c) Find the optimal solution of the following problem using the method of Lagrangean multipliers:

$$\begin{aligned} \text{Minimize } Z &= 2x_1^2 + x_2^2 + x_3^2 - 4x_1 - 2x_2 - 8x_3 \\ \text{Subject to: } &x_1 + x_2 + x_3 = 2, \quad x_1, x_2, x_3 \geq 0 \end{aligned}$$